Ultra-low-power Monostatic Backscatter Platform with Phase-Aware Channel Estimation and System-Level Validation

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***Abstract*— This paper presents a novel phase-aware channel-estimation (CE) method and an experimentally validated hardware–software platform for an ultra-low-power monostatic backscatter link. Conventional phase-stationary least squares (LS) and linear minimum mean square error (LMMSE) estimators incur deterministic bias when residual phase-slope is present across pilot symbols, a backscatter-specific impairment that is particularly consequential in monostatic configurations. To address this limitation, this paper introduces a lifted pilot model augmented with a linearized residual-phase-slope regressor, yielding closed-form phase-aware LS and LMMSE estimators with minimal additional computation. A double-Rician fading model characterizes the cascaded forward and backscatter hops, and a resource-optimal pilot allocation rule is analytically derived to balance CE accuracy against payload efficiency. The prototype integrates a semi-passive tag, a software-defined radio (SDR) reader, and a 2×1 planar Yagi–Uda array providing 7 dBi gain and over 30 dB isolation at 2.4–2.5 GHz. The receiver combines pilot-aided carrier frequency offset (CFO) compensation, the proposed lifted estimators, and zero-forcing (ZF) equalization for symbol recovery over the cascaded backscatter channel. It achieves 500 kbps at 1 m, with tag-side power consumption of 10 µW for the radio-frequency (RF) switch and 11.2 mW including the general-purpose microcontroller unit (MCU), corresponding to 20 pJ/bit and 22.4 nJ/bit, respectively. Measured error vector magnitudes (EVMs) are 2.97% for on-off keying (OOK) and 4.02% for binary phase-shift keying (BPSK). Bit error rate (BER) measurements and over-the-air (OTA) full-color image reconstruction confirm reliable multimedia-capable operation and provide practical co-design guidance for future Internet of Things (IoT) deployments.**

***Index Terms*— Backscatter communication, Internet of Things (IoT), channel estimation, zero-forcing equalization, resource allocation, carrier frequency offset, software-defined radio (SDR), monostatic architecture, planar Yagi–Uda antenna.**

I. INTRODUCTION

THE rapid growth and pervasive adoption of the Internet of Things (IoT) impose unprecedented demands on wireless communication technologies. Meeting these demands calls for solutions that are scalable, energy-efficient, and economically sustainable. Conventional wireless systems depend on active radio-frequency (RF) transceivers that consume significant power, are dependent on batteries, and incur high operational and maintenance costs. As these constraints hinder scaling IoT, the pursuit of ultra-low-power wireless solutions, particularly backscatter communication, has emerged as a central research direction for near-zero-power connectivity. By passively modulating incident or dedicated RF signals, backscatter devices drastically reduce energy consumption and enable battery-free or near-battery-free operation. This transformative capability establishes backscatter as a foundational technology for the expanding IoT and emerging 6G era, both of which are projected to connect devices at unprecedented scale. Such exponential growth further underscores the need for scalable, ultra-low-power communication methods that can meet the connectivity and maintenance requirements of next-generation IoT ecosystems.

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A. Motivation

Most prior backscatter systems have been confined to ultra-low-throughput applications such as radio-frequency identification (RFID) and basic sensing. The emerging demand for multimedia communication—ranging from high-resolution video surveillance and real-time biomedical telemetry to intelligent transportation and augmented-reality services—introduces far more stringent requirements in throughput, latency, and reliability [1], [2], [3]. In particular, smart city infrastructures exemplify the growing demand for multimedia-oriented backscatter connectivity: such capabilities could enhance public safety, optimize traffic flow, and enable personalized healthcare, thereby transforming urban environments. However, enabling such services with backscatter links requires more than increasing the raw data rate; it also requires reliable channel acquisition, interference suppression, and signal recovery under the severe power and hardware constraints.

Meeting these requirements is impeded by several fundamental challenges. Backscatter channels are inherently susceptible to deep fades that degrade reliability; residual carrier phase distortions compromise channel estimation (CE) accuracy; and the absence of dedicated energy sources constrains sustainable operation. Furthermore, large-scale deployment would necessitate robust interference suppression, efficient pilot allocation, and infrastructure capable of integrating backscatter devices with edge and cloud intelligence. These gaps highlight the need for new system designs that move beyond basic identification and enable robust multimedia transmission while operating within strict power constraints.

B. Related Work

Recent research has advanced backscatter beyond identification and sensing along three directions: throughput enhancement, channel modeling, and interference suppression.

1) Throughput and Spectral Efficiency Enhancement: Early backscatter systems employed simple amplitude-shift keying (ASK) or phase-shift keying (PSK) modulation. Building on these schemes, higher-order schemes such as 16-quadrature amplitude modulation (QAM) have been introduced, improving both spectral efficiency and data rates [4], [5]. Recent studies have extended this line of work to millimeter-wave (mmWave) gigabit backscatter front-ends [6], [7]. In addition, optimized high-order modulation strategies based on transfer learning have been proposed to enhance dual-channel multiple-input multiple-output (MIMO) systems [8]. These developments position backscatter as a technology capable of evolving beyond identification and sensing into high-throughput, multimedia communication. As backscatter scales toward multiuser and multimedia-capable operation, waveform and transform-domain techniques developed for modern multiuser wireless systems—such as discrete Fourier transform (DFT)-spread orthogonal frequency division multiplexing (OFDM) with frequency-domain spectrum shaping (FDSS) [9] and integrated Walsh–Hadamard/DFT processing [10]—provide architectural references for future extensions targeting higher spectral efficiency and stronger multiuser interference control.

2) Backscatter Channel Modeling and Estimation: Foundational measurement studies revealed that small-scale fading in backscatter links behaves as the product of two independent channels, effectively doubling the path loss exponent [11]. Subsequent models have characterized spatial fading in both monostatic and bistatic scenarios [12], while retrodirective architectures were shown to alleviate the double-fading problem, restoring single-hop channel characteristics [13]. On the estimation side, approaches such as least squares and DFT-based techniques for large antenna readers [14], reciprocity-driven full-duplex estimation [15], and low-complexity schemes employing reconfigurable surfaces [16] have been developed to improve robustness. These methods form the foundation for acquiring reliable channel state information (CSI) in both ambient and dedicated backscatter systems.

3) Interference Mitigation and Reliable Detection: Another important research direction addresses interference, which is especially critical in monostatic and bistatic architectures where strong direct-link or self-interference can overshadow the weak backscattered signal. Hardware solutions include carrier suppression loops in ultra-high-frequency (UHF) RFID readers [17], tunable isolation structures for co-polarized antennas [18], and enhanced metasurface monostatic designs [19]. On the algorithmic side, interference cancellation techniques have been developed for bistatic networks [20], while ambient backscatter using digital video broadcasting (DVB) signals has demonstrated practical interference rejection [21]. More recent works introduced likelihood-based detectors for distributed MIMO systems [22] and optimized methods to guarantee reliability under channel uncertainty and fading dynamics [23]. These approaches illustrate the complementary role of physical isolation techniques and advanced signal processing in ensuring scalable and reliable backscatter links. For multi-tag extensions, FDSS-based discrete cosine transform (DCT)-spread massive-MIMO OFDM frameworks offer relevant design insights, employing block-diagonalization (BD) precoding for multiuser-interference (MUI) suppression alongside Cholesky-decomposition-based zero-forcing (CD-ZF) and lattice reduction-based minimum mean square error (LR-MMSE) detection for bit error rate (BER) improvement [24].

C. Key Contribution

Although notable advances have been made, important challenges remain unresolved. Realizing reliable, multimedia-oriented backscatter systems requires integrating algorithmic innovation with practical hardware, while simultaneously achieving ultra-low power consumption and high data rates. This work addresses these gaps through the following key contributions:

- A backscatter-specific residual phase-slope impairment is identified in monostatic pilot observations, causing deterministic bias in conventional phase-stationary CE even at high signal-to-noise ratio (SNR) [25], [26]. The proposed phase-aware estimator reformulates the scalar channel unknown as a two-parameter vector within the same normal-equation framework, eliminating this bias while preserving closed-form least squares (LS) and linear minimum mean square error (LMMSE) solutions with minimal additional computation.
- A unified hardware–software framework is established by integrating theoretical modeling, phase-aware CE, and hardware prototyping, closing the gap between analytical estimation methods and practical monostatic backscatter implementation.
- The architecture achieves ultra-low-power operation, high data-rate transmission, and meter-scale communication, with simulations and experimental results validating its performance beyond conventional low-throughput backscatter systems.
- Multimedia-capable monostatic backscatter operation is experimentally demonstrated through BER evaluation and over-the-air (OTA) full-color image reconstruction, confirming the practicality of robust, high-throughput backscatter IoT links.

The remainder of this paper is organized as follows. The overall system architecture is described in Section II. Section III presents the system model, including the channel model and pilot resource allocation. Section IV details the proposed phase-aware CE and equalization framework. Section V describes the hardware implementation. Section VI reports the experimental and numerical results, and Section VII concludes the paper.

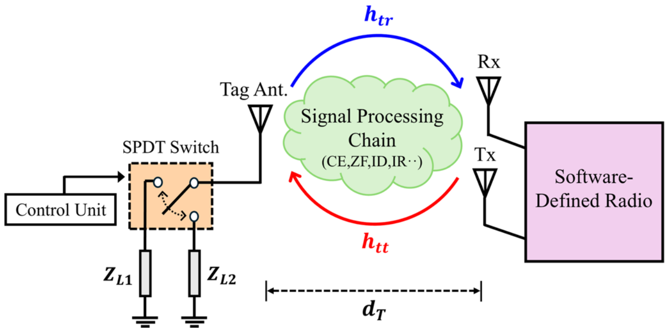


Fig. 1. Block diagram of the proposed monostatic BMC system (d_T: the distance between the tag and the transceiver).

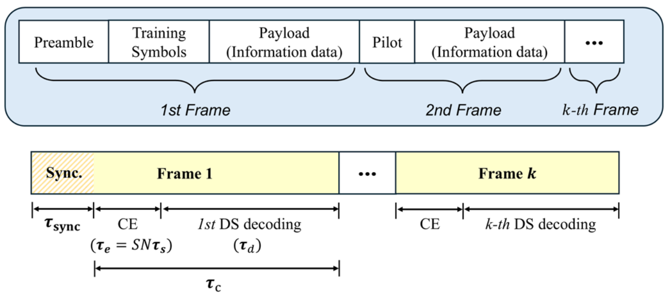


Fig. 2. Frame structure for channel estimation and information decoding.

II. System Overview

To enable robust multimedia communication in backscatter communication systems, a monostatic backscatter multimedia data communication (BMC) system is designed, which integrates custom-designed hardware with a rigorously structured signal-processing pipeline. The full framework of the proposed system is illustrated in Fig. 1, which shows the interaction between the software-defined radio (SDR)-based reader and the semi-passive backscatter tag. The key architectural decision to adopt a monostatic configuration—where both transmitting and receiving antennas are placed close to each other—reduces hardware complexity but introduces critical challenges such as self-interference, limited communication range, and severely attenuated SNR [27]. To address these challenges, a tailored hardware design is developed by constructing a high-gain planar Yagi–Uda antenna with split-ring resonators (SRRs) between the transmit (Tx) and receive (Rx) antennas. In parallel, the signal processing pipeline in software is partitioned into two complementary stages: real-time pre-processing on the SDR and offline post-processing on the host PC. The backscatter tag employs a single-pole double-throw (SPDT) switch toggling between two impedance loads Z_{L1} and Z_{L2}, thereby modulating the incident carrier with either on-off keying (OOK) or binary phase-shift keying (BPSK). The tag's modulation is entirely passive and driven by a low-power control circuit, ensuring energy efficiency [28], [29]. The SDR transceiver transmits a continuous wave (CW) carrier signal and simultaneously captures the backscattered signal via the same reader antenna.

Upon reception, the SDR executes the first stage of signal conditioning (pre-processing). The first operation is DC offset removal, which is essential to eliminate the baseband bias caused by the inherent characteristics of the backscatter process and internal hardware mismatches, especially prominent in monostatic architecture due to strong self-interference. Next, coarse carrier synchronization and phase alignment are applied to suppress linear phase rotation associated with residual carrier-frequency offset and Tx–Rx synthesizer mismatch. A frequency-domain low-pass filter (LPF) based on fast Fourier transform (FFT) is then used to isolate the narrowband backscattered signal from wideband noise and adjacent spectral leakage, yielding a cleaner signal representation for further processing [30].

In the post-processing stage, additional signal refinement and high-level decoding are performed. A second layer of DC offset removal further suppresses any residual baseline distortion. Pilot-aided CE is then conducted using both LS and LMMSE techniques. The proposed lifted LS estimator provides a closed-form, phase-aware estimate of the backscatter channel coefficient using known pilot symbols, while the lifted LMMSE estimator, which has slightly higher computational complexity than the LS method, leverages second-order statistics to minimize estimation errors in noisy environments [31]. The inclusion of both estimators allows for comparative analysis of their estimation accuracy and subsequent impact on the BER performance. Following CE, a one-tap zero-forcing (ZF) equalizer applies the inverse of the estimated effective cascaded channel to compensate for the phase rotation and amplitude scaling induced by the cascaded forward channel and backscatter channel. After equalization, the signal is demodulated using either OOK or BPSK, followed by information decoding (ID) and image reconstruction (IR).

The pipeline establishes a clear partition between hardware-level and baseband-level mitigation: antenna-level isolation suppresses the dominant self-interference, while the baseband stages address residual impairments that hardware cannot fully eliminate, including deterministic baseband bias, carrier-frequency and phase mismatch, and multiplicative distortions arising from the effective channel.

III. System Model

A. Backscatter Channel Model in SISO Systems

In monostatic single-input single-output (SISO) backscatter communication systems, the received baseband signal is affected by two constituent wireless channels: the forward channel from the reader to the tag, denoted h_{tt}, and the backscatter channel from the tag back to the reader, denoted h_{tr}. Since the backscatter process is governed by passive reflection, the effective channel at the receiver is defined as the cascaded multiplicative product $h_{eq} \triangleq h_{tt}h_{tr}$. To characterize this composite fading behavior, a double-Rician fading channel model is adopted for the forward and backscatter channels. This model is motivated by the targeted monostatic indoor deployment, where both reader–tag and tag–reader hops contain dominant line-of-sight (LoS) components. Single-hop

Rayleigh or Rician models collapse the cascaded backscatter channel into one equivalent coefficient, whereas a double-Rayleigh model preserves the product structure but neglects specular power. The double-Rician model captures both effects and includes simpler regimes as limiting cases, reducing to double-Rayleigh fading when both hop Rician factors approach zero and to a deterministic cascaded channel when both approach infinity.

Specifically, $h_{tt} \sim \mathcal{CN}(V_{tt}, \sigma_{tt}^2)$ and $h_{tr} \sim \mathcal{CN}(V_{tr}, \sigma_{tr}^2)$, where V_{tt} and V_{tr} are the deterministic amplitudes of the LoS components, and σ_{tt}^2, σ_{tr}^2 are the diffuse powers. The corresponding Rician K–factors are given by $K_{tt} = |V_{tt}|^2/\sigma_{tt}^2$, $K_{tr} = |V_{tr}|^2/\sigma_{tr}^2$, which quantify the ratio of specular-to-diffuse power in each link [32]. The envelope of the effective channel $r \triangleq |h_{eq}| = |h_{tt}h_{tr}|$ follows a non-trivial probability distribution derived from the product of two independent Rician variables. Following [33, Eq. 6.66], the probability density function (PDF) of r is given by:

$$p_h(r) = \frac{r}{\sigma_{tt}^2\sigma_{tr}^2}\exp(-(K_{tt}+K_{tr}))\sum_{i=0}^{\infty}\sum_{l=0}^{\infty}\frac{1}{(i!\,l!)^2}\left(\frac{rK_{tt}}{2\sigma_{tt}^2}\right)^i \times \left(\frac{rK_{tr}}{2\sigma_{tr}^2}\right)^l \left(\frac{\sigma_{tt}}{\sigma_{tr}}\right)^{i-l} K_{i-l}\left(\frac{r}{\sigma_{tt}\sigma_{tr}}\right), \qquad r \geq 0, \quad (1)$$

where $K_\nu(\cdot)$ is the modified Bessel function of the second kind with order ν. The received instantaneous SNR is defined as:

$$\gamma = \frac{r^2\mathcal{E}_s}{N_0}. \quad (2)$$

Applying the change of variable $r = \sqrt{\gamma N_0/\mathcal{E}_s}$ yields the induced SNR density:

$$f_\gamma(\gamma) = \frac{1}{2}p_h\left(\sqrt{\frac{\gamma N_0}{\mathcal{E}_s}}\right)\sqrt{\frac{N_0}{\mathcal{E}_s\gamma}}, \qquad \gamma > 0, \quad (3)$$

where $\mathcal{E}_s$ denotes the energy of the symbol. In a multiplicative backscatter channel, moderate fading on either hop creates deep fades on h_{eq}, so γ fluctuates much more strongly than in single-hop Rayleigh/Rician fading channels [34]. Consequently, the performance must be measured in expectation over the fading distribution rather than over instantaneous additive white Gaussian noise (AWGN) channels. The average-effective SNR is then obtained from the second moments of the constituent Rician fading:

$$\bar{\gamma}_{\text{eff}} = \mathbb{E}\left[|h_{eq}|^2\right]\frac{\mathcal{E}_s}{N_0} = (\sigma_{tt}^2 + |V_{tt}|^2)(\sigma_{tr}^2 + |V_{tr}|^2)\frac{\mathcal{E}_s}{N_0} = (1+K_{tt})(1+K_{tr})\sigma_{tt}^2\sigma_{tr}^2\frac{\mathcal{E}_s}{N_0}. \quad (4)$$

Given a conditional error probability $P_e(\gamma)$, the corresponding average effective error probability is defined as:

$$\bar{P}_e = \int_0^\infty P_e(\gamma)f_\gamma(\gamma)\,d\gamma = \int_0^\infty P_e\left(\frac{\mathcal{E}_s}{N_0}r^2\right)p_h(r)\,dr. \quad (5)$$

Note that monostatic measurements provide access only to the end-to-end effective channel h_{eq}; therefore, the hop-to-hop correlation $\mathrm{corr}(h_{tt}, h_{tr})$ is not directly identifiable from the received record. While such dependence does not change the averaging structure in (5), it changes the distribution of h_{eq} and hence the induced SNR density f_γ. Accordingly, the independent-product model for f_γ should be replaced by the density implied by the joint hop distribution, or by an empirical estimate obtained from measured γ samples under identical receiver processing. The general average-error expression in (5) is specialized in Section VI-C for OOK and coherent BPSK with their respective conditional error probabilities, yielding the average BER expressions used in the receiver evaluation.

B. Optimal Resource Allocation and Frame Design

Backscatter frames are designed to balance two competing requirements: (i) longer training reduces CE error and improves the effective SNR of the cascaded links, whereas (ii) shorter training increases the payload fraction and thereby the achievable data rate. The tension is acute under cascaded Rician fading because small estimation errors propagate strongly through the product channel and the reader estimates CSI only once per frame. Frame design is posed as an optimization problem that allocates time between pilots and data to satisfy a reliability target with minimum overhead. As shown in Fig. 2, let τ_{sync} denote the preamble duration, τ_e the training or tracking time, and τ_c the per-frame transmission time. In the proposed format, a frame is composed of S pilot slots, each containing a pilot block of N successive symbols. With symbol period τ_s, the total training time is defined as $\tau_e \triangleq SN\tau_s$, and the payload duration within a frame is $\tau_d = \tau_c - \tau_e$. Throughout this subsection, all times are expressed in seconds, so average energies scale linearly with the time durations under constant transmit power [35], [36].

Given the transmitted pilot power p_0, the total pilot energy is defined as $\mathcal{E}_p = p_0\tau_e$. Without committing to a particular estimator, the single-frame Cramér-Rao lower bound (CRLB) is used to obtain a scalar bound on the CE mean square error (MSE), while the full two-parameter Fisher information matrix (FIM) under colored noise is addressed in Section IV-D. The resulting scalar bound is given by

$$\sigma_e^2(\tau_e) \;\geq\; N_0/\mathcal{E}_p. \quad (6)$$

This error is then mapped to the backscattered average-effective SNR by a standard one-frame-inversion model,

$$\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e) \approx \frac{\bar{\gamma}_{\text{eff}}}{1 + \bar{\gamma}_{\text{eff}}\sigma_e^2(\tau_e)/\sigma_h^2}, \quad (7)$$

where $\bar{\gamma}_{\text{eff}}$, in the absence of CE error, is given in (4), and $\sigma_h^2 = \mathbb{E}\left[|h_{eq}|^2\right]$. Substituting (6) into (7) yields a simple monotone improvement law:

$$\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e) \approx \bar{\gamma}_{\text{eff}}\frac{\tau_e}{\tau_e + \alpha}, \qquad \alpha \triangleq \frac{\bar{\gamma}_{\text{eff}}N_0}{p_0\sigma_h^2}. \quad (8)$$

The rational dependence in (8) follows directly from the marginal two-parameter CRLB derived in Section IV-D after eliminating the nuisance ϕ_1 via the Schur complement, which yields the τ_e scaling of the CE error variance. The expression above reveals a fundamental trade-off in frame design: increasing τ_e improves CE accuracy, thereby enhancing the SNR $\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e)$ (hence lowers $\bar{P}_e$); however, it also reduces the fraction of symbols available for payload transmission. Denoting the pilot fraction by $\rho = \tau_e/\tau_c$, the resulting effective spectral efficiency per frame is expressed as:

$$\mathcal{R}(\tau_e) = (1-\rho)\log_2\left(1+\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e)\right)$$
$$= \left(1-\frac{\tau_e}{\tau_c}\right)\log_2\left(1+\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e)\right). \quad (9)$$

Longer training increases reliability but decreases throughput, whereas shorter training improves throughput but degrades reliability. By explicitly quantifying this trade-off between reliability and throughput, the optimal pilot length can be determined by formulating an optimization problem [37]. Specifically, given a target reliability $\bar{P}_{\text{tar}}$ and a minimum spectral efficiency, τ_e is chosen to maximize $\mathcal{R}(\tau_e)$ subject to the constraints imposed by the target BER, total frame duration, and minimum data rate:

$$\begin{aligned}
(\text{P1}):\ &\max_{\tau_e\geq 0}\ \mathcal{R}(\tau_e)\\
\text{s.t.}\ \ &\bar{P}_e\left(\bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e)\right) \leq \bar{P}_{\text{tar}},\\
&\tau_{\text{sync}}+\tau_e \ \leq\ \tau_c,\\
&\left(1-\frac{\tau_e}{\tau_c}\right)R\ \geq\ R_{\text{min}}.
\end{aligned} \quad (10)$$

The objective $\mathcal{R}(\tau_e)$ can be analyzed in closed form using (8)–(9). Let $g(\tau_e) = \bar{\gamma}_{\text{eff}}^{\text{CE}}(\tau_e)$, then $\mathcal{R}(\tau_e) = (1-\rho)\log_2(1+g(\tau_e))$. One verifies that $g'(\tau_e) = \bar{\gamma}_{\text{eff}}\frac{\alpha}{(\tau_e+\alpha)^2} > 0$ and $g''(\tau_e) < 0$. The first-order condition of (P1) on the continuous relaxation is obtained by differentiation:

$$\frac{d\mathcal{R}}{d\tau_e} = -\frac{1}{\tau_c}\log_2(1+g(\tau_e)) + \frac{1-\tau_e/\tau_c}{\ln 2}\cdot\frac{g'(\tau_e)}{1+g(\tau_e)}. \quad (11)$$

The first term is negative and its magnitude grows monotonically with τ_e, while the second term is positive and decreases, which proves unimodality and guarantees a unique stationary point $\hat{\tau}_e \in (0,\tau_c)$ that can be found by a one-dimensional search. Feasibility clips this stationary point to the interval induced by the constraints in (10). Writing the reliability target as an SNR threshold $\Gamma_{\text{tar}} = \bar{P}_e^{-1}(\bar{P}_{\text{tar}})$, the bounds are expressed as:

$$\tau_{\min} = \max\left\{\tau_s, \frac{\alpha\Gamma_{\text{tar}}}{\bar{\gamma}_{\text{eff}}-\Gamma_{\text{tar}}}\right\}, \quad (12)$$

and the corresponding upper bound on the feasible training time is determined by the frame-duration and minimum-rate constraints as:

$$\tau_{\max} = \min\left\{\tau_c-\tau_{\text{sync}}, \tau_c\left(1-\frac{R_{\min}}{R}\right)\right\}. \quad (13)$$

Therefore, the optimal training time can be obtained as:

$$\tau_e^{\star} = \min\{\max\{\hat{\tau}_e, \tau_{\min}\}, \tau_{\max}\}. \quad (14)$$

In implementation, $\tau_e^{\star}$ is converted to an integer pilot length by selecting a pilot placement and quantizing to symbols. Let the placement parameter be $\beta \in \{1,2,\cdots,S\}$, where $\beta = 1$ denotes a single contiguous pilot burst and larger β denotes pilot placement over multiple slots, with one block of N symbols in each selected slot. Let $\hat{\tau}_e$ denote the unique stationary point of $\mathcal{R}(\tau_e)$. The training time realized by N pilot symbols is $\tau_e^{\star} = \beta N\tau_s$. The implementable pilot length is therefore $N^{\star} = \lceil\tau_e^{\star}/(\beta\tau_s)\rceil$, which guarantees that the BER target used to obtain $\tau_e^{\star}$ is not violated. In cases where the frame budget is tight, $N^{\star}$ can be set to:

$$N^{\star} \leftarrow \max\{n \in \mathbb{Z}_+ : \tau_{\text{sync}} + \beta n\tau_s \leq \tau_c\}. \quad (15)$$

Based on (14)–(15), the realized pilot fraction and payload duration are $\rho^{\star} = \tau_e^{\star}/\tau_c$ and $\tau_d^{\star} = \tau_c - \tau_{\text{sync}} - \tau_e^{\star}$.

IV. SOFTWARE FRAMEWORK

After front-end conditioning in the SDR, the received backscatter waveform is stored as complex baseband samples. The post-processing stage on the host PC then executes a deterministic chain that removes residual baseline terms, estimates the effective backscatter channel consistent with the cascaded double-Rician fading, inverts that channel with a single frame equalizer, and finally decodes the information stream. The design goal is to deliver unbiased inputs to the estimators, conserve SNR at low operating levels, and keep each block's assumptions explicit and verifiable.

A. DC Offset Removal

A CW reader uses closely spaced Tx and Rx antennas, which causes strong self-interference at the receiver. Local oscillator (LO) feedthrough and residual in-phase/quadrature (I/Q) imbalance add a constant baseband term [38]. Quasi-static environmental reflectors contribute additional constant power independent of the tag switching. This DC component appears as a constant bias term in the discrete-time complex baseband signal, and if not removed, it can significantly degrade subsequent CE and detection stages.

Let $y[n]$ denote the complex baseband signal obtained from the SDR after pre-processing. Over the operating interval, the received samples are well described by:

$$y[n] = h_{eq}x[n] + d_{\text{DC}} + w[n]. \quad (16)$$

Here, $x[n]$ is the digital baseband sequence corresponding to the tag-modulated signal observed after carrier recovery, d_{DC} is the dataset-wide DC offset term, and $w[n]$ is zero-mean AWGN. The offset evolves on a time scale much longer than a frame; therefore, a single estimate for the entire recording is adopted. Received pilot samples are gathered in the index set $\mathcal{P}_k \subset \{n_0,\cdots,n_{k-1}\}$, and the remaining payload samples in frame k are indexed by disjoint set $\mathcal{D}_k$. The DC estimator relies on a pilot design that enforces a zero-mean tag symbol sequence over $\mathcal{P}_k$, namely $\mathbb{E}[x[n] \mid n \in \mathcal{P}_k] = 0$. This condition is satisfied for both modulation formats by defining $x[n]$ as an antipodal two-level sequence and constructing the pilot with equal occupancy of its two levels over $\mathcal{P}_k$.

In the BPSK configuration, the tag state sequence is antipodal by construction, $x[n] \in \{+1, -1\}$, and the pilot uses an equal number of $+1$ and -1 symbols over $\mathcal{P}_k$. Hence $\sum_{n\in\mathcal{P}_k} x[n] = 0$ and $\mathbb{E}[x[n] \mid n \in \mathcal{P}_k] = 0$ hold exactly.

In the OOK configuration, only two fixed load states are available due to the SPDT constraint. Let the tag-induced coefficient take the two deterministic values x_{l_0} and x_{l_1} with $x_{l_0} \neq x_{l_1}$, so $x_{ph}[n] \in \{x_{l_0}, x_{l_1}\}$. Defining $x_m \triangleq (x_{l_1} + x_{l_0})/2$, $x_k \triangleq (x_{l_1} - x_{l_0})/2$, and $x[n] \triangleq (x_{ph}[n] - x_m)/x_k$ yields $x[n] \in \{+1, -1\}$ with $x_{ph}[n] = x_m + x_k x[n]$ for all n. Let $\bar{h}_{eq}$ denote the effective complex gain that maps the tag-induced coefficient $x_{ph}[n]$ to the received baseband sample and is treated as constant within a frame, and let d_{rx} denote an additive switching-independent bias. The received sample is modeled as $y[n] = \bar{h}_{eq} x_{ph}[n] + d_{rx} + w[n] = \bar{h}_{eq} x_k x[n] + (d_{rx} + \bar{h}_{eq} x_m) + w[n]$. Accordingly, (16) holds with the effective definitions $h_{eq} \triangleq \bar{h}_{eq} x_k$ and $d_{\mathrm{DC}} \triangleq d_{rx} + \bar{h}_{eq} x_m$. Under this normalization, the effective channel gain h_{eq} in (16) absorbs the deterministic OOK differential factor x_k, so that the subsequent pilot observation and lifted regression operate on a single complex scalar corresponding to the double-Rician product channel up to the scalar factor x_k.

The OOK pilot alternates the two load states with equal occupancy over $\mathcal{P}_k$, so that $\sum_{n\in\mathcal{P}_k} x[n] = 0$ and $\mathbb{E}[x[n] \mid n \in \mathcal{P}_k] = 0$ hold exactly. Combined with $\mathbb{E}[w[n]] = 0$, taking the conditional expectation of (16) over the pilot indices yields $\mathbb{E}[y[n] \mid n \in \mathcal{P}_k] = d_{\mathrm{DC}}$, which justifies estimating the dataset-wide DC offset term from the pilot sample mean and removing it prior to CE. Consequently, the DC offset is calculated as the mean of the pilot samples:

$$\hat{d}_{\mathrm{DC}} = \frac{1}{T_k^{\mathrm{tot}}} \sum_{n\in\mathcal{P}_k} y[n], \tag{17}$$

where T_k^{tot} denotes the number of pilot samples in $\mathcal{P}_k$. The estimator is unbiased and efficient; its mean and variance are given by $\mathbb{E}[\hat{d}_{\mathrm{DC}}] = d_{\mathrm{DC}}$, and $\sigma^2(\hat{d}_{\mathrm{DC}}) = \sigma_w^2 / T_k^{\mathrm{tot}}$. The debiased sequence is then expressed as:

$$\tilde{y}[n] = y[n] - \hat{d}_{\mathrm{DC}}, \tag{18}$$

which removes the estimated DC offset and provides the debiased sequence used in the subsequent CE process.

B. Residual Phase-Aware Channel Estimation

Reliable decoding in the monostatic SISO backscatter link requires an accurate estimate of the cascaded effective channel, as even small estimation errors propagate multiplicatively through the two-hop link. When the effective channel gain is unknown, the payload samples are subject to uncompensated phase rotation and amplitude scaling, which degrades the post-equalization SNR and produces an error floor that persists even as payload energy increases. This challenge is especially severe in backscatter systems, where residual phase-slope, primarily arising from reader-side oscillator and phase-locked loop (PLL) mismatch after coarse carrier recovery, appears as a multiplicative phase distortion on the cascaded two-hop observation.

To handle these issues, phase-aware LS and LMMSE estimators are developed that explicitly model the residual intra-frame phase slope as a small linear perturbation, in contrast to conventional estimators that assume phase stationarity over the pilot block. The resulting estimators retain linearity, admit closed-form solutions, and suppress the systematic bias that their phase-stationary counterparts suffer in the presence of residual drift.

After DC offset removal across the entire received signal, the received pilot samples within each coherence frame follow a flat SISO fading channel model. This assumption is supported by 2.45-GHz indoor measurements of the received baseband pilot response. With a sampling rate of 1 MS/s, the dominant multipath energy was concentrated within one sampling interval, and the remaining energy above the -20 dB level was confined within 2 μs. The measured root-mean-square (RMS) delay spreads were 0.722 μs for OOK and 0.543 μs for BPSK. Because both values are below 1 μs, a conservative bound of $\tau_{\mathrm{rms}} \leq 1$ μs is adopted, yielding a coherence bandwidth on the order of 200 kHz under the 0.5 correlation criterion. The small difference between the two delay spreads is attributed to separate acquisition runs and pilot-response extraction effects, since physical propagation delay spread is determined primarily by the reader–tag geometry and indoor multipath environment rather than by the tag modulation format.

The symbol duration at 500 kbps is 2 μs, which exceeds the measured delay spreads; hence, inter-symbol interference caused by frequency-selective fading is negligible under the stated conditions. An FFT-domain narrowband observation window with cutoff frequency $f_L = 200$ kHz is applied only to the branch used for coherence-bandwidth assessment and pilot-aided CE, not to the data-decoding branch. Thus, the channel response over the estimation band can be modeled as flat during each frame, while the data-decoding branch operates at the 500-kbps symbol rate. Any residual distortion is further compensated by the equalization stage in Section IV-C. Nevertheless, the flat-fading approximation may break down if the signal bandwidth exceeds the coherence bandwidth or if Doppler and residual oscillator drift produce excessive intra-frame phase rotation, as quantified in Section IV-B3.

Under the operating conditions, the received pilot samples can therefore be modeled as:

$$\tilde{y}_p[n] = h_{eq} e^{j\phi_1 n} x_p[n] + w[n], \tag{19}$$

where ϕ_1 is a residual phase-slope parameter, introduced by imperfections such as hardware mismatch, slight frequency offsets, or oscillator instability. The exponential term $e^{j\phi_1 n}$ models a small linear phase drift across the pilot symbols. $x_p \triangleq (A_s - \xi_i) x_0$ denotes the known pilot symbols, where A_s

is the load-independent antenna structural mode, ξ_i denotes the selected load reflection coefficient, and x_0 is the underlying pilot waveform. This separation confines load-dependent effects to ξ_i while keeping A_s load-invariant. The assumption that h_{eq} remains constant over the pilot interval is justified by the following operating conditions. In the BMC system, the receiver gain was fixed after instrument warm-up, and the measurements were conducted in a controlled indoor environment maintained at room temperature (25 °C). In addition, CE was confined to steady-state pilot samples within the optimized CE interval, whose duration is much shorter than the channel coherence time. Repeated pilot responses across frames also exhibited highly consistent end-to-end behavior, supporting the validity of the single-scalar approximation for h_{eq} over the pilot block.

1) Lifted Least Squares Estimation Method: Assuming that the residual phase-slope is small over a short pilot interval, it can be approximated as $e^{j\phi_1 n} \approx 1 + j\phi_1 n$, which follows from the first-order Taylor series expansion. Introducing the lifted parameters $\theta_0 \triangleq h_{eq}$ and $\theta_1 \triangleq j\phi_1 h_{eq}$, the received samples are approximated by the following linear form,

$$\tilde{y}_p[n] \approx \theta_0 x_p[n] + \theta_1 n x_p[n] + w[n]. \tag{20}$$

Stacking the pilot samples over $n \in \mathcal{P}_k$, let the observation vector $\mathbf{y_p} = \left[\tilde{y}_p[n]\right]_{n\in\mathcal{P}_k} \in \mathbb{C}^{|\mathcal{P}_k|\times 1}$ and the pilot vector $\mathbf{x_p} = \left[x_p[n]\right]_{n\in\mathcal{P}_k} \in \mathbb{C}^{|\mathcal{P}_k|\times 1}$. The design matrix is then written as:

$$\mathbf{X_p} \triangleq \left[\mathbf{x_p}\ \mathbf{n} \odot \mathbf{x_p}\right], \tag{21}$$

with $\mathbf{X_p} \in \mathbb{C}^{|\mathcal{P}_k|\times 2}$, $\mathbf{n} = [n]_{n\in\mathcal{P}_k}$, and $\odot$ denoting the elementwise product. The linear model with $\boldsymbol{\theta} = [\theta_0\ \theta_1]^T$ is written as:

$$\mathbf{y_p} = \mathbf{X_p}\boldsymbol{\theta} + \mathbf{w}. \tag{22}$$

The phase-aware LS problem is formulated as follows:

$$\widehat{\boldsymbol{\theta}}_{\mathrm{LS}} = \arg\min_{\boldsymbol{\theta}} \left\|\mathbf{y_p} - \mathbf{X_p}\boldsymbol{\theta}\right\|_2^2. \tag{23}$$

From the first-order stationarity condition, the solution is obtained by the normal equations in closed form,

$$\widehat{\boldsymbol{\theta}}_{\mathrm{LS}} = \left(\mathbf{X_p^H X_p}\right)^{-1}\mathbf{X_p^H y_p}, \tag{24}$$

where $\mathbf{X_p}^H$ is the Hermitian transpose of $\mathbf{X_p}$. To evaluate this efficiently and to expose the trade-offs of the designed pilots, five sufficient frame-level statistics are introduced: $s_0 \triangleq \sum_{n\in\mathcal{P}_k}\left|x_p[n]\right|^2$, $s_1 \triangleq \sum_{n\in\mathcal{P}_k} n\left|x_p[n]\right|^2$, $s_2 \triangleq \sum_{n\in\mathcal{P}_k} n^2\left|x_p[n]\right|^2$, $t_0 \triangleq \sum_{n\in\mathcal{P}_k} x_p^*[n]\,\tilde{y}_p[n]$, $t_1 \triangleq \sum_{n\in\mathcal{P}_k} n x_p^*[n]\,\tilde{y}_p[n]$, where s_0 is the pilot energy, s_1 quantifies the time centering of that energy, s_2 quantifies its temporal spread, t_0 is the matched-filter between the pilot and the received samples, and t_1 is the same correlation weighted by the sample index to capture the small linear phase variation across the pilot burst.

With these definitions, the Gram matrix and right-hand side reduce to

$$\mathbf{X_p^H X_p} = \begin{bmatrix} s_0 & s_1 \\ s_1 & s_2 \end{bmatrix}, \qquad \mathbf{X_p^H y_p} = \begin{bmatrix} t_0 \\ t_1 \end{bmatrix}, \tag{25}$$

which are the only quantities required to obtain the closed-form estimator. The estimator is identifiable when the Gram matrix is positive definite, which in this notation is the single condition

$$\Delta \triangleq s_0 s_2 - s_1^2 > 0. \tag{26}$$

Under this condition, the solution of the normal equations is obtained by inverting the 2×2 matrix:

$$\begin{bmatrix} \hat{\theta}_0 \\ \hat{\theta}_1 \end{bmatrix} = \frac{1}{\Delta}\begin{bmatrix} s_2 & -s_1 \\ -s_1 & s_0 \end{bmatrix}\begin{bmatrix} t_0 \\ t_1 \end{bmatrix}. \tag{27}$$

Consequently, the LS estimate of the effective channel h_{eq} is obtained from the first component as:

$$\hat{h}_{\mathrm{LS}} \triangleq \left(\widehat{\boldsymbol{\theta}}_{\mathrm{LS}}\right)_0 = \frac{s_2 t_0 - s_1 t_1}{\Delta}. \tag{28}$$

From (27), the small residual phase-slope component can be isolated from the normal-equation solution. A diagnostic estimate is therefore defined by normalizing $\hat{\theta}_1$ by $\hat{\theta}_0$ and taking the imaginary part,

$$\hat{\phi}_1 = \Im\left\{\frac{\hat{\theta}_1}{\hat{\theta}_0}\right\} = \Im\left\{\frac{s_0 t_1 - s_1 t_0}{s_2 t_0 - s_1 t_1}\right\}. \tag{29}$$

The rationale for taking the imaginary part follows directly from the lifted parameterization: by construction, $\theta_1 = j\phi_1\theta_0$, so their ratio $\theta_1/\theta_0 = j\phi_1$ is purely imaginary. Accordingly, $\Im\{\hat{\theta}_1/\hat{\theta}_0\}$ recovers $\hat{\phi}_1$ directly and simultaneously suppresses noise-induced perturbations that fall on the real axis.

The residual phase-slope component merits separate analysis, as even a small residual slope disrupts phase stationarity, degrades coherent combining, and can inflate the phase contribution to estimation error. If this phase is ignored and the conventional LS estimator $\tilde{h}_{\mathrm{OLS}} = t_0/s_0$ is used, a first-order expansion of $e^{j\phi_1 n}$ yields

$$\tilde{h}_{\mathrm{OLS}} \approx h_{eq} + j\phi_1 h_{eq}\frac{s_1}{s_0} + \frac{1}{s_0}\sum_{n\in\mathcal{P}_k} x_p^*[n]w[n], \tag{30}$$

so the estimate is systematically biased by the term $j\phi_1 h_{eq}\, s_1/s_0$ whenever the pilot indices are non-centered in time. That bias would be carried into the equalizer as a combined amplitude-and-phase error and would not disappear even at high SNR. The proposed two-parameter LS formulation removes this issue because the second column $\mathbf{n} \odot \mathbf{x_p}$ explicitly models the small linear phase across the pilots, and the cross terms in the closed-form solution above cancel the effect of s_1 for any pilot placement that satisfies $\Delta > 0$.

The estimation accuracy follows directly from standard LS error covariance analysis. With zero-mean noise of variance σ_w^2,

$$\mathrm{Cov}\left(\widehat{\boldsymbol{\theta}}_{\mathrm{LS}} - \boldsymbol{\theta}\right) = \sigma_w^2\left(\mathbf{X_p^H X_p}\right)^{-1} = \frac{\sigma_w^2}{\Delta}\begin{bmatrix} s_2 & -s_1 \\ -s_1 & s_0 \end{bmatrix}, \tag{31}$$

and the LS error variance is written as:

$$\sigma_{e,\mathrm{LS}}^2 = \frac{\sigma_w^2 s_2}{\Delta}. \tag{32}$$

These expressions expose the trade-off with frame design. For a fixed pilot energy s_0, centering the pilot indices so that $s_1 = 0$ maximizes Δ and minimizes $\sigma_{e,\mathrm{LS}}^2$. Increasing the temporal span of the pilots increases s_2 and improves the conditioning through $\Delta = s_0 s_2 - s_1^2$, provided s_1 remains close to zero. These relations guide the choice of pilot locations when allocating training time inside a frame.

To characterize when the pilot Gram determinant Δ becomes small, note that a pure translation of the pilot indices does not change Δ. Let $n' = n + c$ for an integer shift c, while keeping the pilot samples $x_p[n]$ unchanged. Under this translation, $s_0' = s_0$, $s_1' = s_1 + c s_0$, and $s_2' = s_2 + 2 c s_1 + c^2 s_0$. Substituting into $\Delta' \triangleq s_0' s_2' - (s_1')^2$ yields $\Delta' = \Delta$. Therefore, within-frame placement shifts do not cause Δ to approach zero. For fixed s_0, only the index spread inside the pilot set $\mathcal{P}_k$ and the normalized energy distribution across $\mathcal{P}_k$ affect Δ. Define the normalized non-negative weights $\Omega_n \triangleq |x_p[n]|^2 / s_0$ over $n \in \mathcal{P}_k$, with $\sum_{n \in \mathcal{P}_k} \Omega_n = 1$. The energy-weighted mean index and variance can be written as $\mu_\Omega = s_1 / s_0$, $\sigma_\Omega^2 = s_2 / s_0 - (s_1 / s_0)^2$. Then the determinant admits the exact factorization $\Delta = s_0^2 \sigma_\Omega^2$. Therefore, for fixed nonzero pilot energy s_0, $\Delta \to 0$ holds if and only if $\sigma_\Omega^2 \to 0$, which occurs only when the effective pilot support collapses to an almost-single index in an energy-weighted sense. This regime is promoted by insufficient pilot length, tight clustering of pilot indices, or strong energy concentration $|x_p[n]|^2$ on a narrow subset of $\mathcal{P}_k$. In contrast, spreading pilot indices and avoiding energy concentration keeps σ_Ω^2 away from zero, which keeps Δ bounded away from zero.

Numerical stability can be assessed through the condition number of the pilot Gram matrix. Introducing the centered index $\tilde{n} \triangleq n - \mu_\Omega$, it follows that $\sum_{n \in \mathcal{P}_k} \tilde{n} |x_p[n]|^2 = 0$, so the two regressors $\mathbf{x}_\mathbf{p}$ and $\tilde{\mathbf{n}} \odot \mathbf{x}_\mathbf{p}$ are orthogonal under the energy inner product. The corresponding centered Gram matrix becomes diagonal with diagonal entries proportional to s_0 and $s_0 \sigma_\Omega^2$, which yields the closed-form expression for the condition number $\kappa_2 = \max\{\sigma_\Omega^2, \sigma_\Omega^{-2}\}$. For any prescribed conditioning threshold $\kappa_{\mathrm{th}} > 1$, the numerical ill-conditioned regime is equivalently characterized by $\kappa_2 > \kappa_{\mathrm{th}}$, that is, $\sigma_\Omega^2 > \kappa_{\mathrm{th}}$ or $\sigma_\Omega^2 < \kappa_{\mathrm{th}}^{-1}$, which links the numerical stability directly to the energy-weighted index spread of the pilot.

2) Lifted Linear Minimum Mean Square Error Estimation Method: When second-order channel statistics are available, the estimator is obtained by minimizing the MSE over affine functions of the observation vector $\mathbf{y}_\mathrm{p}$:

$$(\mathbf{M}, \mathbf{c})^\star = \arg \min_{\mathbf{M} \in \mathbb{C}^{|\mathcal{P}_k| \times 2}, \mathbf{c} \in \mathbb{C}^2} \mathbb{E}\left[\left\| \boldsymbol{\theta} - \left(\mathbf{M}^H \mathbf{y}_\mathbf{p} + \mathbf{c} \right) \right\|^2 \right], \tag{33}$$

where the expectation is over the channel, the residual phase term, and the receiver noise. A Gaussian prior is adopted to summarize slow temporal statistics across frames: $\boldsymbol{\theta} \sim \mathcal{CN}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Lambda}_{\boldsymbol{\theta}})$, where $\boldsymbol{\mu}_{\boldsymbol{\theta}} = [\mu_{\theta_0}\ \mu_{\theta_1}]^T$ and $\boldsymbol{\Lambda}_{\boldsymbol{\theta}} = \mathrm{diag}(\sigma_{\theta_0}^2, \sigma_{\theta_1}^2)$, with $\mu_{\theta_1} \approx 0$, since the residual phase slope is assumed to fluctuate around zero. Under this linear-Gaussian model, the resulting LMMSE estimator is the posterior mean,

$$\begin{aligned}\widehat{\boldsymbol{\theta}}_{\mathrm{LM}} = \boldsymbol{\mu}_{\boldsymbol{\theta}} + \boldsymbol{\Lambda}_{\boldsymbol{\theta}} \mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \left(\sigma_w^2 \mathbf{I}_{|\mathcal{P}_k|} + \mathbf{X}_\mathbf{p} \boldsymbol{\Lambda}_{\boldsymbol{\theta}} \mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \right)^{-1} \\ \times \left(\mathbf{y}_\mathbf{p} - \mathbf{X}_\mathbf{p} \boldsymbol{\mu}_{\boldsymbol{\theta}} \right).\end{aligned} \tag{34}$$

Starting from the lifted model (22) with (21), the normal equations depend only on the Gram matrix $\mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \mathbf{X}_\mathbf{p}$, already obtained in (25), and on the data-prior correlation vector $\mathbf{X}_\mathbf{p}^{\boldsymbol{H}} (\mathbf{y}_\mathbf{p} - \mathbf{X}_\mathbf{p} \boldsymbol{\mu}_{\boldsymbol{\theta}})$. Using s_0, s_1, s_2, t_0, and t_1, already defined for the LS estimator, the expressions can be written as:

$$\mathbf{X}_\mathbf{p}^{\boldsymbol{H}} (\mathbf{y}_\mathbf{p} - \mathbf{X}_\mathbf{p} \boldsymbol{\mu}_{\boldsymbol{\theta}}) = \begin{bmatrix} t_0 - \mu_{\theta_0} s_0 - \mu_{\theta_1} s_1 \\ t_1 - \mu_{\theta_0} s_1 - \mu_{\theta_1} s_2 \end{bmatrix}, \tag{35}$$

which separates the information from the received matched filters (t_0, t_1) and the subtraction of what would be expected from the prior $\boldsymbol{\mu}_{\boldsymbol{\theta}}$. To keep the algebra compact and to make the role of pilot energy, pilot spread, prior, and noise explicit, introduce the scalars: $b_{00} \triangleq \sigma_{\theta_0}^{-2} + \sigma_w^{-2} s_0$, $b_{11} \triangleq \sigma_{\theta_1}^{-2} + \sigma_w^{-2} s_2$, $b_{01} \triangleq \sigma_w^{-2} s_1$, $a_0 \triangleq \sigma_w^{-2} t_0 + \sigma_{\theta_0}^{-2} \mu_{\theta_0}$, $a_1 \triangleq \sigma_w^{-2} t_1 + \sigma_{\theta_1}^{-2} \mu_{\theta_1}$, and the positive determinant is defined as:

$$\Delta_B \triangleq b_{00} b_{11} - b_{01}^2 > 0. \tag{36}$$

The coefficients b_{00} , b_{11} , b_{01} collect, respectively, prior precision and noise–weighted pilot moments into a two-by-two normal matrix, while a_0, a_1 collect the prior–data correlations. With these definitions the posterior mean of the LMMSE problem reduces to a single 2×2 inversion,

$$\begin{aligned}\widehat{\boldsymbol{\theta}}_{\mathrm{LM}} &= \left(\boldsymbol{\Lambda}_{\boldsymbol{\theta}}^{-1} + \sigma_w^{-2} \mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \mathbf{X}_\mathbf{p} \right)^{-1} \left(\sigma_w^{-2} \mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \mathbf{y}_\mathbf{p} + \boldsymbol{\Lambda}_{\boldsymbol{\theta}}^{-1} \boldsymbol{\mu}_{\boldsymbol{\theta}} \right) \\ &= \frac{1}{\Delta_B} \begin{bmatrix} b_{11} & -b_{01} \\ -b_{01} & b_{00} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}.\end{aligned} \tag{37}$$

Consequently, the LMMSE estimate of the effective channel gain is obtained from the first component,

$$\hat{h}_{\mathrm{LM}} = \left(\widehat{\boldsymbol{\theta}}_{\mathrm{LM}} \right)_0 = \frac{b_{11} a_0 - b_{01} a_1}{\Delta_B}. \tag{38}$$

The posterior error covariance follows from the same normal matrix,

$$\begin{aligned}\mathrm{Cov}\left(\widehat{\boldsymbol{\theta}}_{\mathrm{LM}} - \boldsymbol{\theta} \mid \mathbf{y}_\mathbf{p} \right) &= \left(\boldsymbol{\Lambda}_{\boldsymbol{\theta}}^{-1} + \sigma_w^{-2} \mathbf{X}_\mathbf{p}^{\boldsymbol{H}} \mathbf{X}_\mathbf{p} \right)^{-1} \\ &= \frac{1}{\Delta_B} \begin{bmatrix} b_{11} & -b_{01} \\ -b_{01} & b_{00} \end{bmatrix},\end{aligned} \tag{39}$$

and the LMMSE posterior error variance, given by the first diagonal entry of the posterior error covariance matrix, is expressed as:

$$\sigma_{e,\mathrm{LM}}^2 = \frac{b_{11}}{\Delta_B}. \tag{40}$$

The posterior variance b_{11} / Δ_B is governed by s_0, s_1, s_2, prior precision, and noise variance through Δ_B. While the cross term s_1 enters Δ_B through $-b_{01}^2$ and can increase the posterior variance under non-centered placement ($s_1 \neq 0$), the lifted LMMSE removes the first-order deterministic phase-slope bias that would otherwise contaminate a scalar LMMSE [39]. For comparison, the conventional scalar LMMSE estimator that ignores the second column of $\mathbf{X_p}$ solves the unlifted problem $\mathbf{y_p} = h_{eq}\mathbf{x_p} + \mathbf{w}$ and yields the familiar shrinkage,

$$\tilde{h}_{\mathrm{OLM}} = \frac{\sigma_{\theta_0}^2 s_0}{\sigma_{\theta_0}^2 s_0 + \sigma_w^2 s_0}\frac{t_0}{s_0} + \frac{\sigma_w^2}{\sigma_{\theta_0}^2 s_0 + \sigma_w^2}\mu_{\theta_0}. \tag{41}$$

If a small linear phase persists across the pilots and the pilot indices are not centered, a first-order expansion shows that $\tilde{h}_{\mathrm{OLM}}$ inherits an s_1-dependent deterministic bias component proportional to $j\phi_1 h_{eq} s_1/s_0$, scaled by the LMMSE shrinkage factor in (41). The proposed lifted LMMSE removes this first-order bias by jointly estimating the effective channel coefficient and the residual phase-slope component, while preserving Bayesian shrinkage toward the calibrated prior for short pilot blocks or low-SNR regimes with minimal additional computational complexity.

3) Linearization Validity under Channel Estimation Bias Constraint: Although the proposed LS and LMMSE estimators admit closed-form solutions under the linearized per-frame model, their validity must be assessed against the exact nonlinear observation model in (19). The subsequent bias analysis quantifies the approximation error introduced by the first-order expansion and verifies that the residual CE bias remains small within the derived admissible range for both estimators.

Let $\mathbf{G}$ denote a common 2×2 normal matrix used by each estimator, and define it as:

$$\mathbf{G} \triangleq \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}, \tag{42}$$

$$\mathbf{G} = \begin{cases} \mathbf{X_p^H X_p} = \begin{bmatrix} s_0 & s_1 \\ s_1 & s_2 \end{bmatrix}, & \text{LS}, \\ \mathbf{X_p^H X_p} + \sigma_w^2 \boldsymbol{\Lambda}_{\boldsymbol{\theta}}^{-1} = \begin{bmatrix} s_0 + \lambda_0 & s_1 \\ s_1 & s_2 + \lambda_1 \end{bmatrix}, & \text{LMMSE}, \end{cases} \tag{43}$$

with $\lambda_i \triangleq \sigma_w^2/\sigma_{\theta_i}^2$ denoting the noise-to-prior variance ratio for the *i*-th lifted parameter. Comparing the exact and linearized models via the first-order Taylor expansion with remainder $e^{j\phi_1 n} = 1 + j\phi_1 n + b[n]$, where the Taylor remainder satisfies $|b[n]| \leq (|\phi_1|^2 n^2)/2$, isolates a deterministic sample-wise mismatch $m_n = h_{eq} x_p[n] b[n]$ that propagates through the same normal equations. Thus, the resulting bias of the lifted estimators can be written as:

$$\mathbb{E}\big[\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\big] = \mathbf{G}^{-1}\mathbf{X_p^H m}, \tag{44}$$

where $\mathbf{m} = [m_n]_{n \in \mathcal{P}_k} \in \mathbb{C}^{|\mathcal{P}_k| \times 1}$ stacks the sample-wise mismatch terms. Bounding $|b[n]|$ and using $\mathbf{G}$ yield the upper bound on CE bias due to linearization mismatch as follows:

$$\frac{\big|\mathbb{E}[\hat{\theta}_0] - \theta_0\big|}{|h_{eq}|} \leq \frac{|\phi_1|^2}{2}\frac{s_2(g_{22} + |s_1| n_{\max})}{\det \mathbf{G}}, \tag{45}$$

Algorithm 1 Phase-Aware Lifted Channel Estimation

Input: $\{\tilde{y}_p[n]\}_{n \in \mathcal{P}_k}$, $\{x_p[n]\}_{n \in \mathcal{P}_k}$, σ_w^2, mode $\mathfrak{C} \in \{\text{LS, LMMSE}\}$. If LMMSE: prior $(\boldsymbol{\mu}_\theta, \boldsymbol{\Lambda}_\theta)$.
Initialization: Set $s_i \leftarrow 0$ for $i = 0, 1, 2$ and $t_j \leftarrow 0$ for $j = 0, 1$.
for each $n \in \mathcal{P}_k$ **do**
  $s_i \leftarrow s_i + n^i |x_p[n]|^2$, $i = 0, 1, 2$.
  $t_j \leftarrow t_j + n^j x_p^*[n]\tilde{y}_p[n]$, $j = 0, 1$.
end for
if $\mathfrak{C}$ = LS **then**
  $\Delta \leftarrow s_0 s_2 - s_1^2$.
  $\hat{\theta}_0 \leftarrow (s_2 t_0 - s_1 t_1)/\Delta$.
  $\hat{\theta}_1 \leftarrow (s_0 t_1 - s_1 t_0)/\Delta$.
  $\hat{h}_{\mathrm{LS}} \leftarrow \hat{\theta}_0$.
else if $\mathfrak{C}$ = LMMSE **then**
  $b_{00} \leftarrow \sigma_{\theta_0}^{-2} + \sigma_w^{-2} s_0$, $b_{11} \leftarrow \sigma_{\theta_1}^{-2} + \sigma_w^{-2} s_2$.
  $b_{01} \leftarrow \sigma_w^{-2} s_1$.
  $a_i \leftarrow \sigma_w^{-2} t_i + \sigma_{\theta_i}^{-2}\mu_{\theta_i}$, $i = 0, 1$.
  $\Delta_B \leftarrow b_{00} b_{11} - b_{01}^2$.
  $\hat{\theta}_0 \leftarrow (b_{11} a_0 - b_{01} a_1)/\Delta_B$.
  $\hat{\theta}_1 \leftarrow (b_{00} a_1 - b_{01} a_0)/\Delta_B$.
  $\hat{h}_{\mathrm{LM}} \leftarrow \hat{\theta}_0$.
end if
$\hat{\phi}_1 \leftarrow \Im\{\hat{\theta}_1/\hat{\theta}_0\}$.
Output: $\hat{h} \in \{\hat{h}_{\mathrm{LS}}, \hat{h}_{\mathrm{LM}}\}$, $\hat{\phi}_1$.

where $n_{\max} \triangleq \max_{n \in \mathcal{P}_k} |n|$ denotes the largest absolute pilot index within the $\mathcal{P}_k$, and (45) is valid for both the LS and LMMSE through the single choice of $\mathbf{G}$.

With a target fraction $\eta \in (0,1)$, (45) yields the explicit admissible region for the residual phase-slope:

$$|\phi_1| \leq \sqrt{\frac{2\eta \det \mathbf{G}}{s_2(g_{22} + |s_1| n_{\max})}}. \tag{46}$$

For time-centered pilot indices, $s_1 = 0$, so the $|s_1| n_{\max}$ term vanishes and Δ is maximized for stable inversion. Under this condition, the admissible phase-slope bound simplifies to

$$|\phi_1| \leq \sqrt{\frac{2\eta g_{11}}{s_2}}, \qquad g_{11} = \begin{cases} s_0, & \text{LS}, \\ s_0 + \lambda_0, & \text{LMMSE}. \end{cases} \tag{47}$$

Under the centered consecutive indexing $n = -(N-1)/2, \cdots, (N-1)/2$, constant-modulus pilots satisfy $s_0 = N$ and $s_2 = \sum n^2 = N(N^2 - 1)/12$. Substituting into (47) then yields the following pilot-length constraint:

$$|\phi_1| \leq \sqrt{\frac{24\eta c_\lambda}{(N^2 - 1)}}, \qquad c_\lambda = \begin{cases} 1, & \text{LS}, \\ 1 + \dfrac{\lambda_0}{s_0}, & \text{LMMSE}. \end{cases} \tag{48}$$

For a measured $|\phi_1|$ and the same target fraction η, solving (48) for N gives the largest admissible pilot length that still satisfies the η-bias criterion:

$$N_{\max}(\phi_1; c_\lambda) = \left\lfloor \sqrt{1 + \frac{24\eta c_\lambda}{|\phi_1|^2}} \right\rfloor. \quad (49)$$

Equations (46)–(49) certify the validity of the linearized estimators by ensuring that the normalized residual CE bias remains below the prescribed tolerance η. They also provide explicit design relations that link the admissible residual phase-slope to the pilot length for both the LS and LMMSE estimators.

For practical assessment, evaluating (48) with the system parameters of this work yields an admissible per-sample phase slope on the order of 10^{-2} rad/sample under centered placement, corresponding to a frequency offset tolerance of several hundred Hz at the deployed symbol rate. This margin substantially exceeds the residual phase drift of reference-locked synthesizer architectures in quasi-static indoor environments, confirming that the first-order linearization holds with large margin under the targeted operating conditions; experimental quantification is provided in Section VI.

When the linearized per-frame phase model is used outside its certified regime, the neglected higher-order Taylor remainder produces a deterministic sample-wise model-mismatch term in the lifted observation model. Since this term is not a zero-mean additive disturbance for a given pilot support, its projection through the LS normal equations introduces a nonzero estimator bias. The CE accuracy must then be characterized by the full MSE rather than by the variance term alone, namely:

$$\mathrm{MSE_{LS}} = \mathbb{E}\left[\left|\hat{h}_{\mathrm{LS}} - \mathbb{E}[\hat{h}_{\mathrm{LS}}]\right|^2\right] + \left|\mathbb{E}[\hat{h}_{\mathrm{LS}}] - h_{eq}\right|^2. \quad (50)$$

Here, the first term is the variance of the LS estimate, whereas the second term is the squared deterministic bias induced by the linearization mismatch. The variance term retains the inverse-overhead scaling characterized by the marginal two-parameter CRLB, whereas the bias term is controlled by the admissible residual phase-slope conditions derived above. Within the admissible region, the prescribed η-criterion bounds this bias contribution, so the MSE remains variance-dominated and the effective-SNR mapping based on the marginal two-parameter CRLB scaling with τ_e remains valid. Outside this region, the mismatch-induced bias can become comparable to or larger than the variance term, producing an SNR-independent CE error floor even when the noise-induced variance decreases with increasing pilot energy.

The same projection framework can also be extended to an unmodeled quadratic phase term $\phi_2 n^2$. With the curvature mismatch $b_2[n] = j\phi_2 n^2$, substituting $m_n = h_{eq} x_p[n] b_2[n]$ into (44) and applying the η-criterion gives the general bound

$$|\phi_2| \le \eta \cdot \frac{\Delta}{|s_2^2 - s_1 s_3|} \triangleq |\phi_2|_{\lim}, \quad (51)$$

where $s_3 \triangleq \sum_{n\in\mathcal{P}_k} n^3 |x_p[n]|^2$ extends the pilot moment definition. For the deployed pilot configuration and $\eta = 0.1$, the centered placement yields $|\phi_2|_{\lim} = 8.34 \times 10^{-5}$ rad/sample^2, while the non-centered placement tightens the bound to 4.38×10^{-5} rad/sample^2 across the pilot block. Fig. 3 sweeps hypothetical $|\phi_2|$ and compares the two-parameter $\{1, n\}$ estimator against the $\{1, n, n^2\}$ control. Below the admissible bounds, $|\phi_2|_{\lim}$, both estimators remain near their respective variance floors. Beyond either bound, the two-parameter curves exhibit a mismatch-dominated degradation that steepens with $\bar{\gamma}_{\mathrm{eff}}$, whereas the three-parameter model shifts the degradation knee to larger $|\phi_2|$ values. At $|\phi_2|_{\lim}$, the $\{1, n, n^2\}$ control is consistently about 5.0–5.6 dB better than $\{1, n\}$; at the non-centered bound, the gap is 4.2–5.9 dB, confirming basis truncation of the unmodeled n^2 term as the dominant mechanism under both pilot placement conditions.

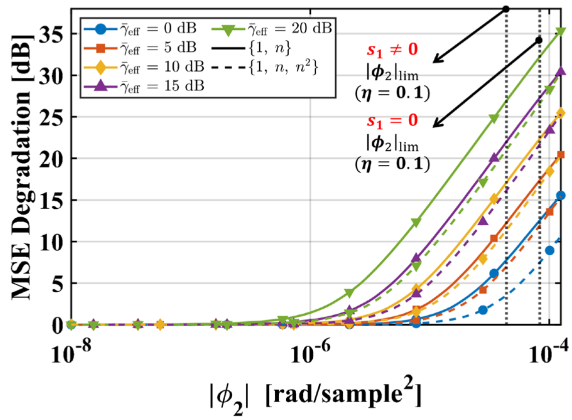


Fig. 3. MSE degradation of the two-parameter ($\{1, n\}$) and three-parameter ($\{1, n, n^2\}$) lifted estimators as a function of the unmodeled quadratic phase coefficient $|\phi_2|$. The vertical dotted lines indicate the admissible bounds, $|\phi_2|_{\lim}$ at $\eta = 0.1$ for centered ($s_1 = 0$) and non-centered ($s_1 \neq 0$) pilot placement.

The following remarks address how the proposed framework extends to practical deployment scenarios, focusing on a) multi-tag scalability and b) mobility tolerance with frame-design guidelines.

a) Multi-Tag Extension: When M_t tags share the same carrier, the superposed signal at the reader generally prevents joint identification of the per-tag parameters $(h_{eq,k}, \phi_{1,k})$ without structured pilot access. Time-division pilot scheduling restores identifiability by allowing only tag k to modulate in slot k, while the other tags hold a matched load ($\xi \approx 0$); the samples on $\mathcal{P}_k$ then conform to the single-tag model in (19). The phase-slope associated with tag k can be decomposed as $\phi_{1,k} = \phi_{1,\mathrm{com}} + \delta\phi_{1,k}$, where $\phi_{1,\mathrm{com}}$ is the reader-side common component largely mitigated by CFO compensation, and $\delta\phi_{1,k}$ captures tag-specific jitter and parasitic mismatch. Consequently, the lifted design matrix (21), the Gram structure (25), and the closed-form LS and LMMSE solutions (28), (38) apply per tag without modification. The total pilot overhead scales as $\tau_e^{(M_t)} = \sum_{k=1}^{M_t} N_k \tau_s$ under contiguous per-tag pilot placement, and the per-tag allocation can be guided by (P1) subject to the same frame-budget constraints. Thus, at the estimator level, the required modification is structured multi-tag pilot access, such as time-division multiple access (TDMA), while the per-tag estimator remains unchanged.

TABLE I
MOBILITY-DEPENDENT FRAME PARTITIONING GUIDELINE FOR THE PROPOSED SYSTEM

$\|v_D\|$ [m/s]	Scenario	$\|f_D\|$ [Hz]	$\Delta\phi_{\text{data}}$ [rad]	$\beta^\star$
≤ 0.95	Quasi-static	≤ 15.51	≤ 0.101	1
1.0	Normal walk	16.33	0.105	2
2.0	Fast walk	32.67	0.210	3
5.0	Jogging	81.67	0.526	6

b) Mobility Tolerance & Frame Design Guidelines: The system is designed for static and quasi-static indoor deployments. To quantify the velocity range over which the frame-wise phase-stationarity assumption remains valid, the Doppler-induced phase slope is compared with the centered-index admissible bound in (48). For a monostatic backscatter link, the backscattered component experiences a two-way Doppler shift under worst-case radial motion, given by $|f_D| = 2|v_D|f_c/c$. The corresponding per-sample phase increment is $|\phi_1|_D = 2\pi|f_D|\tau_s$, and requiring $|\phi_1|_D$ to remain within the admissible phase-slope bound provides a sufficient condition for the lifted estimator to remain valid over the pilot aperture $T_p = N^\star\tau_s$. For the adopted frame parameters, a separate and more restrictive condition governs the data block: the phase rotation accumulated over the payload duration $T_d = N_d\tau_s$ must remain within a phase-rotation budget ϕ_{flat}, namely $\Delta\phi_{\text{data}} = 2\pi|f_D|T_d \le \phi_{\text{flat}}$. Here, $\phi_{\text{flat}} = 0.1$ rad is adopted as a conservative phase-rotation budget, chosen consistently with the tolerance level $\eta = 0.1$ used throughout the paper. Solving this constraint for $|v_D|$ yields the critical velocity threshold $v_{D,\max} = \phi_{\text{flat}}c/(4\pi f_c T_d)$. With $f_c = 2.45$ GHz, $\tau_s = 2$ µs, and $N_d = 512$, this gives $v_{D,\max} \approx 0.95$ m/s, confirming that the targeted quasi-static indoor scenario remains within the allowable phase-rotation budget. Beyond $v_{D,\max}$, the frame can be partitioned into $\beta^\star$ subframes with independent CE updates per subframe, where $\beta^\star = \lceil |v_D|/v_{D,\max} \rceil$, subject to the same frame-budget constraints. Table I lists the required $\beta^\star$ for representative velocities exceeding this threshold.

Beyond the range covered by Table I, the approximately linear growth of $\beta^\star$ with $|v_D|$ increases the per-frame training overhead proportionally. Consequently, the payload fraction $1 - (\tau_{\text{sync}} + \beta^\star\tau_e^\star)/\tau_c$ shrinks toward zero and the effective throughput collapses. The per-frame partitioning ceases to be feasible once $\tau_{\text{sync}} + \beta^\star\tau_e^\star$ approaches τ_c; in this regime, the quasi-static observation model underlying (19) is no longer compatible with the frame-budget constraint in (P1), and the receiver would require symbol-level decision-directed (DD) tracking or shorter coherence frames.

C. Zero-Forcing Equalization

In each frame, the optimal designated pilot interval of duration τ_e provides an estimated scalar channel gain $\hat{h}$ for the effective channel h_{eq}. The subsequent payload samples of that same frame are then equalized by a single complex gain, defined as the ZF equalizer $p_{ZF} \triangleq \hat{h}^{-1}$, and applied symbol-by-symbol to form the normalized channel sequence before the decision over payload indices only. Under the flat SISO assumption at the symbol rate, the data samples are expressed as $y_d = h_{eq}x_d + w_d$, and the equalized symbols are $z_d \triangleq p_{ZF}y_d$. Using the data model, the equalized stream can be decomposed as:

$$z_d = x_d + \left(\frac{h_{eq}}{\hat{h}} - 1\right)x_d + \frac{w_d}{\hat{h}}. \tag{52}$$

This decomposition consists of a desired term identical to the transmitted symbols, a residual symbol distortion proportional to the normalized estimation mismatch $(h_{eq} - \hat{h})/\hat{h}$, and a noise component amplified by the factor $1/|\hat{h}|$. In the proposed monostatic backscatter setting, the end-to-end channel response within a frame is modeled as a single complex scalar, so a one-tap ZF equalizer is sufficient to compensate for the corresponding amplitude scaling and phase rotation without modifying the symbol timing or signal bandwidth [40].

The decomposition in (52) further reveals the low-SNR sensitivity of the ZF equalizer: as $|\hat{h}|$ approaches the deep-fade regime characteristic of the cascaded double-Rician fading channel, the noise term $w_d/\hat{h}$ is amplified inversely with the estimated channel magnitude, which motivates the comparative evaluation against the error-variance-aware (EA) MMSE equalizer in the subsequent analysis. Since the estimation-error power $\sigma_e^2(\tau_e)$ is governed by the allocated pilot duration and the selected estimator, the ZF sensitivity in (52) directly links equalization performance to the resource-allocation variables in (6)–(14).

D. Fisher Information Matrix and CRLB under Post-processing Colored Noise

To justify the scalar CRLB used in (6) under the same receiver pipeline, the FIM is derived for the joint estimation of the complex effective channel h_{eq} and the residual phase-slope parameter ϕ_1 in the presence of post-processing colored noise.

Let the stacked pilot observation vector after receiver-side pre-processing be written as:

$$\tilde{\mathbf{y}}_{\mathbf{p}} = h_{eq}\mathbf{D}(\phi_1)\mathbf{x}_{\mathbf{p}} + \mathbf{w}, \tag{53}$$

where $\mathbf{D}(\phi_1) \triangleq \operatorname{diag}\left(e^{j\phi_1 n}\right)_{n\in\mathcal{P}_k}$ models the intra-pilot linear phase drift in (19) and $\mathbf{w} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_w)$ is proper complex Gaussian noise whose generally non-diagonal covariance $\mathbf{C}_w$ captures the effect of the receiver pre-processing. In particular, the pre-processing can be represented as a fixed receiver-side linear operator $\mathbf{F}$ that includes the DC offset removal and low-pass filtering, so that $\mathbf{w} = \mathbf{F}\mathbf{v}$ for the raw noise vector $\mathbf{v}$ and $\mathbf{C}_w \triangleq \mathbb{E}[\mathbf{w}\mathbf{w}^{\boldsymbol{H}}] = \mathbf{F}\mathbf{C}_v\mathbf{F}^{\boldsymbol{H}}$. Therefore, $\mathbf{C}_w$ is parameter-independent with respect to (h_{eq}, ϕ_1) because $\mathbf{F}$ is receiver-side and does not depend on the unknown channel parameters, while $\mathbf{C}_v$ is determined by the receiver front-end and residual leakage statistics under fixed gain and fixed measurement conditions. This parameter-independence allows the Fisher information to follow directly from the Slepian-Bangs formula for proper complex Gaussian observations [41], [42], [43].

Define the auxiliary vectors $\mathbf{s} \triangleq \mathbf{D}(\phi_1)\mathbf{x_p}, \mathbf{g} \triangleq \mathbf{n} \odot \mathbf{s}$, and the quadratic forms $q_0 \triangleq \mathbf{s}^H \mathbf{C}_w^{-1}\mathbf{s}, q_1 \triangleq \mathbf{s}^H \mathbf{C}_w^{-1}\mathbf{g}, q_2 \triangleq \mathbf{g}^H \mathbf{C}_w^{-1}\mathbf{g}$. For the real parameterization of the two unknown parameters (h_{eq}, ϕ_1) as $\boldsymbol{\vartheta} \triangleq [\Re\{h_{eq}\}, \Im\{h_{eq}\}, \phi_1]^T$, the FIM can be written in closed form as:

$$\begin{aligned}\boldsymbol{J}(\boldsymbol{\vartheta}) &= 2\Re\left\{\left[\frac{\partial(h_{eq}\mathbf{s})}{\partial\vartheta_i}\right]^H \mathbf{C}_w^{-1}\left[\frac{\partial(h_{eq}\mathbf{s})}{\partial\vartheta_j}\right]\right\} \\ &= 2\begin{bmatrix} q_0 & 0 & -\Im\{h_{eq}q_1\} \\ 0 & q_0 & \Re\{h_{eq}q_1\} \\ -\Im\{h_{eq}q_1\} & \Re\{h_{eq}q_1\} & |h_{eq}|^2 q_2 \end{bmatrix}. \end{aligned} \tag{54}$$

The matrix CRLB is $\mathrm{Cov}(\boldsymbol{\vartheta}) \succcurlyeq \boldsymbol{J}(\boldsymbol{\vartheta})^{-1}$, since ϕ_1 is a nuisance parameter for the rate-overhead trade-off, the relevant bound for (6) is the marginal CRLB for estimating h_{eq} when ϕ_1 is unknown. Eliminating ϕ_1 via the Schur complement yields the corresponding lower bound on the complex-channel MSE:

$$\mathbb{E}\left[\left|\hat{h} - h_{eq}\right|^2\right] \geq \sigma_{\hat{h}}^2, \tag{55}$$

$$\sigma_{\hat{h}}^2 \triangleq \mathrm{tr}(\mathbf{S}^{-1}) = \frac{2q_0 - (|q_1|^2/q_2)}{2q_0[q_0 - (|q_1|^2/q_2)]}, \tag{56}$$

where $\mathbf{S}$ is the 2×2 Schur complement of $\boldsymbol{J}(\boldsymbol{\vartheta})$ associated with the nuisance coordinate ϕ_1. This bound is the rigorous counterpart of the scalar CE error variance used in (6)–(9), because the rate-overhead optimization requires a scalar lower bound on the estimation accuracy of the single complex coefficient h_{eq}, while treating ϕ_1 as a nuisance phase-drift parameter. In the special case where the post-processed noise is effectively white, $\mathbf{C}_w = \sigma_w^2\mathbf{I}$, and the coupling term is negligible, the bound reduces to $\sigma_{\hat{h}}^2 = \sigma_w^2/\|\mathbf{x_p}\|^2$, recovering the familiar inverse-energy scaling. Under the constant-power pilot assumption in Section III-B, $\|\mathbf{x_p}\|^2$ scales proportionally with the pilot energy $\mathcal{E}_p$, which yields the $1/(p_0\tau_e)$ scaling that motivates the scalar CRLB form adopted in (6).

Beyond the inverse-energy scaling, the rational dependence in (8) follows from the marginal two-parameter CRLB associated with the joint pilot-domain estimation of h_{eq} and ϕ_1. Under the pilot observation model used to form the two-parameter Fisher information, eliminating ϕ_1 via the Schur complement yields the marginal information for h_{eq} in the form $q_0 - |q_1|^2/q_2$. The information-loss term $|q_1|^2/q_2$ is non-negative and does not exceed q_0 because $|q_1|^2 \leq q_0 q_2$ holds under the $\mathbf{C}_w^{-1}$ inner product, so nuisance coupling through ϕ_1 can only reduce the information relative to the decoupled case.

To expose the τ_e-dependence, consider the constant-statistics pilot extension used to vary τ_e at fixed per-sample pilot power. Increasing τ_e appends additional pilot samples processed through the same fixed receiver chain. Consequently, the receiver-side noise-coloring structure remains fixed, while the dimensions of $\mathbf{C}_w$ and the pilot-domain regressors increase by concatenating samples with identical per-sample statistics. Under this construction, the pilot quadratic form q_0 accumulates additively with the pilot support and scales proportionally to $\mathcal{E}_p$. In contrast, since $\mathbf{g}$ includes the pilot index vector $\mathbf{n}$, the mixed and quadratic forms scale with higher-order moments of the support, yielding $|q_1| \propto p_0\tau_e^2$ and $q_2 \propto p_0\tau_e^3$ for the same pilot-duration sweep. Therefore, there exist τ_e-independent constants $k_0 > 0$, $k_2 > 0$, and k_1 such that $q_0 = k_0p_0\tau_e$, $q_1 = k_1p_0\tau_e^2$, and $q_2 = k_2p_0\tau_e^3$. Substituting these scaling factors into the Schur complement shows that the marginal information remains proportional to $p_0\tau_e$. Inserting the resulting CRLB scaling into the effective-SNR mapping gives $\bar{\gamma}_{\mathrm{eff}}^{\mathrm{CE}}(\tau_e) \approx \bar{\gamma}_{\mathrm{eff}}\tau_e/(\tau_e + \alpha)$, with α collecting the τ_e-independent coupling and noise-coloring constants. This establishes the mapping from the marginal two-parameter CRLB obtained by Schur-complement elimination of ϕ_1, rather than from a scalar bound that implicitly treats ϕ_1 as known.

V. Hardware Implementation

A. Tag Design and Modulation Schemes

The backscatter tag plays a critical role in encoding information onto an incident carrier wave through a low-power load modulation technique. The proposed tag employs a single antenna interfaced with a controllable SPDT switch to alternate between two distinct complex loads, Z_{L1} and Z_{L2}, enabling binary modulation. The reflection behavior of the tag is governed by the impedance mismatch between the antenna impedance Z_a and the selected load Z_L, resulting in a reflection coefficient ξ_i defined by:

$$\xi_i = \frac{Z_{Li} - Z_a^*}{Z_{Li} + Z_a}, \qquad i = 1, 2, \tag{57}$$

where Z_a^* denotes the complex conjugate of the antenna impedance. The modulation scheme is implemented by toggling between two distinct impedance states, each corresponding to a unique load reflection coefficient. For OOK modulation, one of the impedances is matched to the antenna impedance so that $\xi_i \approx 0$. The other impedance is deliberately mismatched to induce strong reflection with $|\xi_i| \approx 1$. This amplitude-based scheme is straightforward to implement and well-suited for energy-constrained tags. For BPSK modulation, the two impedance states are chosen so that their reflection coefficients satisfy $\xi_1 = -\xi_2$. This yields constant envelope backscatter with a phase separation of π.

To realize this modulation scheme, an ultra-low-power SPDT switch is designed and implemented to enable efficient impedance control for binary modulation in the backscatter tag. The switch employs the HMC221BE device from Analog Devices, which offers a typical insertion loss of 0.7 dB, contributing to the overall low loss performance of the circuit. The fabricated circuit, consuming only 10 µW, is constructed on a compact 40 mm × 25 mm printed circuit board (PCB) using a 0.2-mm-thick FR-4 substrate, which has a relative dielectric

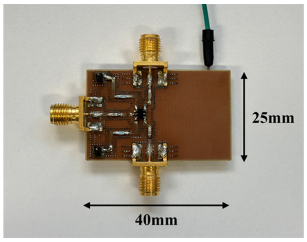


Fig. 4. Designed semi-passive tag for load modulation.

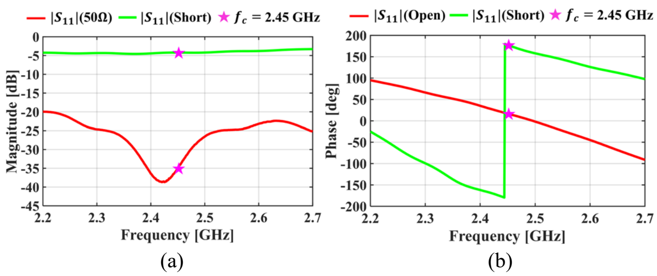


Fig. 5. Measured $|S_{11}|$ of the proposed load modulator (semi-passive tag) (a) magnitude and (b) phase.

constant (ε_r) of 4.14 and loss tangent (tan δ) of 0.02 as shown in Fig. 4. The reflection characteristics of the proposed switch were measured using a calibrated vector network analyzer (VNA), and the results are shown in Fig. 5. The measured reflection magnitude $|S_{11}|$ exhibited clear discrimination between the two load states as shown in Fig. 5(a). When terminated with a standard 50 Ω load representing the absorptive mode, the $|S_{11}|$ reached approximately –35.0 dB at the center frequency of 2.45 GHz, indicating minimal reflection consistent with a matched termination. In contrast, the short-circuit termination yielded a magnitude of –4.2 dB, corresponding to increased reflection. These results confirm the intended impedance contrast between the two states, necessary for reliable amplitude-based modulation. Furthermore, Fig. 5(b) shows the respective phase response under open and shorted load conditions. At 2.45 GHz, the measured reflection phase was approximately 16.8° in the shorted state and 177.5° in the open state, yielding a differential phase of 160.7°.

Accordingly, further analysis is required to identify the non-idealities leading to the deviation from the ideal 180° phase shift and complete reflection. Specifically, the –4.2 dB reflection in the reflective state is attributed to the insertion loss of the SPDT switch (~1.1 dB), losses in SMA connectors and cables (~1.5 dB), inherent insertion loss in the short termination itself, and parasitic capacitance arising from the switch layout and measurement setup. At the operating frequency, the non-negligible reactance of capacitive elements shifts the effective load impedance away from the ideal short/open states, further reducing the reflected amplitude when combined with practical loss mechanisms. As for the phase deviation, the short-to-open reflection phase difference fails to reach a perfect 180° because parasitic capacitances and inductances within the switch circuit introduce signal distortion. Even with a measured ~161° differential, robust antipodal modulation is maintained under practical backscatter conditions. These measurements confirm that the proposed switching circuit delivers sufficient phase and amplitude contrast for robust OOK and BPSK modulation in the backscatter tag.

Notably, the phase-aware estimator remains free of phase-contrast-induced bias despite the measured 160.7° phase difference. The normalization algorithm shown in (16)–(18) maps the two physical tag states to an equivalent binary representation $x[n] \in \{+1, -1\}$ by absorbing the magnitude asymmetry and phase deviation from 180° into the effective complex channel coefficient h_{eq}. After DC removal, the pilot observation model in (19) and the lifted regression in (20) therefore operate on the normalized sequence, so hardware non-ideality in the static phase contrast appears as a constant complex factor within h_{eq}. In the lifted sufficient statistics, this constant factor enters t_0 and t_1 only through the same scalar channel coefficient and the Gram moments, and therefore does not introduce an additional index-dependent phase-slope component. The ratio mapping in (29) further removes this constant factor because h_{eq} appears as a common multiplicative term in both θ_0 and θ_1. After normalization, the residual effect of the nonideal phase contrast is a reduction in modulation separation, $|x_1 - x_0|^2 \propto \sin^2(\Delta\bar{\psi}/2)$. At $\Delta\bar{\psi} =$ 160.7°, this corresponds to an equivalent SNR penalty of approximately 0.12 dB relative to an ideal 180° contrast, which is common to all estimator variants and negligible at the operating SNR.

In addition, the two absorption paths—switching-dependent distortion into h_{eq} and switching-independent bias into d_{DC}—preserve the equal-occupancy structure of the antipodal pilot sequence, so the zero-mean condition remains valid under practical hardware impairments.

B. Tx/Rx High Isolation Antenna Design

The proposed reader antenna subsystem, shown in Figs. 6 and 7, is designed to (i) maximize the directive gain toward the tag for strengthening the backscattered signal and (ii) provide high Tx–Rx port isolation via the integrated SRR structure. In the proposed platform, the SDR transmits CW signals and simultaneously receives the tag's modulated response on closely spaced ports. The short antenna separation induces strong electrical coupling between the transmit and receive feeds, allowing carrier leakage into the receive path. This leakage reduces the effective SNR by forcing the front-end to operate with limited gain, generating a strong deterministic component after down-conversion, and distorting the statistics used for DC-offset removal and pilot-aided CE. Consequently, a high-isolation antenna system is essential to ensure reliable operation of the entire signal processing chain. The detailed design parameters of the proposed antenna are listed in Table II.

The SRR array is integrated between the transmit and receive regions of the antenna board to suppress the dominant near-field electric coupling. An SRR stores electric energy in the gaps and magnetic energy in the loops. At resonance, the structure exhibits a high surface impedance to tangential electric fields,

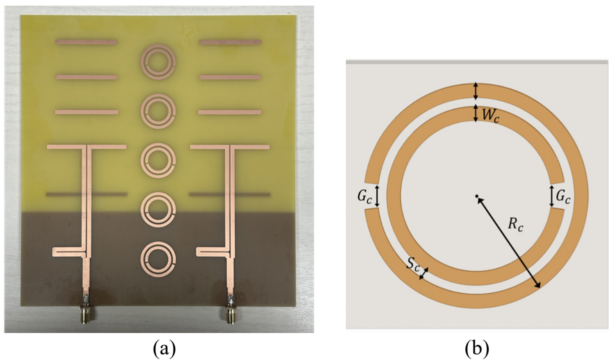


Fig. 6. (a) Fabricated 2×1 planar Yagi–Uda antenna array and (b) geometry of the SRR for isolation improvement.

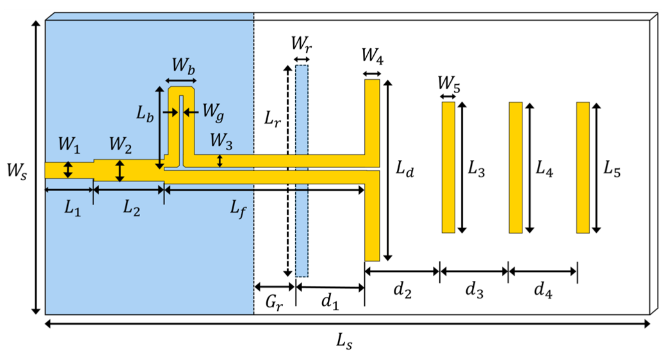


Fig. 7. Configuration of the proposed antenna.

TABLE II
DESIGN PARAMETERS OF ANTENNA & SRR (UNIT: MM)

W_s	76	W_b	5.5	L_3	37	L_r	48
W_1	3	W_g	0.5	L_4	36	d_1	28
W_2	4	W_r	2.5	L_5	36	d_2	20.5
W_3	2.5	L_s	170	L_b	17	d_3	20.5
W_4	2.5	L_1	11	L_d	46.5	d_4	20.5
W_5	2.5	L_2	17	L_f	63.75	G_r	8.75
G_c	1	R_c	11.5	W_c	1.5	S_c	0.5

effectively suppressing lateral field lines that could otherwise couple energy from the transmit feed into the receive feed. The magnitude of $|S_{21}|$ exhibited a deep transmission notch, and $|S_{11}|$ exhibited the companion resonance around 2.4–2.5 GHz as shown in Fig. 8. The resonators are first tuned at the unit cell level, then realized as an array on FR-4 substrate and placed between the two apertures. The measured S-parameters in Fig. 9 showed that incorporating the resonators reduced the inter-port coupling $|S_{21}|$ to below −30 dB, which is approximately 10 dB better than the case without resonators. At the same time, the input match $|S_{11}|$ remained better than −10 dB, confirming that the isolation enhancement did not come at the expense of matching or bandwidth.

The inherent inter-port coupling of the 2×1 planar Yagi–Uda array without the SRR structure was $|S_{21}| = -23.63$ dB at 2.45 GHz. On the SDR side, the USRP N200 reader employs a WBX

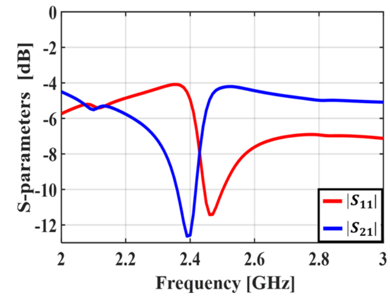


Fig. 8. Frequency response of the designed SRR: $|S_{11}|$ & $|S_{21}|$.

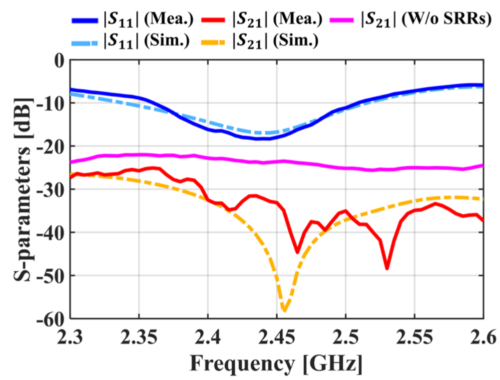


Fig. 9. Simulated and measured S-parameters of the proposed antenna: $|S_{11}|$ & $|S_{21}|$.

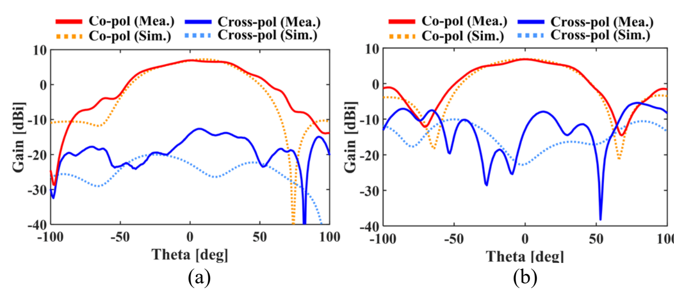


Fig. 10. Simulated and measured gain radiation patterns at 2.45 GHz for (a) $\phi = 0°$ in xz-plane and (b) $\phi = 90°$ in yz-plane.

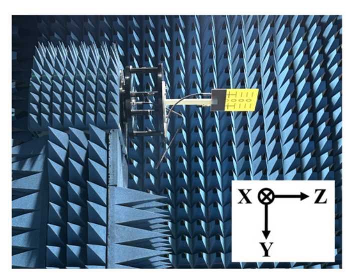


Fig. 11. The antenna measurement setup in the anechoic chamber.

wideband transceiver daughterboard with separate TX/RX and RX2 RF ports. In the implemented full-duplex configuration, transmission and reception are routed through separate RF ports, so the dominant measured leakage path is the antenna-level Tx–Rx coupling rather than port sharing inside the SDR. The SRR array is therefore specifically designed to suppress this antenna-level coupling. With the SRR structure integrated, the measured $|S_{21}|$ was reduced to below −30 dB, an improvement of approximately 6.4 dB, while $|S_{11}|$ remained better than −10 dB. The current Tx–Rx element spacing is approximately 20 mm, determined by the compact board footprint (76 mm × 170 mm) and the co-planar end-fire alignment required by the monostatic link geometry. Although additional isolation could be obtained through physical ground separation, increased element spacing, or perpendicular antenna orientation, the measured >30 dB isolation with the SRR structure provides sufficient suppression for the proposed receiver pipeline: The receiver-side DC-removal stage described in Section V-C suppresses the dominant deterministic leakage component, while the host-side DC-removal stage further reduces residual offset. Perpendicular antenna orientation would fundamentally alter the co-polarized end-fire radiation pattern and degrade the directive gain required for the monostatic backscatter link. Further isolation enhancement through ground separation or increased element spacing remains a promising direction for future designs targeting higher-order modulation or extended range.

While the achieved isolation effectively suppresses carrier leakage into the receive path, isolation by itself does not guarantee reliable operation in the monostatic backscatter link. To counteract the substantial double-hop path loss and the low backscattered signal power, the reader and the tag should use high-gain antennas directed along their LoS to maximize link reliability. To achieve this, planar Yagi–Uda antennas are adopted for both the tag and the reader. The antenna's geometry concentrates illumination onto the tag, enhances the front-to-back ratio, and selectively receives energy from the desired direction, thereby directly improving the SNR and increasing the communication distance. As shown in Fig. 10, the measured radiation patterns demonstrated a realized gain of 7 dBi at 2.45 GHz, with good agreement to simulations in both principal planes, thereby validating the array's end-fire radiation. The experimental performance characterization was carried out in an anechoic chamber, as depicted in Fig. 11.

C. Software-Defined Radio as the Reader: Pre-Processing

The reader employs the SDR platform in which the field-programmable gate array (FPGA) provides unified control of the transmit and receive signal paths, as illustrated in Fig. 12. During data transmission, the digital controller assembles the frame sequence and drives the digital up-converter (DUC) and digital-to-analog converter (DAC), which in turn feed the RF chain [44], [45].

The first operation removes the large DC component that is inherent to a monostatic backscatter reader. In a homodyne receiver, the LO frequency is identical to the RF carrier frequency due to direct conversion. Any leakage of the LO signal or the RF carrier into the mixer's input path results in a self-mixing component, producing a deterministic DC term in the baseband signal. At the same time, the low-noise amplifier (LNA) before the mixer can further amplify this leakage component. This produces a large DC spike in the spectrum and a constant bias in the time domain. The FPGA estimates and

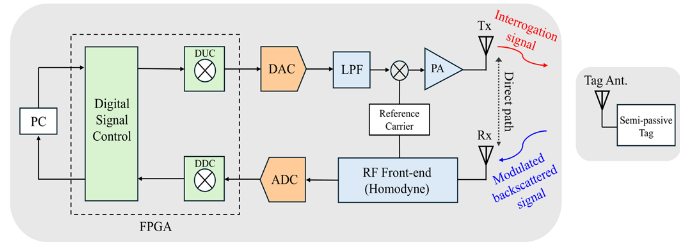


Fig. 12. Block diagram of the SDR reader implementing pre-processing through digital signal control on the FPGA.

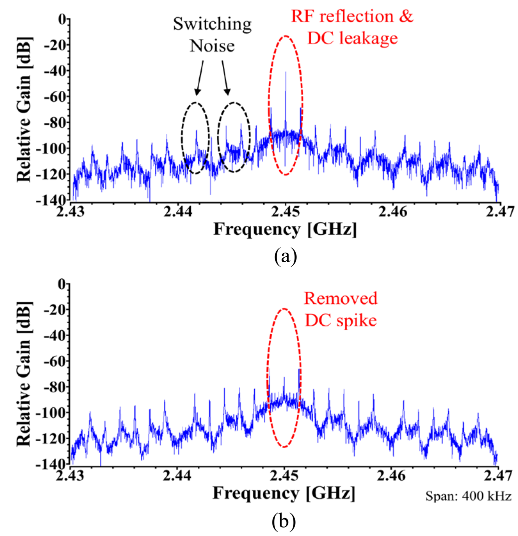


Fig. 13. Frequency spectrum of the receiver during data reception at the center frequency of 2.45 GHz: (a) before and (b) after DC offset removal.

removes this component over the full recording, thereby suppressing the carrier line around the center frequency and recentering the constellation. During data reception, the spectrum prior to correction is shown in Fig. 13(a), where the carrier leakage and other spurious peaks are visible near 2.45 GHz. After the initial processing stage of the received data, the DC spike is suppressed as shown in Fig. 13(b). The remaining narrow spectral harmonics originate from the tag switch. As the data rate grows, steeper current and voltage transitions in the switch generate stronger harmonic content and induce nonlinear mixing. These effects manifest as small peaks symmetrically located around the carrier frequency.

After DC offset suppression, the complex baseband coming out of the analog-to-digital converter (ADC) still exhibits a rapid constellation rotation. In the proposed system, this rotation is caused by a small frequency mismatch of the LO carrier between the carrier transmitter illuminating the tag and the receiver. Although the Tx and Rx are disciplined to the same reference clock, their PLL frequency synthesizers operate independently. This mismatch introduces a carrier frequency offset (CFO) Δf that imprints a linearly growing phase term across every frame and destroys the phase reference required by coherent detection. Residual CFO imposes a continuous phase rotation as illustrated in Fig. 14(a). To compensate for this

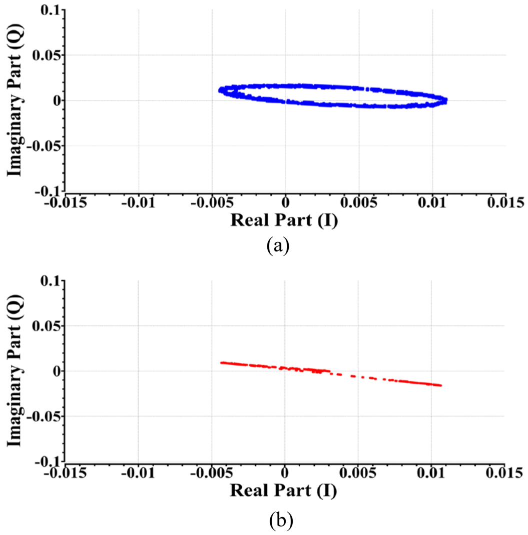


Fig. 14. Constellation diagram of the received data: (a) before and (b) after CFO compensation for enabling coherent detection.

effect, the raw discrete-time complex baseband signal is modeled as follows:

$$y_0[n] = \left(h_{eq} x_b[n] + d_0\right) e^{j(2\pi\Delta f n T_s + \Phi_0)} + w_0[n], \tag{58}$$

where $x_b[n]$ is the backscattered sample, d_0 is the deterministic leakage term, $w_0[n]$ is circular complex Gaussian noise, T_s is sampling period, and Φ_0 is an arbitrary initial phase. After sampling, the residual carrier phase factor appears as a common complex rotation that multiplies both the desired backscatter component and the direct-path leakage.

The receiver computes the first-lag autocorrelation of the raw samples, $u[n] = y_0[n] y_0^*[n-1]$. The average statistic $\bar{u}[n]$ is then obtained over a short window of W consecutive samples starting at index n_0, which isolates the CFO-induced phase increment. Under the standard assumption that adjacent data samples are nearly uncorrelated at lag-one, the data-dependent cross terms vanish, yielding $\mathbb{E}[u[n]] \approx |d_0|^2 e^{j2\pi\Delta f T_s}$. Hence, the phase of $\bar{u}[n]$ directly reveals the CFO phase advance $2\pi\Delta f T_s$, independent of the unknown data symbols and the constant leakage amplitude. The CFO is estimated as follows:

$$\widehat{\Delta f} = \frac{1}{2\pi T_s} \arg(\bar{u}[n]), \tag{59}$$

where $\arg(\cdot)$ denotes the phase angle of a complex variable in radians. With the estimated $\widehat{\Delta f}$, the received samples are multiplied by a complex sinusoidal wave of opposite rotation. The corrected signal is thus expressed by:

$$y[n] = y_0[n] e^{-j\left(2\pi\widehat{\Delta f}(n-n_0)T_s + \widehat{\Phi}_0\right)}, \tag{60}$$

where $\widehat{\Phi}_0 = \arg(y_0[n_0])$ serves as a reference for the absolute phase within the frame. This derotation procedure achieves carrier synchronization by transforming the constellation from a rotating ring into a phase-aligned line, as shown in Fig. 14(b).

After CFO compensation, the remaining phase trajectory within a pilot block is well approximated by a linear model

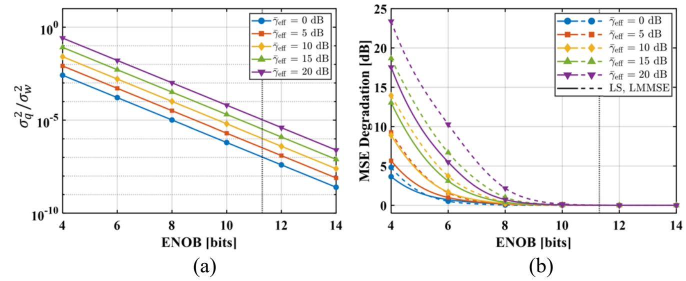


Fig. 15. Impact of ADC resolution on CE performance across $\bar{\gamma}_{\text{eff}}$: (a) quantization-to-thermal noise power ratio σ_q^2/σ_w^2 versus ENOB, and (b) MSE degradation of the lifted LS and LMMSE estimators relative to the infinite-precision baseline versus ENOB. The vertical dotted line indicates the measured operating ENOB of the deployed ADC.

Algorithm 2 End-to-End Backscatter Baseband Processing

Input: $\{y_0[n]\}$, $\{x_p[n]\}_{n\in\mathcal{P}_k}$, $\mathcal{P}_k$, T_s, W, n_0, S, σ_w^2, prior $(\boldsymbol{\mu}_\theta, \boldsymbol{\Lambda}_\theta)$

Initialization: Set $\hat{b}[n] \leftarrow 0$, $\forall n$, and $\widehat{\Phi}_0 \leftarrow \arg(y_0[n_0])$.

$\bar{u} \leftarrow \frac{1}{W}\sum_{n=n_0}^{n_0+W-1} y_0[n] y_0^*[n-1]$; $\widehat{\Delta f} \leftarrow \frac{\arg(\bar{u})}{2\pi T_s}$.

$y[n] \leftarrow y_0[n] e^{-j[2\pi\widehat{\Delta f}(n-n_0)T_s + \widehat{\Phi}_0]}$, $\forall n$.

$y[n] \leftarrow \text{FFT-LPF}(y[n]; f_L)$, $\forall n$.

$\hat{d}_{DC} \leftarrow \frac{1}{T_k^{\text{tot}}}\sum_{n\in\mathcal{P}_k} y[n]$; $\tilde{y}[n] \leftarrow y[n] - \hat{d}_{DC}$, $\forall n$.

for each frame k **do**

  $(\hat{h}, \hat{\phi}_1) \leftarrow$ **Alg. 1** $(\{\tilde{y}_p[n]\}_{n\in\mathcal{P}_k}, \{x_p[n]\}_{n\in\mathcal{P}_k}, \sigma_w^2,$

  $\mathfrak{C}, (\boldsymbol{\mu}_\theta, \boldsymbol{\Lambda}_\theta))$.

  for each $n \in \mathcal{D}_k$ **do**

    $z_d[n] \leftarrow \tilde{y}[n]/\hat{h}$.

    $\hat{b}[n] \leftarrow 1\{\Re\{z_d[n]\} \geq 0\}$.

  end for

end for

Output: Decoded bits $\{\hat{b}[n]\}$

---

because the only remaining terms governing phase evolution—namely, the residual CFO and slow intra-frame drift—both vary linearly over the short pilot interval. The residual CFO contributes a constant rate phase term $-2\pi(\Delta f - \widehat{\Delta f})T_s n$, while the intra-frame drift $\varphi[n] \approx \varphi[n_0] + \varphi'[n_0](n - n_0)$ is locally linear within the coherence window. Hence, the total residual phase $\phi_0 + \phi_1(n - n_0)$ accurately represents post-CFO behavior, confirming that the linear phase-slope model remains valid for CE. Following compensation, the data stream conforms to the pre-processing model in (16). Consequently, CFO correction stabilizes the constellation, allowing subsequent CE and ZF equalization to operate on a phase-stationary record, which in turn enables reliable decoding. The improvement is clearly visible in the observed constellation diagram of Fig. 14(b).

The final hardware consideration in the receive chain is the finite resolution of the ADC. The deployed converter provides a signal-to-noise-and-distortion ratio (SINAD) of 69.8 dB, corresponding to an effective number of bits (ENOB) of 11.3 bits. In the monostatic configuration, the ADC input is dominated by direct-path leakage rather than the backscattered

TABLE III
KEY EXPERIMENTAL PARAMETERS FOR REPRODUCIBILITY

Parameter	Value
Reader & RF front-end	
Operating carrier frequency, f_c	2.45 GHz
Reader RF output power[a], p_0	−8 dBm
SDR platform	USRP N200 w/ WBX daughterboard
Baseband sampling rate	1 MS/s
Reader excitation waveform	CW
Antenna subsystem	
Reader antenna	2×1 planar Yagi–Uda array w/ SRRs
Antenna and SRR geometry	Provided in Table II
Tag subsystem	
Tag controller MCU	ATmega328P
Tag RF switch	HMC221BE
Tag modulation schemes	OOK, BPSK
Frame structure and signaling	
Data rate, R	500 kbps
Symbol duration, τ_s	2 μs
Pilot length, $N^\star$	120 symbols
Payload data-block length, N_d	512 symbols
Pilot fraction, $\rho^\star$	0.190
Pilot–payload frame duration[b]	1.264 ms
Receiver processing	
CFO estimation window, W	240 samples
CE-branch FFT-LPF cutoff, f_L	200 kHz
Operating conditions	
Reader–tag distance, d_T	1 m
Measurement environment	Indoor, 25 °C

[a] p_0 denotes the conducted RF output power measured at the WBX daughterboard output port, not the SDR DC power consumption.

[b] The pilot–payload frame duration excludes preamble and synchronization overhead and is computed as $N^\star\tau_s + N_d\tau_s = 1.264$ ms.

component, producing a composite crest factor of only 0.31 dB prior to DC offset removal. Following removal, the OOK baseband crest factor is 3.01 dB, obviating the need for ENOB back-off and preserving the available converter resolution for subsequent processing stages. To assess whether quantization compromises CE performance, Fig. 15 evaluates the quantization-to-thermal noise power ratio σ_q^2/σ_w^2 and the MSE degradation of both lifted estimators as a function of ENOB. At the operating ENOB of approximately 11 bits, σ_q^2/σ_w^2 remains below 10^{-5} for all evaluated $\bar{\gamma}_{\text{eff}}$, placing the quantization floor at least 50 dB below the thermal noise as shown in Fig. 15(a). Correspondingly, Fig. 15(b) confirms that the MSE degradation relative to an infinite-precision baseline stays below 0.05 dB across the entire SNR range for both the lifted LS and LMMSE. At this power ratio, the effective noise after quantization is indistinguishable from the thermal-noise-only case, and the

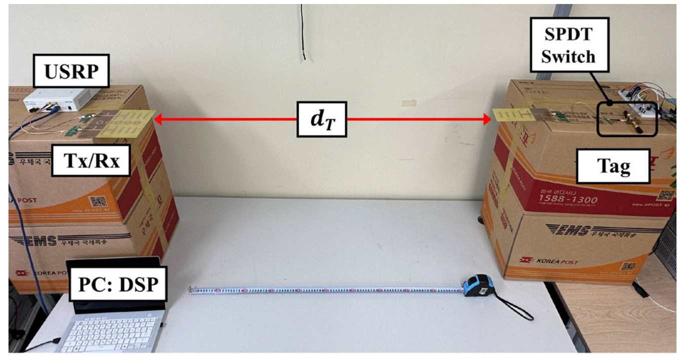


Fig. 16. Indoor experimental setup of the proposed BMC system: $d_T = 1$ m.

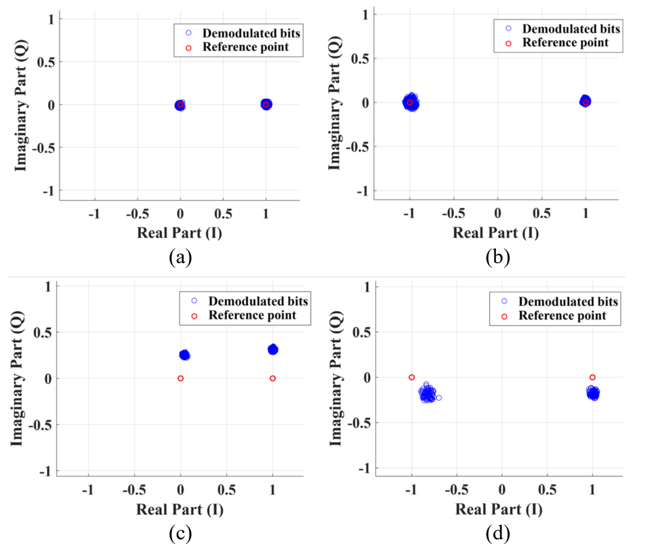


Fig. 17. Constellation diagram after the signal processing chain for OOK and BPSK with and without integrated SRRs: (a) OOK with SRRs, (b) BPSK with SRRs, (c) OOK without SRRs, and (d) BPSK without SRRs.

Gauss–Markov and affine-optimality conditions underlying the lifted LS and LMMSE remain an accurate approximation at the operating ENOB. The complete end-to-end baseband processing chain is summarized in Algorithm 2.

VI. EXPERIMENTAL RESULTS

This section evaluates the end-to-end signal processing chain in the indoor setup of Fig. 16. The key experimental parameters that govern the measurements throughout this section are summarized in Table III. First, the error vector magnitude (EVM) after the full signal-processing chain was evaluated for OOK and BPSK. EVM is defined as the RMS distance between the equalized samples and the ideal constellation, normalized by the reference symbol power and expressed as a percentage.

$$\text{EVM}_{\text{RMS}}[\%] = 100\sqrt{\frac{\frac{1}{N}\sum_{k=1}^{N}|y_{ks}-s_{ks}|^2}{\frac{1}{N}\sum_{k=1}^{N}|s_{ks}|^2}}. \tag{61}$$

Here, s_{ks} denotes the reference symbol and y_{ks} the equalized sample at the decision instant after the full processing chain. With the SRRs integrated at the antenna, the measured EVM was 2.97% for OOK and 4.02% for BPSK. These results reflect improved Tx–Rx isolation in the monostatic reader. The slightly larger EVM for BPSK is consistent with its phase sensitivity because residual phase error directly perturbs the decision boundary, whereas OOK is primarily amplitude-driven. Without the SRRs, the EVM rose to 34% for OOK and 22% for BPSK. In this configuration, transmit leakage at the antenna dominates the receiver input, leaving a residual carrier at baseband even after digital suppression; the impact becomes more severe for higher-order modulation schemes where tighter amplitude and phase accuracy are required and the symbol-error probability increases significantly for a given EVM. The constellation diagrams of the proposed BMC system with and without SRRs are shown in Fig. 17.

A. Experimental Validation: CFO Estimation and Compensation

As a prerequisite to the end-to-end performance evaluation, the CFO estimator assumptions are validated on measured payload records under the adopted pulse-shaping and sampling, and the estimator bias is compared between nonzero-mean OOK pilots and zero-mean BPSK pilots. The CFO estimator extracts $\widehat{\Delta f}$ from the phase advance of the lag-one autocorrelation statistic. Let $y_0^{\mathrm{K}}[n]$ and $y_0^{\mathrm{B}}[n]$ denote the OOK and BPSK raw baseband records used for the corresponding evaluation, respectively. For compact notation, let $y_0[n]$ denote the selected record used for the corresponding autocorrelation evaluation, either before or after DC offset removal, and the normalized lag-L autocorrelation is defined as $C_L \triangleq \mathbb{E}[y_0[n]y_0^*[n-L]]/\mathbb{E}[|y_0[n]|^2]$. The key modeling step assumes that adjacent data samples are nearly uncorrelated at lag one. This ensures that correlation leakage does not dominate the window-averaged product, $\frac{1}{N}\sum_n y_0[n]y_0^*[n-1]$, allowing its phase to primarily track the CFO-induced rotation. This assumption is directly validated under the adopted pulse shaping and sampling by computing $|C_L|$ over multiple lags on the measured payload for both OOK and BPSK. As illustrated in Fig. 18, the lag-one magnitude $|C_1|$ was small for both modulations, with values of approximately −22 dB for OOK and −22.7 dB for BPSK. Moreover, DC offset removal changes the lag-one magnitude only marginally, by about 1.1 dB for OOK and 0.13 dB for BPSK. These results support the estimator premise that lag-one correlation leakage is weak in the measured data record and that the estimator statistic is not dominated by adjacent sample correlation under the adopted receiver processing.

The remaining modulation-dependent estimator bias is instead governed by the deterministic component that enters the same autocorrelation phase statistic through the interaction between the pilot mean and the DC offset removal. In the processing chain, DC offset removal subtracts the estimate $\hat{d}_{\mathrm{DC}}$ obtained from the pilot average. This operation is intrinsically well matched to a zero-mean BPSK pilot, for which the pilot average isolates the dataset-wide DC offset, thereby reducing deterministic leakage into the autocorrelation product and yielding a smaller signed bias and absolute error. In contrast, an

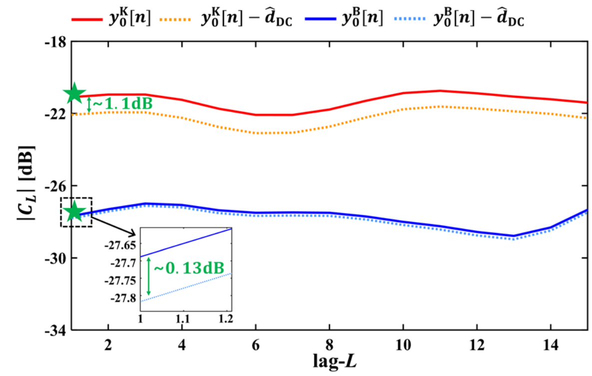


Fig. 18. Validation of the lag-1 uncorrelated-sample assumption for the CFO estimator under the adopted pulse shaping and sampling. The markers indicate the lag-1 value $|C_1|$.

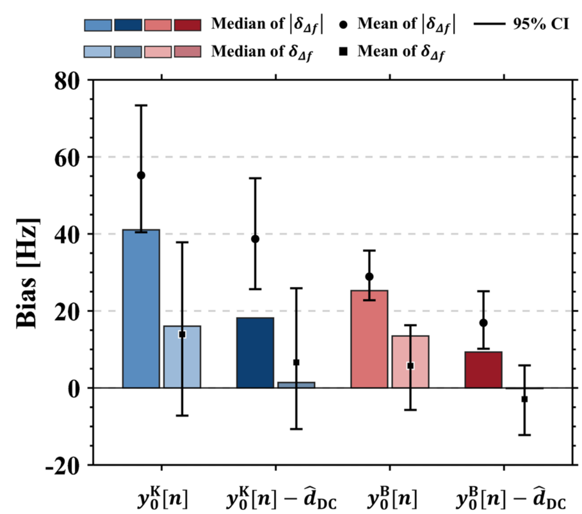
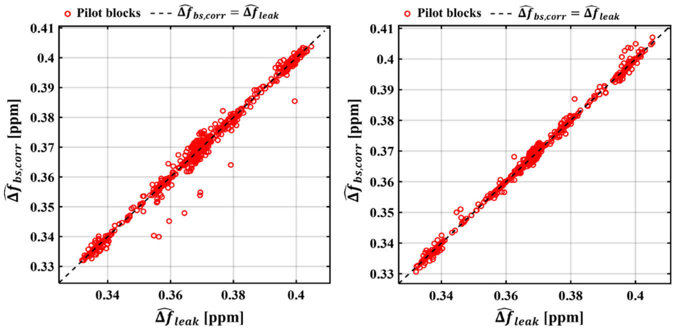


Fig. 19. Signed and absolute bias of the proposed CFO estimator for nonzero-mean OOK pilots and zero-mean BPSK pilots, with 95% confidence intervals (CIs).

OOK pilot is nonzero-mean at the waveform level, so the pilot average contains a signal-dependent component in addition to the DC offset. This limits how completely DC can be separated and leaves a larger residual deterministic component in $y_0[n]$, which propagates into the autocorrelation phase statistic and increases CFO estimator bias. This mechanism is quantitatively confirmed in Fig. 19, which reports the proposed CFO estimation bias $\delta_{\Delta f} \triangleq \widehat{\Delta f} - \Delta f_{\mathrm{ref}}$ and the corresponding absolute bias $|\delta_{\Delta f}|$ for nonzero-mean OOK and zero-mean BPSK, evaluated before and after DC offset removal on the selected baseband record $y_0[n]$. Here, Δf_{ref} denotes a fixed reference CFO obtained from the entire measured dataset. Before DC offset removal, the mean CFO bias was 13.92 Hz for OOK and 5.76 Hz for BPSK, and the mean absolute error was 55.22 Hz for OOK and 28.91 Hz for BPSK. After DC offset removal, the mean CFO bias decreased to 6.63 Hz for OOK and to −2.91 Hz for BPSK, and the mean absolute error decreased to 38.7 Hz for OOK and 16.92 Hz for BPSK.

Strong direct leakage refers to the operating regime in which the deterministic leakage term d_0 in (58) dominates the



(a) (b)

Fig. 20. Comparison between the leakage-derived CFO and the backscatter-correlated CFO estimates under strong direct leakage: (a) OOK with mean error of -6.0×10^{-4} ppm and root-mean-square error (RMSE) of 2.7×10^{-3} ppm, and (b) BPSK with mean error of 1.1×10^{-4} ppm and RMSE of 1.2×10^{-3} ppm.

received baseband magnitude over the pilot interval. Using (58), define the normalized backscatter-to-leakage ratio as $\tilde{\rho}[n] \triangleq |h_{eq} x_b[n]|/|d_0|$, and assume $\tilde{\rho}[n] \ll 1$ throughout the pilot window. Then the raw sample can be written as $y_0[n] = d_0 e^{j\tilde{\Phi}[n]}(1 + \tilde{\rho}_s[n]) + w_0[n]$ with $\tilde{\rho}_s[n] = h_{eq} x_b[n]/d_0$ and $\tilde{\Phi}[n] = 2\pi\Delta f n T_s + \Phi_0$. The one-lag product $u[n]$ therefore takes the form

$$u[n] = |d_0|^2 e^{j2\pi\Delta f T_s}(1 + \tilde{\epsilon}[n]) + \tilde{\eta}[n], \qquad (62)$$

where $\tilde{\epsilon}[n]$ collects the backscatter-dependent cross terms and satisfies $|\tilde{\epsilon}[n]| = O(\tilde{\rho}[n]) + O(\tilde{\rho}[n-1])$ where $O(\cdot)$ is big O notation, and $\tilde{\eta}[n]$ collects noise-dependent terms. Under strong direct leakage, $\tilde{\epsilon}[n]$ is uniformly small over the pilot, so the phase increment of $u[n]$ is governed by the leakage phasor increment, yielding a CFO estimate that converges to the residual rotation $2\pi\Delta f T_s$ up to an $O(\tilde{\rho})$ perturbation. Crucially, the derotation computed from $u[n]$ removes the same carrier rotation experienced by the backscattered component, even in the presence of switching-induced spectral replicas and tag-state-dependent load pulling. This follows from the multiplicative structure of (58), where the residual CFO appears as a common complex rotation multiplying both the leakage and the backscatter components after sampling.

Consider generic linear pilot correlation for pilot alignment:

$$\tilde{v}[n] = \sum_{m=0}^{M_p-1} y_0[n+m]\tilde{c}^*[m], \qquad (63)$$

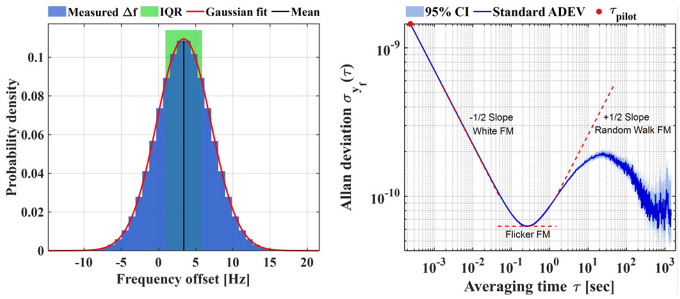
with a known pilot template $\tilde{c}[m]$, where M_p denotes the number of pilot samples used in the correlation window. Substituting (58) shows that the common CFO factor separates as $\tilde{v}[n] = e^{j2\pi\Delta f n T_s}\bar{v}[n]$ where $\bar{v}[n]$ absorbs the complete internal structure induced by switching harmonics, cyclostationary amplitude and phase variations, and any load-pulling effects through $x_b[n]$ and the linear combining. Taking the one-lag product of the correlated sequence yields

$$\tilde{v}[n]\tilde{v}^*[n-1] = e^{j2\pi\Delta f T_s}\bar{v}[n]\bar{v}^*[n-1], \qquad (64)$$

so the CFO contribution remains an exact multiplicative factor independent of the detailed composition of $\bar{v}[n]$. Consequently, any estimator that extracts the global phase increment from either the raw $u[n]$ or a one-lag product formed after pilot correlation estimates the same $\widehat{\Delta f}$, and derotation using this estimate removes the same carrier rotation from the backscattered component.

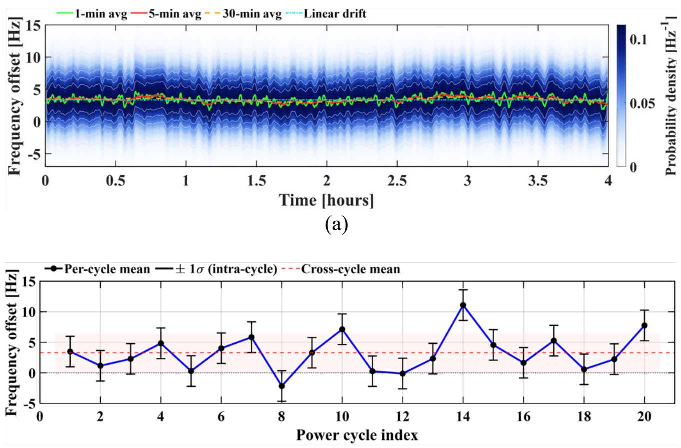
Fig. 20 validates this equivalence in measured data by plotting the leakage derived estimate $\widehat{\Delta f}_{leak}$ and the pilot correlated estimate $\widehat{\Delta f}_{bs,corr}$, both extracted from $y_0[n]$ prior to derotation, and reported in parts per million (ppm) referenced to the 10 MHz system reference clock (1 ppm $\equiv$ 10 Hz). The point cloud collapses tightly onto this identity line over the full observed CFO range, indicating that the two estimators track the same block-to-block CFO variation rather than merely agreeing at a single operating point. The same near-identity geometry is observed in Fig. 20(a) for OOK and Fig. 20(b) for BPSK, which is consistent with a common residual carrier-rotation factor that multiplies the received baseband before any pilot correlation step. Small departures from the identity line appear as isolated outliers and modest spread around the diagonal, which is expected from finite sample averaging and noise contamination in individual pilot blocks, not from an additional tag-dependent carrier rotation. Overall, the diagonal alignment in ppm directly supports that switching-induced spectral replicas and tag-dependent pulling effects are absorbed into the internal correlated term while the removed rotation governed by $\widehat{\Delta f}_{leak}$ remains the same rotation experienced by the backscattered component, thereby preserving a coherent phase reference under strong leakage conditions.

A possible ambiguity is whether the pre-compensation nonzero phase slope stems from the CFO of independently locked PLLs or from oscillator phase noise when $\Delta f \approx 0$. To distinguish these two possibilities, a continuous four-hour recording comprising 59.9×10^6 pilot blocks was collected from the monostatic reader at f_c. The probability density of the CFO estimate $\widehat{\Delta f}$ obtained from (59) is shown in Fig. 21(a). The distribution is well-approximated by a Gaussian with mean $\mu_{\widehat{\Delta f}} = 3.39$ Hz and standard deviation $\sigma_{\widehat{\Delta f}} = 3.64$ Hz, with mean absolute error (MAE) of 4.08 Hz. The observed nonzero mean over 59.9×10^6 per-pilot-block estimates indicates the presence of a deterministic bias component in the frequency offset. When $\mu_{\widehat{\Delta f}}$ is close to zero, the intra-pilot-block phase slope is typically dominated by oscillator and PLL-induced frequency instability; in contrast, the present measurement shows $\mu_{\widehat{\Delta f}} \neq 0$ and therefore indicates an additional deterministic mismatch between the Tx and Rx synthesizers. This offset is primarily attributable to residual steady-state lock errors of the two PLLs, which can settle to slightly different frequencies through independent feedback dynamics even under a shared reference clock. The 1st-to-99th-percentile interval $[-5.08, 11.87]$ Hz corresponds to a fractional deviation on the order of 10^{-9} at f_c, confirming that the shared-reference architecture suppresses the inter-PLL offset to the sub-parts-per-billion (ppb) level. To assess the impact of this offset on the



(a) (b)

Fig. 21. Statistical characterization of the CFO estimation under a shared reference clock with independent PLLs: (a) probability density of $\widehat{\Delta f}$ with interquartile range (IQR) shading, and (b) Allan deviation of the fractional-frequency fluctuation.



(a)

(b)

Fig. 22. Temporal stability of the measured frequency offset: (a) four-hour time series showing per-frame $\widehat{\Delta f}$ with 1-, 5-, and 30-min moving averages and a linear drift fit, and (b) power-cycle dependence of $\widehat{\Delta f}$ showing per-cycle mean with intra-cycle $\pm 1\sigma$ error bars, the cross-cycle mean, and a shaded band indicating cross-cycle mean $\pm 1\sigma$.

backscatter receiver, the measured mean $\mu_{\widehat{\Delta f}} = 3.39$ Hz indicates a deterministic frequency offset of the same magnitude; the corresponding uncompensated phase accumulation in one pilot block is 5.11×10^{-3} rad (0.293°). Even under the worst-case 99th-percentile offset of 11.87 Hz, the accumulation reaches only 17.9×10^{-3} rad (1.03°). Both values are negligible relative to the BPSK decision boundary of $\pi/2$, and equally negligible for the antipodal OOK decision statistic under the normalization in (16)–(18). Nonetheless, because this offset produces a nonzero intra-pilot-block phase slope, the CFO compensation in (60) is applied so that the lifted estimators operate on the post-compensation residual rather than the full offset.

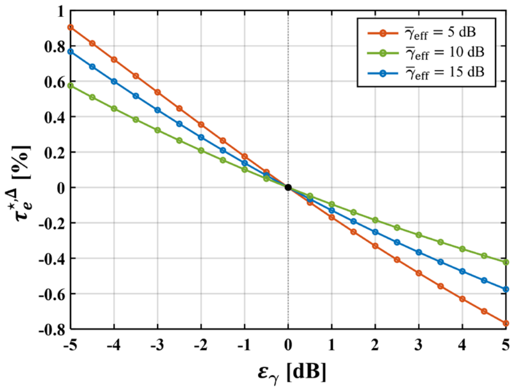
The Allan deviation $\sigma_{y_f}(\tau)$ of the fractional-frequency fluctuation $y_f = \widehat{\Delta f}/f_c$ was evaluated over 3,650 averaging times from the pilot block interval $\tau_{\text{pilot}} = 240$ µs to 1,440 s. The measured $\sigma_{y_f}(\tau)$ curve in Fig. 21(b) is consistent with the dependence expected from a reference-locked PLL synthesizer: a $\tau^{-1/2}$ roll-off at short τ (White FM), an intermediate region consistent with flicker FM, and a $\tau^{+1/2}$ rise at long τ indicative of random-walk FM driven by slow environmental perturbations. This behavior confirms that the stochastic component of the per-block frequency offset is oscillator and PLL-origin. Combined with the persistent nonzero mean $\mu_{\widehat{\Delta f}}$, the measurements therefore support a decomposition of the intra-pilot-block phase slope into (i) a deterministic CFO bias of 3.39 Hz attributable to independent PLL lock offsets and (ii) a stochastic fluctuation component of oscillator/PLL origin whose time-scale dependence is characterized by $\sigma_{y_f}(\tau)$.

The temporal stability of $\widehat{\Delta f}$ over the four-hour session is shown in Fig. 22(a). The hourly mean varied from +3.11 Hz to +3.53 Hz while the hourly standard deviation remained within 3.63 Hz to 3.65 Hz, suggesting that both the deterministic bias and the stochastic fluctuation level remained approximately constant over the measurement interval. A linear regression yielded a drift rate of 0.03 Hz/h, accumulating only 0.12 Hz across the full recording. The reproducibility across 20 independent power cycles is examined in Fig. 22(b). The cross-cycle mean was 3.29 Hz with a cycle-to-cycle standard deviation of 3.14 Hz, while the intra-cycle standard deviation remained tightly bounded at 2.50 ± 0.01 Hz. The wider cross-cycle spread is expected, since each power-on event forces both synthesizers to re-acquire lock from a different initial state, whereas the tight intra-cycle bound confirms that the stochastic fluctuation level is reproducible within each session.

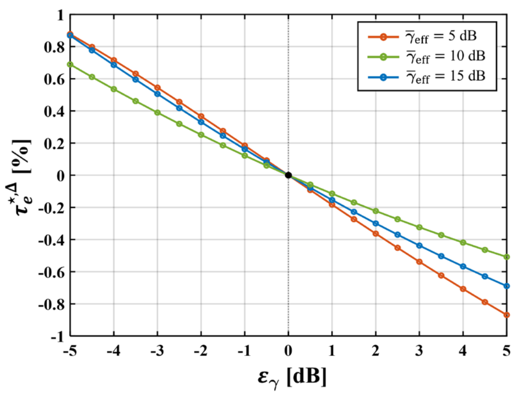
B. Optimal Pilot Resource Strategy Verification

The practical applicability of the proposed framework depends not only on its analytical optimality but also on its robustness to parameter mismatch, since the closed-form solution in (14) relies on the pre-frame $\bar{\gamma}_{\text{eff}}$ estimate, which may deviate from the true channel condition in realistic deployments. A sensitivity analysis is therefore conducted to quantify the impact of SNR estimation errors on both the optimal training time $\tau_e^\star$ and the resulting BER performance. Let $\hat{\gamma} = \bar{\gamma}_{\text{eff}} 10^{\varepsilon_\gamma/10}$ denote the estimated SNR, where ε_γ represents the SNR estimation error in dB. The pilot duration is then computed based on $\hat{\gamma}$ rather than the true $\bar{\gamma}_{\text{eff}}$, leading to a suboptimal operating point. A key theoretical observation is that $\tau_e^\star$ depends solely on $\bar{\gamma}_{\text{eff}}$ through the rate maximization criterion in (9), and is completely independent of the Rician K-factor. This follows directly from the optimization objective in (P1), where the ergodic spectral efficiency $\mathcal{R}(\tau_e)$ is determined by the mean channel gain rather than the instantaneous fading statistics. Specifically, the effective SNR after CE in (8) and the rate function in (9) involve only $\bar{\gamma}_{\text{eff}}^{\text{CE}}$ and the CE overhead parameter α, neither of which depends on K. The pilot duration shift remains invariant across different K-factor values for a given SNR estimation error. However, the BER variation, defined as $\bar{P}^{\Delta} \triangleq \bar{P}_e(\hat{\gamma}) - \bar{P}_e(\bar{\gamma}_{\text{eff}})$ in dB, explicitly depends on K-factor through the expectation over h_{eq} in (5).

For the numerical K-factor sweep, a symmetric double-Rician setting is used, where $K_{tt} = K_{tr} \triangleq K$, so that the effect

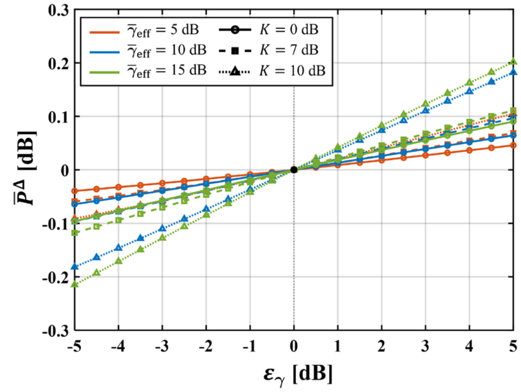


(a)

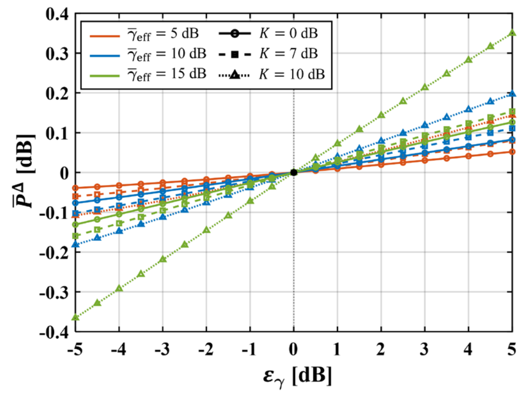


(b)

Fig. 23. Optimal pilot duration shifts $\tau_e^{\star,\Delta}$ versus SNR estimation error ε_γ, evaluated at $\bar{\gamma}_{\mathrm{eff}} \in \{5,10,15\}$ dB: (a) OOK and (b) BPSK.



(a)



(b)

Fig. 24. Sensitivity of BER variation $\bar{P}^\Delta$ versus SNR estimation error ε_γ, evaluated at $\bar{\gamma}_{\mathrm{eff}} \in \{5,10,15\}$ dB and $K \in \{0,7,10\}$ dB: (a) OOK and (b) BPSK.

of the common K-factor can be isolated as a single sweep variable while preserving the cascaded product structure of the channel. The simulation evaluates OOK and BPSK across $\bar{\gamma}_{\mathrm{eff}} \in \{5, 10, 15\}$ dB and $\varepsilon_\gamma \in [-5, +5]$ dB, while $K \in \{0, 7, 10\}$ dB is applied only for the BER-variation analysis, using 10^6 double-Rician fading realizations. The normalized pilot shift $\tau_e^{\star,\Delta}$ decreases monotonically with ε_γ : underestimation yields conservative allocation with increased pilot overhead, while overestimation results in insufficient pilot resources. Higher $\bar{\gamma}_{\mathrm{eff}}$ produces smaller $\left|\tau_e^{\star,\Delta}\right|$, reflecting the diminishing slope of $\tau_e^\star$ in the high-SNR regime. The worst-case pilot deviation was $\left|\tau_e^{\star,\Delta}\right|_{\max} = 0.842\%$ for OOK and 0.818% for BPSK, both occurring at $\bar{\gamma}_{\mathrm{eff}} = 5$ dB as shown in Fig. 23. The corresponding maximum $\bar{P}^\Delta$ was 0.188 dB for OOK and 0.328 dB for BPSK at $K = 10$ dB, consistent with Fig. 24. The slightly larger $\bar{P}^\Delta$ in BPSK is attributed to the steeper slope of its BER curve at the same $\bar{\gamma}_{\mathrm{eff}}$, which amplifies the BER sensitivity to perturbations in the $\bar{\gamma}_{\mathrm{eff}}$, caused by pilot mis-allocation. The BER variation $\bar{P}^\Delta$ exhibits asymmetric behavior: underestimation of the pre-frame SNR leads to conservative pilot allocation, which slightly improves CE accuracy and yields negligible $\bar{P}^\Delta$ or even a BER improvement. In contrast, overestimation under-allocates pilot resources and causes measurable degradation. Higher K-factors amplify the relative $\bar{P}^\Delta$, as reduced fading severity makes the link more sensitive to suboptimal pilot allocation. These results confirm that the proposed resource allocation maintains robust performance under ± 5 dB SNR estimation error, with $\left|\tau_e^{\star,\Delta}\right|_{\max} < 1\%$ of the frame duration and $\bar{P}^\Delta < 0.35$ dB across all evaluated scenarios.

Having established that the optimal pilot duration $\tau_e^\star$ is robust to SNR estimation errors, an equally important question is whether the temporal placement of the allocated pilots impacts estimation accuracy under residual phase drift. To this end, the super-frame $[\mathbf{P}_{N^\star/\beta}][\mathbf{D}_{N_d}] \times \beta$ is constructed, where $\mathbf{P}_{N^\star/\beta}$ denotes a pilot block of $N^\star/\beta$ symbols and $\mathbf{D}_{N_d}$ denotes a data block of N_d payload symbols; each sub-frame is repeated β times so that the total pilot budget $N^\star$ remains constant for all β, and the pilot energy is held fixed $N^\star p_0$. When $\beta = 1$, all pilots are contiguous at the frame start, recovering the proposed frame design; as β increases, the pilots are progressively dispersed across the frame. The pilot budget is $N^\star = 120$, obtained by solving (P1) under $N_d = 512$, a value dictated by the memory constraint of the tag microcontroller unit (MCU), and the specified system parameters, which yields a pilot overhead of $\rho^\star \approx 19\%$. This value additionally admits thirteen divisors, enabling a fine-grained sweep of β between 1 and 30. Each pilot block yields an LS estimate of h_{eq}; data symbols between consecutive blocks are equalized through linear

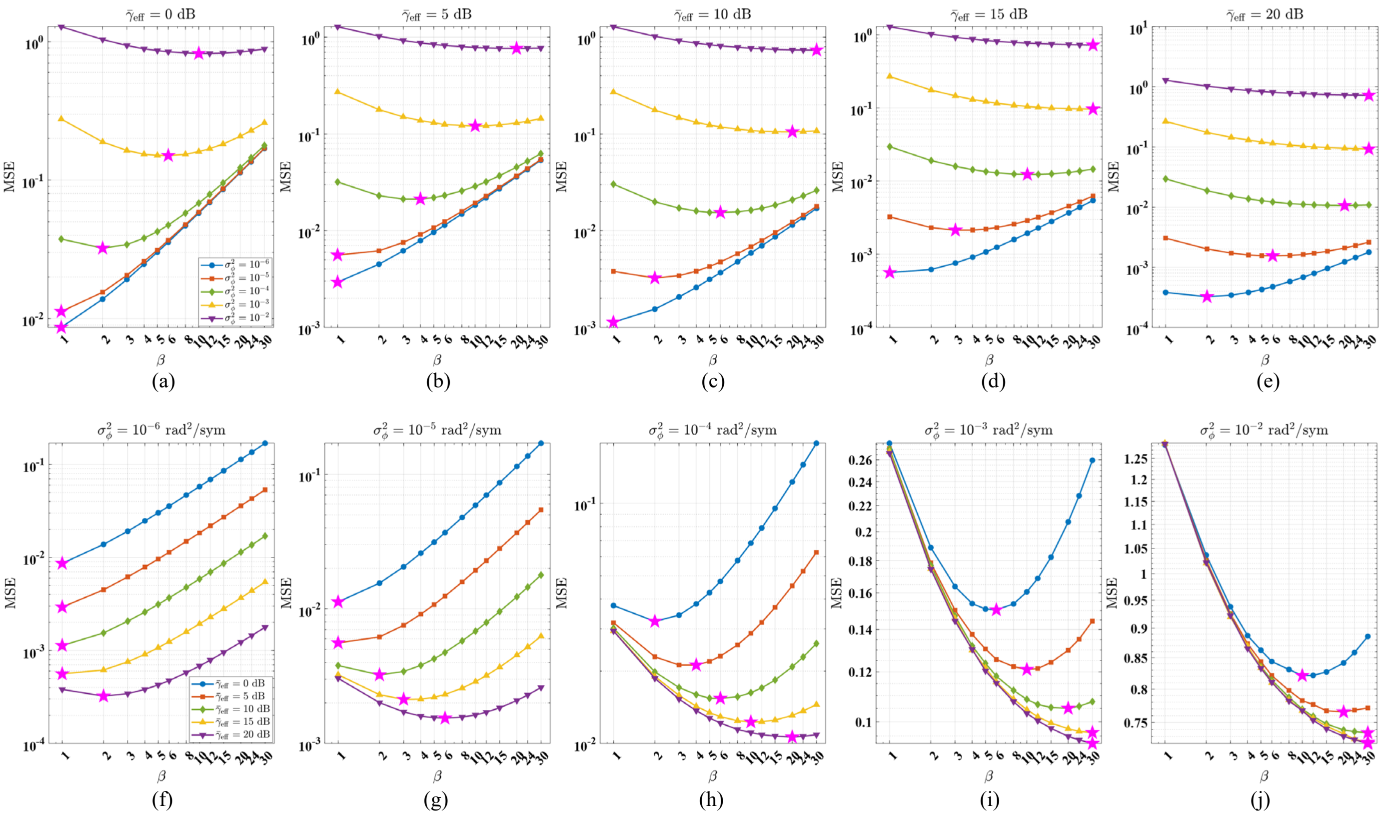


Fig. 25. MSE versus pilot dispersion parameter β under varying per-symbol phase drift variance σ_ϕ^2 and $\bar{\gamma}_{\text{eff}}$: (a) $\bar{\gamma}_{\text{eff}} = 0$ dB, (b) $\bar{\gamma}_{\text{eff}} = 5$ dB, (c) $\bar{\gamma}_{\text{eff}} = 10$ dB, (d) $\bar{\gamma}_{\text{eff}} = 15$ dB, (e) $\bar{\gamma}_{\text{eff}} = 20$ dB, (f) $\sigma_\phi^2 = 10^{-6}$ rad^2/sym, (g) $\sigma_\phi^2 = 10^{-5}$ rad^2/sym, (h) $\sigma_\phi^2 = 10^{-4}$ rad^2/sym, (i) $\sigma_\phi^2 = 10^{-3}$ rad^2/sym, and (j) $\sigma_\phi^2 = 10^{-2}$ rad^2/sym.

interpolation of adjacent estimates. The lifted LS estimator in (28) is adopted for this study because it is prior-free: the lifted LMMSE in (38) requires the calibrated prior covariance $\boldsymbol{\Lambda}_\theta$, which is determined from contiguous pilot observations as described in Section VI-C; when the pilot budget is redistributed into β blocks of $N^\star/\beta$ symbols, the per-block sample count becomes insufficient for reliable prior calibration, particularly at large β. The LS baseline therefore isolates the pure effect of pilot placement without conflating it with prior mismatch.

To emulate intra-frame phase impairments of varying severity, a per-symbol random phase increment is applied on the effective channel, with increment variance σ_ϕ^2, and the MSE is evaluated across a wide range of $(\sigma_\phi^2, \bar{\gamma}_{\text{eff}})$. The measured MSE as a function of β is shown in Fig. 25(a)–(j). The star marker on each curve denotes the optimal $\beta^\star$. Two competing mechanisms shape every curve: finer partitioning improves phase tracking by shortening the interpolation gap, whereas the reduced per-block pilot count $N^\star/\beta$ amplifies estimation noise. The resulting $\beta^\star$ increases monotonically in both σ_ϕ^2 and $\bar{\gamma}_{\text{eff}}$, since faster drift demands more frequent updates and higher SNR exposes phase drift as the dominant residual error. At the lowest drift rates the channel is quasi-static and $\beta^\star = 1$, whereas under severe drift $\beta^\star$ saturates at the maximum value and the MSE curves converge across SNR, indicating that the accumulated phase rotation within each data segment exceeds the tracking capability of piecewise-linear interpolation.

In the proposed BMC platform, the raw per-frame frequency offset characterized in Figs. 21 and 22 has a mean of only 3.39 Hz with an intra-cycle standard deviation of 2.50 Hz, corresponding to a fractional deviation on the order of 10^{-9} at f_c. The per-frame CFO compensation in (59)–(60) removes the bulk of this offset, and any residual phase slope within a pilot block is absorbed by the ϕ_1 coefficient of the lifted estimator. The remaining stochastic phase perturbation attributable to oscillator jitter is substantially smaller, placing the operating point well within the quasi-static regime on the simulation grid and inside the admissible $|\phi_1|$ region certified by (46)–(47). In this operating regime, phase drift does not constitute the dominant residual impairment, so dispersing pilots in time provides little additional benefit over a contiguous block with the same total pilot energy. Accordingly, the MSE penalty of the contiguous allocation ($\beta = 1$) relative to $\beta^\star$ is negligible. The single-parameter optimization in (P1) therefore yields a near-optimal pilot configuration without explicit dispersion control; time-dispersed placement becomes beneficial only at drift rates far exceeding those measured in the proposed platform.

To certify that (P1) admits a unique maximizer when the CE error variance accounts for the joint estimation of h_{eq} and the nuisance parameter ϕ_1, the unimodality of $\mathcal{R}(\tau_e)$ is established

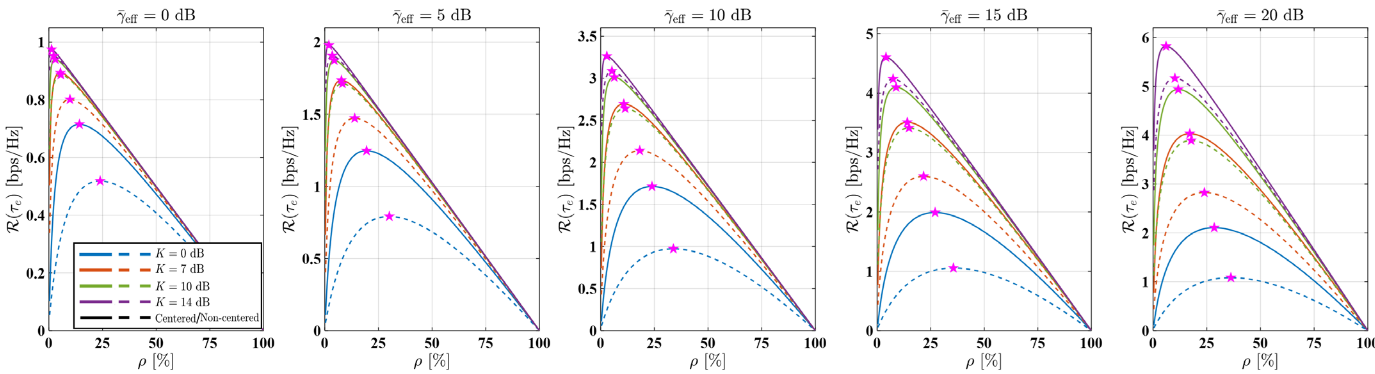


Fig. 26. Spectral efficiency $\mathcal{R}(\tau_e)$ versus pilot overhead ρ for the joint (h_{eq}, ϕ_1) estimation framework, evaluated across Rician K-factors at a given $\bar{\gamma}_{\rm eff}$: (a) $\bar{\gamma}_{\rm eff} = 0$ dB, (b) $\bar{\gamma}_{\rm eff} = 5$ dB, (c) $\bar{\gamma}_{\rm eff} = 10$ dB, (d) $\bar{\gamma}_{\rm eff} = 15$ dB, (e) $\bar{\gamma}_{\rm eff} = 20$ dB.

directly under the marginal two-parameter CRLB obtained by eliminating ϕ_1 via the Schur complement in (56). As discussed in Section IV-D, there exist τ_e-independent constants k_0, k_1, k_2 such that the pilot-domain quadratic forms scale as q_0, q_1, and q_2. Under the Schur-complement structure, the coupling-direction eigenvalue of $\mathbf{S}$ scales as $2(q_0 - |q_1|^2/q_2) \propto \tau_e$, and the perpendicular-direction eigenvalue $2q_0$ shares the same linear τ_e-scaling. Hence, both eigenvalues of $\mathbf{S}$ scale linearly with τ_e, and $\sigma_{\hat{h}}^2$ in (56) consequently scales as $1/\tau_e$. The factor $(k_0 - k_1^2/k_2)$ is strictly positive under the identifiability condition $\Delta > 0$ in (26) and is τ_e-independent; it equals k_0 for centered pilot indices ($s_1 = 0$) and is strictly less than k_0 for non-centered placement. Substituting the resulting CE error variance $\sigma_e^2(\tau_e) \propto 1/\tau_e$ into the one-frame inversion model (7) yields $g(\tau_e) = \bar{\gamma}_{\rm eff}\tau_e/(\tau_e + \alpha')$, where α' generalizes the scalar α in (8) by absorbing all τ_e-independent quantities including the Schur-complement correction from the marginal two-parameter CRLB, and remains strictly positive. Direct differentiation gives $g'(\tau_e) = \bar{\gamma}_{\rm eff}\alpha'/(\tau_e + \alpha')^2 > 0$ and $g''(\tau_e) = -2\bar{\gamma}_{\rm eff}\alpha'/(\tau_e + \alpha')^3 < 0$, confirming that g is strictly increasing and strictly concave on $(0, \tau_c)$. The spectral efficiency $\mathcal{R}(\tau_e)$ in (9) is the product of a strictly decreasing linear factor and a strictly increasing, strictly concave logarithmic factor, with boundary values $\mathcal{R}(0) \geq 0$ and $\mathcal{R}(\tau_c) = 0$. At any interior stationary point $\hat{\tau}_e$ satisfying $\mathcal{R}'(\hat{\tau}_e) = 0$, the second derivative is strictly negative because both the linear penalty and the concavity of g contribute terms of uniform sign. This guarantees at most one stationary point on $(0, \tau_c)$, which is necessarily the unique global maximizer. The Rician K-factor enters α' only through σ_h^2, which is τ_e-independent, so the unimodality holds for all K. Non-centered placement reduces $(k_0 - k_1^2/k_2)$ relative to k_0, thereby increasing α', which shifts $\rho^\star$ upward and lowers the peak $\mathcal{R}^\star$ by the information penalty consumed by nuisance coupling through ϕ_1.

The same structural property extends to the lifted LMMSE under arbitrary prior mis-specification. Scaling the calibrated prior covariance as $\mathbf{\Lambda}'_\theta(\varepsilon_p)$ modifies the regularization terms to λ'_i in the normal matrix defined in (43). Because λ'_0 and λ'_1 are additive constants independent of τ_e, the pilot moments s_0, s_2 carrying the τ_e-dependence are unaffected, and the determinant Δ_B in (36) retains the same polynomial dependence on τ_e as in the nominal case. The posterior variance $\sigma_{e,\rm LM}^2 = b_{11}/\Delta_B$ in (40) therefore preserves a strictly decreasing and strictly convex dependence on τ_e for every ε_p, which upon substitution into (7) yields $g'_{\rm LM} > 0$ and $g''_{\rm LM} < 0$ for all $\varepsilon_p \in \mathbb{R}$. The second-derivative negativity at every stationary point of $\mathcal{R}(\tau_e)$ follows by the same mechanism, so no violation was observed over the tested prior-mismatch range. These analytical conclusions are validated numerically in Fig. 26, which sweeps $\mathcal{R}(\tau_e)$ across $K \in \{0, 7, 10, 14\}$ dB and $\bar{\gamma}_{\rm eff} \in \{0, 5, 10, 15, 20\}$ dB for both centered and non-centered pilot placement under the joint (h_{eq}, ϕ_1) estimation framework with the actual FIM. Every evaluated configuration exhibits a single peak, and an exhaustive grid search over the full range of $\bar{\gamma}_{\rm eff}$, K, and ε_p identifies zero violations, confirming that the one-dimensional search in (14) suffices to determine $\tau_e^\star$ under all tested operating conditions.

C. Quantifying Lifted Estimator and Equalizer Performance under Robustness Constraints and Practical Trade-offs

The estimation accuracy of the lifted LS and LMMSE estimators is evaluated in terms of MSE, as shown in Fig. 27. When $\bar{\gamma}_{\rm eff} > 5$ dB, both estimators achieve MSE values below 10^{-4}, demonstrating that reliable CSI can be obtained for backscatter links. In the low and moderate $\bar{\gamma}_{\rm eff}$ regime, the LMMSE estimator in (38) outperforms the LS estimator in (28) by up to approximately one order of magnitude. This advantage arises from the shrinkage effect, where the LMMSE effectively combines the LS with prior channel statistics to suppress the variance of the pilot matched-filter statistic, thereby reducing estimation error for the same pilot energy. Under the lifted linearized model, the LS estimator is unbiased, and as $\bar{\gamma}_{\rm eff}$ increases, its error decreases monotonically with noise suppression, whereas the LMMSE error gradually saturates due to the residual bias induced by Bayesian shrinkage toward the prior mean. Consequently, at high $\bar{\gamma}_{\rm eff}$ or with sufficiently long

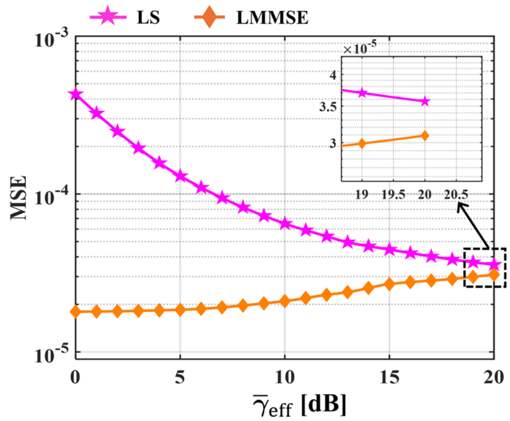


Fig. 27. MSE of the lifted LS and LMMSE estimators over varying $\bar{\gamma}_{\mathrm{eff}}$.

TABLE IV
DECOMPOSITION OF CHANNEL ESTIMATION MSE INTO FIVE ERROR COMPONENTS AT THE OPERATING POINT

SNR	0 dB		10 dB		20 dB	
Error Comp.	**MSE**	**Cont. [%]**	**MSE**	**Cont. [%]**	**MSE**	**Cont. [%]**
(i) Finite-length noise	3.96×10^{-4}	99.9%	3.96×10^{-5}	99.7 %	3.96×10^{-6}	98.2 %
(ii) Residual CFO	6.88×10^{-8}	< 0.1 %	6.88×10^{-8}	0.2 %	6.88×10^{-8}	1.7 %
(iii) Leakage-induced colored noise	3.80×10^{-9}	< 0.01 %	3.80×10^{-9}	< 0.01 %	3.80×10^{-9}	< 0.1 %
(iv) DC-removal bias	1.88×10^{-9}	< 0.01 %	1.90×10^{-10}	< 0.01 %	1.89×10^{-11}	< 0.01 %
(v) Linearization error	1.19×10^{-16}	< 0.01 %	1.19×10^{-16}	< 0.01 %	1.19×10^{-16}	< 0.01 %

pilot sequences, the LMMSE weight approaches unity and both estimators converge, making the LS estimator an equally effective yet computationally simpler alternative.

Thus, while the MSE provides a useful aggregate metric, the conventional form, $\mathbb{E}\left[\left|\hat{h}-h_{eq}\right|^2\right]$, conflates two distinct contributions: (i) amplitude mismatch and (ii) phase ambiguity. When these sources are mixed, it can be unclear whether poor performance arises from magnitude error or residual phase rotation, and even a constant phase offset can artificially increase the reported MSE despite accurate amplitude estimation. To address this, the error can be decomposed into separate magnitude and phase components. Specifically, let $\hat{r} \triangleq \left|\hat{h}\right|$, and $\Delta\psi \triangleq \arg(\hat{h}) - \arg(h_{eq})$ so that $\Delta\psi$ represents the residual phase difference between the estimate and the true channel. By the law of cosines, the instantaneous squared error can be expressed as:

$$\left|\hat{h}-h_{eq}\right|^2 = (\hat{r}-r)^2 + 4\hat{r}r\sin^2\left(\frac{\Delta\psi}{2}\right). \quad (65)$$

Taking expectations and using linearity of the expectation operator, the following decomposition is obtained:

$$\mathbb{E}\left[\left|\hat{h}-h_{eq}\right|^2\right] = \underbrace{\mathbb{E}[(\hat{r}-r)^2]}_{\text{Magnitude term}} + \underbrace{\mathbb{E}\left[4\hat{r}r\sin^2\left(\frac{\Delta\psi}{2}\right)\right]}_{\text{Phase term}}. \quad (66)$$

Notice that the relationship holds identically for every realization of $(r, \hat{r}, \Delta\psi)$, and therefore also under expectation. The decomposition preserves overall comparability with the conventional MSE while providing diagnostic insight: for instance, improvements in synchronization or phase tracking manifest as a reduction in the phase component, whereas pilot energy, pilot length, and estimator design primarily influence the magnitude component [46].

The total analytical MSE can be further decomposed into five physical error components to identify which mechanism fundamentally limits the estimation accuracy. Table IV presents the CE MSE budget into five impairment-driven components at the reported operating points. The finite-length noise term (i), corresponding to the estimation variance in (32), dominates the analytical budget at 98.2–99.9% across all SNR conditions, confirming that the lifted estimator reaches its information-theoretic performance bound. Terms (i) and (iv) decrease with increasing SNR because they originate from noise-driven estimation variance, whereas terms (ii), (iii), and (v) are SNR-independent because their underlying mechanisms—oscillator jitter, Tx–Rx self-interference, and model linearization—are decoupled from the receive noise power. Despite a modest growth in the relative share of the SNR-independent terms at higher SNRs, all four non-dominant terms remain collectively below 2%. Each is rendered negligible by the preceding design algorithms:

- [Term (ii)] Residual CFO: The lifted two-parameter model absorbs the deterministic CFO drift, leaving only stochastic oscillator jitter at 17.6 dB below term (i) at $\bar{\gamma}_{\mathrm{eff}} = 20$ dB.
- [Term (iii)] Leakage-induced colored noise: The SRR isolation and DC removal suppress Tx–Rx self-interference below 0.1%.
- [Term (iv)] DC-removal bias: The zero-mean pilot sequence structurally decouples the DC estimation error from the channel estimate, confining it below 0.01%.
- [Term (v)] Linearization error: The linearization remainder is bounded with a 52.3 dB admissibility margin relative to the centered pilot bound in (47), rendering it negligible at $O(10^{-16})$.

Whether the SRR-induced change in leakage statistics compromises the proposed estimator is assessed through a matched-SNR evaluation. The lifted CE is computed under the measured $\mathbf{C}_w$ with and without SRRs while $\bar{\gamma}_{\mathrm{eff}}$ is equalized at each operating point, isolating the noise-coloring effect from the SNR gain that the SRR provides. The resulting $\Delta\mathrm{MSE} \triangleq \mathrm{MSE}_{\mathrm{w/\,SRR}} - \mathrm{MSE}_{\mathrm{w/o\,SRR}}$ is presented in Fig. 28. The difference remains bounded within $|\Delta\mathrm{MSE}| \leq 0.035$ dB across 0–20 dB, with $|\Delta\mathrm{MSE}| < 0.004$ dB for $\bar{\gamma}_{\mathrm{eff}} \leq 10$ dB where thermal noise dominates, and a monotonic increase at higher SNR consistent with the growing relative contribution of the colored leakage residual as the thermal floor recedes. The

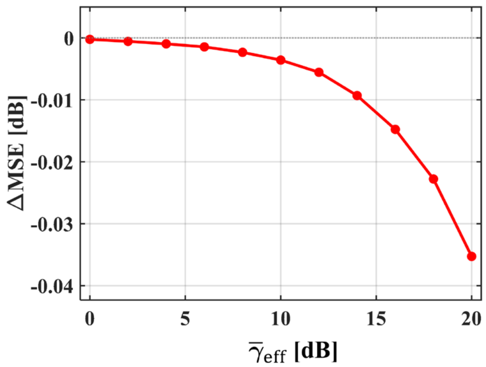


Fig. 28. Matched-SNR CE-MSE difference of the lifted estimator with and without SRRs.

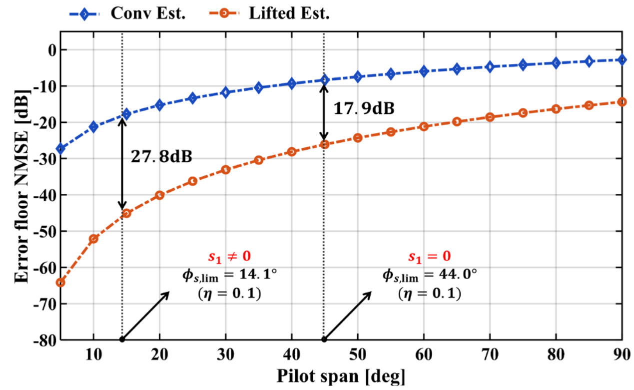


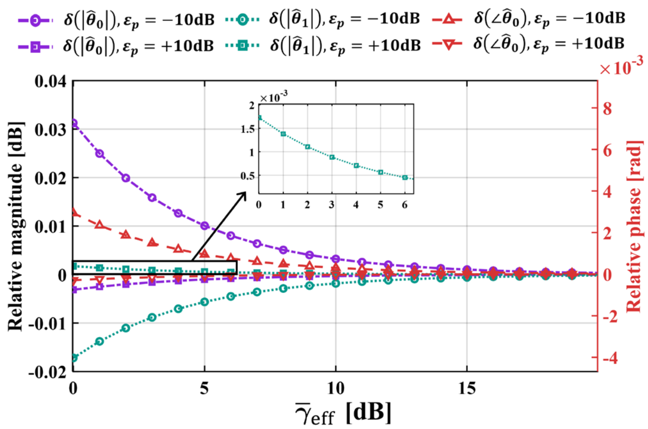
Fig. 29. NMSE floor versus pilot phase span ϕ_s for the conventional scalar and lifted estimators. Vertical lines indicate the admissible pilot phase span at $\eta = 0.1$ for non-centered ($s_1 \neq 0$) and centered pilot placement ($s_1 = 0$).

worst-case 0.035 dB at $\bar{\gamma}_{\text{eff}} = 20$ dB is quantitatively consistent with the leakage-induced colored noise term (iii) remaining below 0.1 % of the total MSE budget, confirming that the parameter-independence of $\mathbf{C}_w$ and the deterministic leakage suppression by DC removal jointly render the lifted estimator insensitive to the SRR-induced covariance change at matched-SNR.

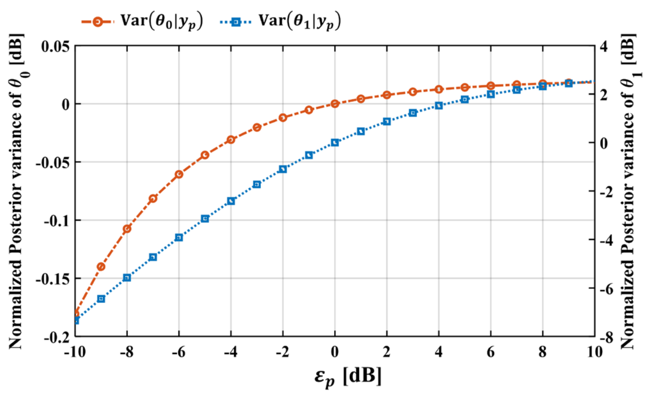
To isolate the impact of residual phase variation within a coherence frame, an ablation is conducted by disabling the phase term and reverting to the conventional scalar estimator that assumes a phase-stationary pilot model. The conventional estimator used in the ablation is the scalar CE solution, which corresponds to fitting a single regressor column $\mathbf{x}_{\mathbf{p}}$ only. This ablation intentionally removes the residual phase term from the estimator and also removes the additional lifted regressor used by the proposed two-parameter formulation. Under this condition, the conventional scalar estimator becomes structurally biased because the assumed phase-stationary pilot model no longer matches the observations. This bias is inherited by the subsequent ZF equalizer through the inverse channel gain, and it does not vanish as SNR increases. Therefore, as SNR grows, the stochastic noise term decreases while the deterministic bias remains, producing the normalized mean square error (NMSE) floor driven by the residual phase slope and the pilot placement through the ratio s_1/s_0. To report the residual phase variation, the residual phase-slope estimate is obtained from the lifted solution by normalizing the first-order coefficient by $\hat{\theta}_0$ and taking the imaginary part, which yields $\hat{\phi}_1$ in rad per sample. The cumulative phase drift across the pilot is then summarized by the pilot phase span $\phi_s \triangleq |\hat{\phi}_1|(n_{\max} - n_{\min})$. The quantity ϕ_s directly captures the total phase rotation from the first to the last pilot sample within a coherence frame and directly matches the model mismatch mechanism. The high-SNR NMSE floor as a function of the pilot phase span for the conventional ablation and the proposed estimator is illustrated in Fig. 29. The conventional ablation exhibits a clear error floor whose level increases monotonically with pilot phase span, consistent with the deterministic bias mechanism. In contrast, the proposed estimator suppresses the structural floor by explicitly estimating the residual phase coefficient jointly with the channel through the lifted two-parameter model. The pilot phase span validity threshold $\phi_{s,\text{lim}}$, derived from the Taylor remainder bias constraint, can also be interpreted from the result. Using the bias bound criterion in (45)–(47) with a target bias tolerance of 10%, the resulting admissible pilot phase span depends on the pilot centering strategy. For non-centered pilot indices ($s_1 \neq 0$), the full bound in (46) retains the $|s_1| n_{\max}$ cross term in the denominator, yielding the more restrictive threshold $\phi_{s,\text{lim}} = 14.1°$. For centered pilot indices ($s_1 = 0$), the cross term is removed, and the simplified bound in (47) relaxes the threshold to $\phi_{s,\text{lim}} = 44.0°$. At these two bounds, the lifted estimator provided 27.8 dB ($s_1 \neq 0$) and 17.9 dB ($s_1 = 0$) of NMSE suppression relative to the conventional estimator. Beyond each respective threshold the conventional estimator becomes increasingly mismatch-limited, while the proposed estimator maintains a substantially lower asymptotic NMSE, demonstrating quantitative removal of the error floor across both pilot placement strategies, consistent with the bias mechanism identified in Section IV-B.

The proposed lifted LMMSE estimator depends on the prior mean and diagonal prior covariance $\mathbf{\Lambda}_\theta = \text{diag}(\sigma_{\theta_0}^2, \sigma_{\theta_1}^2)$ defined on the lifted unknown vector $\boldsymbol{\theta}$. To avoid any test-set leakage, these hyperparameters are determined using pilot-only observations and are fixed before evaluating any payload-dependent metric. Specifically, a calibration set of pilot windows $\mathcal{K}_c$ is formed. For each $k \in \mathcal{K}_c$, a pilot-only LS estimate $\hat{\boldsymbol{\theta}}_{\text{LS},k}$ is computed from the known pilot regressors $\mathbf{X}_{\mathbf{p},k}$ and the received pilot samples $\mathbf{y}_{\mathbf{p},k}$. The $\boldsymbol{\mu}_{\boldsymbol{\theta}}$ is set to the empirical average of $\{\hat{\boldsymbol{\theta}}_{\text{LS},k}\}_{k \in \mathcal{K}_c}$, and the $\sigma_{\theta_0}^2$ and $\sigma_{\theta_1}^2$ are set from the corresponding empirical second-order moments across the same set. The σ_w^2 is estimated from pilot residual energy. This procedure uses only known pilot symbols and their received samples; it does not use payload bits, decoded decisions, or evaluation labels.

Robustness to diagonal prior mismatch is evaluated by scaling the calibrated covariance as $\mathbf{\Lambda}'_\theta(\varepsilon_p) = 10^{\varepsilon_p/10}\mathbf{\Lambda}_\theta$. The sensitivity reports the relative deviations of $|\hat{\theta}_0|$, $|\hat{\theta}_1|$, and the phase angle of $\angle\hat{\theta}_0$ under $\varepsilon_p = \pm 10$ dB, referenced to the



(a)



(b)

Fig. 30. Sensitivity of the lifted LMMSE estimator to diagonal prior variance mis-specification: (a) relative deviations of the estimated coefficients and phase angle versus $\bar{\gamma}_{\text{eff}}$, and (b) normalized posterior variances in dB versus ε_p.

nominal prior $\varepsilon_p = 0$ dB, as illustrated in Fig. 30(a). The deviations decrease smoothly as $\bar{\gamma}_{\text{eff}}$ increases and exhibit no numerical irregularities, indicating stable estimator behavior across the tested SNR range. The larger deviation under $\varepsilon_p = -10$ dB is expected, since reducing the prior variance increases prior precision and strengthens shrinkage toward $\boldsymbol{\mu}_{\boldsymbol{\theta}}$, whereas $\varepsilon_p = +10$ dB weakens regularization and moves the solution toward an LS-like regime. Notably, the $\varepsilon_p = +10$ dB deviations remain negligible across the entire SNR range, confirming that an over-estimated prior gracefully degrades toward the data-driven LS solution without introducing estimation bias. Fig. 30(b) further shows the normalized posterior variances $\text{Var}(\theta_0|\mathbf{y}_p)$ and $\text{Var}(\theta_1|\mathbf{y}_p)$ as a function of ε_p. For the θ_0, the total variation over the full ± 10 dB mismatch range is bounded within 0.21 dB, indicating that the pilot data information s_0/σ_w^2 dominates over the prior contribution in the posterior covariance. The nuisance coefficient θ_1 exhibits a wider variation of approximately 9.5 dB; this behavior is consistent with the lifted information structure. Specifically, the data contribution to the posterior precision is $(1/\sigma_w^2)\mathbf{G}$, so the diagonal terms s_0 and s_2 quantify the data information for θ_0 and θ_1, respectively, while s_1 controls their coupling. Under the adopted centered placement, s_1 is small, making the coupling weak; consequently, variations

TABLE V

HOST COMPUTATIONAL BURDEN OF SCALAR VS. LIFTED CHANNEL ESTIMATORS ($N^* = 120$)

Metric	LS		LMMSE	
	Scalar	Lifted	Scalar	Lifted
*Arithmetic operations per CE call**				
Real mul.	720	1,210	725	1,221
Real add.	720	1,205	723	1,211
Real div.	2	4	2	4
Total ops	**1,442**	**2,419**	**1,450**	**2,436**
Op count overhead	—	1.68 ×	—	1.68 ×
Estimator working memory (real scalars)				
Runtime accum.**	3	7	3	7
Stored params.***	0	0	4	7
Total memory	**3**	**7**	**7**	**14**
*Measured host latency (50,000-trial average)*****				
Latency (μs)	2.122	3.109	2.202	3.355
CPU kcycles	~11	~16	~11	~17
Measured overhead	—	1.47 ×	—	1.52 ×
Frame budget				
CE / frame	—	—	—	0.265 %
Marginal overhead /frame	—	—	—	0.091%

*Upper bound: 1 complex mul. = 4 real mul. + 2 real add.; all statistics at runtime. All variants $O(N)$.

**s_0, s_1, s_2 (real), t_0, t_1 (complex); post-loop intermediates reuse these registers.

***$\sigma_{\theta_0}^2$, $\sigma_{\theta_1}^2$, σ_w^2 (real), μ_{θ_0}, μ_{θ_1} (complex).

****CPU: Intel(R) Core(TM) i5-14600K, 5.0 GHz sustained clock. CE on host, FPGA unmodified. τ_c = 1.264 ms; worst-case shown.

in $\text{Var}(\theta_1|\mathbf{y}_p)$ under ϵ_p do not significantly transfer to θ_0, which remains strongly dominated by the pilot data term s_0/σ_w^2. However, since θ_1 serves solely as a nuisance variable for residual phase-slope absorption and is not used in the subsequent equalization stage, this higher sensitivity does not propagate to the demodulation performance. Both curves vary smoothly and monotonically over the full ±10 dB range, confirming numerical stability under prior mis-specification.

To verify that the lifted two-parameter formulation does not introduce a prohibitive computational penalty and to clarify the practical trade-off between the two proposed estimators, Table V quantifies the per-call arithmetic cost, working memory, and measured host latency of all four estimator variants under a conservative upper-bound convention in which every pilot sample is treated as a general complex symbol and all sufficient statistics are computed at runtime. The additional column $\mathbf{n} \odot \mathbf{x_p}$ in (21) introduces three extra per-sample accumulations (s_1, s_2, t_1) while preserving the 2×2 closed-form solution in (27), yielding an analytical overhead of $1.68\times$ at N^* with $O(N)$ complexity preserved for all four variants. The memory increment is four real registers, exactly the state required by the second regressor column. Profiling on the host confirms a measured overhead of $1.47\times\sim1.52\times$, in which the lifted LMMSE yields the worst-case CE of 0.265% of the frame duration and the marginal lifting cost below 0.1% of the frame budget. Since the overhead is a bounded constant factor arising solely from three additional scalar accumulations per sample, it does not alter the $O(N)$ complexity class, the estimator structure, or the closed-form solvability. The lifted two-parameter formulation therefore preserves the computational profile of the scalar baseline and introduces no operationally significant burden at the pilot length of this system.

Between the two lifted variants, the LMMSE incurs only a small incremental cost over the LS, namely seven additional real scalars of working memory and seventeen additional arithmetic operations per CE call, corresponding to a measured host-latency increase of 0.246 µs that remains below 0.1% of the frame budget. The practical selection between the two is therefore governed by the operating regime rather than by complexity: the LMMSE delivers the MSE advantage reported above when reliable prior statistics are available and the operating SNR is low to moderate, whereas the LS remains the robust default at high SNR, where the two estimators converge, and under deployment conditions in which prior calibration is impractical.

To quantify the statistical and implementation gap, the joint maximum-likelihood (ML) estimator that searches ϕ_1 by maximizing coherent accumulation and computes h_{eq} in closed form conditioned on $\hat{\phi}_1$ is implemented as a reference, using the same received pilot vector and index centering. Fig. 31 reports the bias loss Δ_h and variance loss Δ_v of the lifted estimators relative to joint ML for $\hat{h}$ and $\hat{\phi}$ across $\bar{\gamma}_{\text{eff}}$, with the lifted LMMSE parameterized by the prior-to-data precision ratio $\Lambda_{\text{eff}}^{-1} \triangleq \lambda_0/s_0$, which quantifies the relative strength of the calibrated prior with respect to the pilot-domain data information; the lifted LS, being prior-free, appears as a single curve. For $\hat{h}$, the lifted LS bias loss is small and nearly SNR-invariant, confirming that its bias is dominated by deterministic surrogate mismatch. The lifted LMMSE bias loss decreases with $\bar{\gamma}_{\text{eff}}$ as shrinkage diminishes, with larger $\Lambda_{\text{eff}}^{-1}$ yielding higher bias due to stronger regularization. For $\hat{\phi}$, both methods show a comparatively large bias loss from estimating ϕ_1 indirectly through $\hat{\theta}_1/\hat{\theta}_0$, with the LMMSE curves ordered by $\Lambda_{\text{eff}}^{-1}$. For $\hat{h}$ variance loss, the lifted LS incurs the largest penalty, peaking at moderate $\bar{\gamma}_{\text{eff}}$; the lifted LMMSE lies consistently below, with larger $\Lambda_{\text{eff}}^{-1}$ providing stronger suppression at low $\bar{\gamma}_{\text{eff}}$ and all curves converging toward LS as the posterior becomes data-dominated. For $\hat{\phi}$, variance loss under the lifted LS reaches approximately 38 dB at low $\bar{\gamma}_{\text{eff}}$ due to ratio-mapping amplification; the lifted LMMSE substantially reduces this loss across all $\Lambda_{\text{eff}}^{-1}$ settings.

Table VI confirms that the complexity gap is structural. The lifted estimators scale as $O(N)$ through a 2×2 closed-form solve, whereas the joint ML estimator scales as $O(NN_{\text{grid}})$; on the same platform, this yields orders-of-magnitude higher fixed-point multiplication and addition counts.

To distinguish whether the observed estimation gains arise from phase-slope-aware modeling or from pilot design, the lifted LMMSE was benchmarked against a nonlinear joint ML estimator that searches over ϕ_1 without first-order slope linearization, and a DD least-mean-squares (DD-LMS) tracker

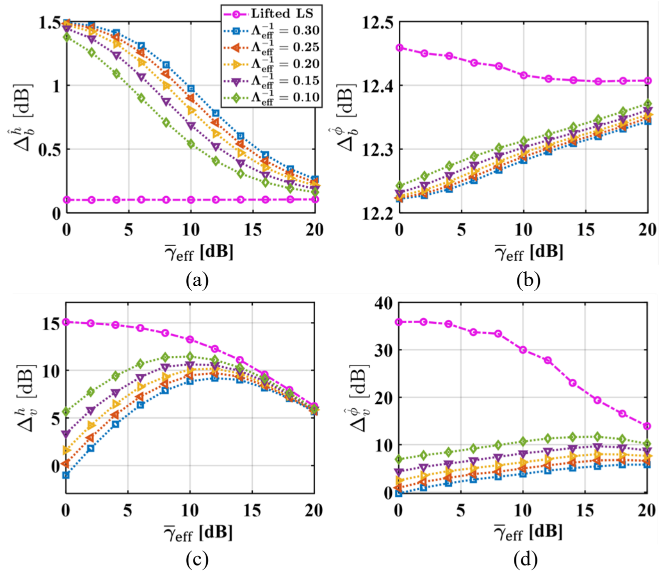


Fig. 31. Bias and variance losses of the lifted estimators relative to the joint ML estimator: bias loss for (a) $\hat{h}$ and (b) $\hat{\phi}$; variance loss for (c) $\hat{h}$ and (d) $\hat{\phi}$. The lifted LS is prior-independent; the lifted LMMSE is parameterized by the prior-to-data precision ratio $\Lambda_{\text{eff}}^{-1}$.

TABLE VI
IMPLEMENTATION COST COMPARISON PER EVENT FOR LIFTED ESTIMATORS AND JOINT ML

Platform Cost	Lifted LS	Lifted LMMSE	Joint ML
Fixed-point multiplications	2.694×10^4	2.714×10^4	2.736×10^7
Fixed-point additions	2.358×10^4	2.378×10^4	2.735×10^7
Complexity gap ML over the lifted method	1.016×10^3 mul., 1.160×10^3 add.	1.008×10^3 mul., 1.150×10^3 add.	—
FPGA cycles/time on USRP N200 Spartan 3A DSP	842 cycles, 8.42×10^{-6} s	849 cycles, 8.49×10^{-6} s	855,038 cycles, 8.55×10^{-3} s
MCU ATmega328P cycles/time	2.930×10^5 cycles, 1.831×10^{-2} s	2.952×10^5 cycles, 1.845×10^{-2} s	3.010×10^8 cycles, 1.881×10^1 s
ML grid storage feasibility for Q15 complex table	—	—	Storage required: 2.15×10^8 bits BRAM available: 1.55×10^6 bits Storage excess: $1.39\times10^2\times$

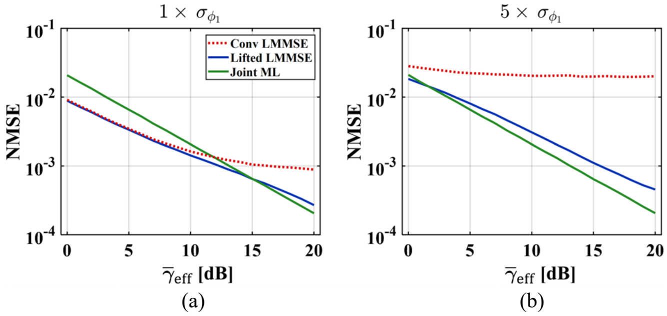


Fig. 32. CE NMSE of the conventional LMMSE, lifted LMMSE, and joint ML (without linearization) estimators: (a) measured operating drift and (b) 5 × drift stress condition.

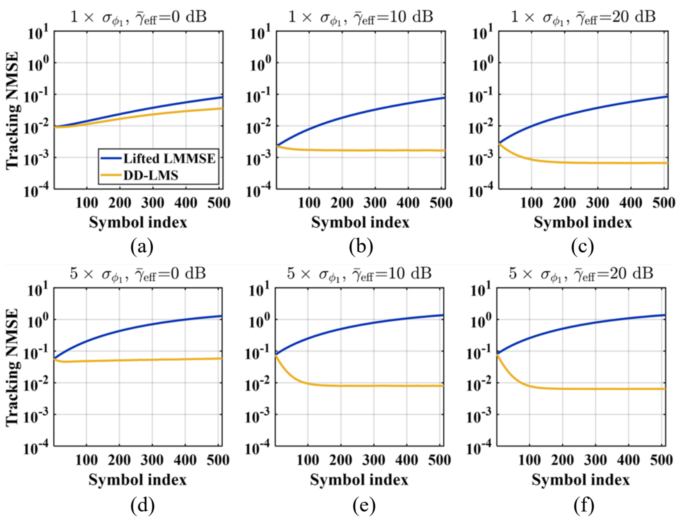


Fig. 33. Data-phase channel tracking NMSE versus symbol index for the pilot-only lifted LMMSE estimate and the DD-LMS adaptive tracker, at $\bar{\gamma}_{\text{eff}} \in \{0, 10, 20\}$ dB: (a), (b), (c) measured operating drift and (d), (e), (f) 5 × drift stress.

that updates the channel estimate symbol-by-symbol during the data phase using detected symbols as virtual pilots. The conventional LMMSE serves as the baseline. Performance is considered at the measured operating point as well as a five-fold drift stress condition. The CE NMSE results in Fig. 32 show that the gain is primarily driven by phase-slope modeling. Under the measured drift condition, the conventional LMMSE exhibits a high-SNR floor due to the bias induced by uncompensated phase rotation, whereas the lifted LMMSE continues to improve by incorporating the phase slope into the regressor, providing more than 5 dB of gain at $\bar{\gamma}_{\text{eff}} = 20$ dB. The nonlinear joint ML is less effective at low SNR, where the likelihood surface along the slope dimension is weakly conditioned and the regularizing effect of the prior precision $\sigma_{\theta_1}^{-2}$ becomes beneficial, but it becomes superior at higher SNR once the nonlinear optimization can fully exploit the exact phase model. Under the 5 × drift stress condition, the conventional floor increases substantially, the lifted LMMSE gain exceeds 16 dB at $\bar{\gamma}_{\text{eff}} \approx 20$ dB, and the crossover point shifts to lower SNR, consistent with the expected growth of linearization mismatch.

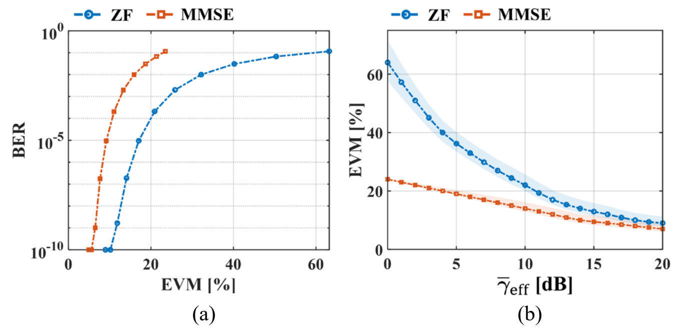


Fig. 34. Noise-enhancement penalty of ZF versus EA MMSE equalization using the estimated $\text{Var}(\hat{h})$: (a) BER–EVM characteristic and (b) EVM performance where shaded area denotes the 90% CI propagated from the closed-form posterior covariance.

The data-decoding phase comparison in Fig. 33 highlights a different trade-off. The pilot-only lifted LMMSE produces a fixed estimate after the pilot interval that does not rely on data decisions, whereas the DD-LMS tracker, initialized from the lifted estimate, continues to adapt and can further reduce the tracking NMSE at moderate-to-high SNR where symbol decisions are reliable. At low SNR, however, decision errors degrade the update direction and limit the attainable tracking gain, as observed at 0 dB under both drift conditions. The benefit of DD tracking is therefore inherently SNR-dependent and requires continuous per-symbol adaptation, while the pilot-only lifted estimator is determined entirely by the pilot observations and the assumed phase-slope statistics. This trade-off supports the pilot-only lifted formulation as the operating design point for the targeted low-power quasi-static indoor deployment, where the dominant improvement originates from explicitly modeling the residual phase-slope rather than from pilot design alone.

In a single-tap flat fading frame, the equalizer is determined from the per-frame $\hat{h}$, and the baseline equalizer is the one-tap ZF. To quantify the noise-enhancement penalty under channel uncertainty, an EA MMSE equalizer is additionally considered by incorporating the estimated error variance $\text{Var}(\hat{h})$ obtained from the proposed estimators. Using the same $\hat{h}$ per-frame, the EA MMSE equalizer is written as:

$$p_{\text{MMSE}} = \frac{\hat{h}^*}{\left|\hat{h}\right|^2 + (\sigma_w^2/\mathcal{E}_s) + \sigma_{\hat{h}}^2}. \tag{67}$$

This expression is exactly a positive real shrinkage of ZF,

$$p_{\text{MMSE}} = \beta_{SH} p_{ZF}, \qquad \beta_{SH} = \frac{\left|\hat{h}\right|^2}{\left|\hat{h}\right|^2 + (\sigma_w^2/\mathcal{E}_s) + \sigma_{\hat{h}}^2}, \tag{68}$$

where the shrinkage factor β_{SH} satisfies $0 < \beta_{SH} \leq 1$, so that both equalizers apply the same phase correction dictated by $\hat{h}$, while the EA MMSE shrinkage mainly limits magnitude growth and suppresses noise amplification. The EVM is evaluated using the manuscript definition in (61), and since the EVM is determined directly by the second-moment dispersion of the post-equalizer error, β_{SH} produces a clear EVM reduction by limiting noise enhancement. In contrast, the BER under the adopted hard-decision receiver is dominated by decision-boundary crossings after per-frame normalization. Since β_{SH} is a positive real scalar, the geometry of the decision boundary and the sign structure of the decision statistic are largely preserved, so many samples remain on the same side of the boundary even when the dispersion level changes. This mechanism explains why the EVM gap can be large while the BER gap remains small in Fig. 34(a). At 0 dB, the ZF equalizer yielded an EVM of 63.2%, compared with 23.5% for EA MMSE, yet the BER values remained comparable at approximately 1.16×10^{-1}. Similarly, at 10 dB, the EVM was reduced to 20.89% for ZF and 11.05% for EA MMSE, while the BER differed only marginally and remained on the order of 10^{-4}, as shown in Fig. 34(b). At higher SNR, both equalizers reached near-zero BER over the tested range, whereas the EVM gap can persist because the EVM captures residual dispersion even when decision errors vanish. The shaded bands in Fig. 34(b) show the 90% confidence interval (CI) obtained by propagating the closed-form posterior covariance through each equalizer. The ZF CI is consistently wider than the EA MMSE CI, confirming that β_{SH} suppresses not only the mean EVM but also its frame-to-frame variability induced by channel estimation uncertainty.

The selection of the ZF equalizer remains consistent with the design target of reliable high-throughput data recovery under strict simplicity and fixed-point robustness constraints. The curves in Fig. 34 indicate that once $\hat{h}$ stabilizes under the proposed CE methods, the EA MMSE offers only marginal BER improvement relative to the ZF across the operating SNR range, despite the EVM advantage. Implementing the EA MMSE also requires stable tracking of σ_w^2, $\mathcal{E}_s$, and $\sigma_{\hat{h}}^2$. Any mismatch perturbs β_{SH} and introduces SNR-dependent amplitude shrinkage, which in turn tightens the requirements on gain management for consistent downstream thresholds under per-frame normalization. The ZF is parameter-free once $\hat{h}$ is available, integrates directly into the existing per-frame processing chain, and avoids additional control dependencies, which is aligned with the platform-driven constraints of the implemented receiver.

To analyze BER performance, the BER expressions for OOK and BPSK under the double-Rician fading channel are derived from (5). For coherent OOK with perfect CSI, the conditional error probability on the instantaneous SNR is $P_e^{\mathrm{OOK}}(\gamma) = Q\left(\sqrt{\gamma/2}\right)$, where $Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{+\infty} \exp\left(-\frac{t^2}{2}\right) dt$ denotes the Gaussian Q-function. The average BER is then [47]

$$\bar{P}_b^{\mathrm{OOK}} = \int_0^{\infty} Q\left(\sqrt{\frac{\gamma}{2}}\right) f_\gamma(\gamma) d\gamma \,. \tag{69}$$

Because the product-Rician envelope density $p_h(r)$ in (1) is expressed as an infinite double series, (69) does not admit a direct closed-form. Applying Craig's representation of the Gaussian Q-function and interchanging the integration order

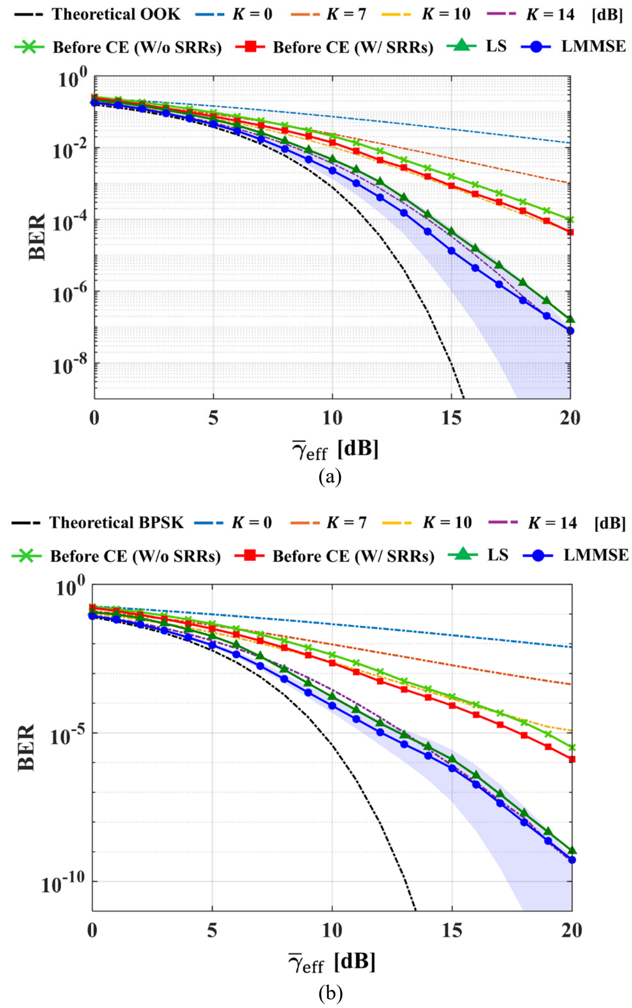


Fig. 35. BER performance versus $\bar{\gamma}_{\mathrm{eff}}$ under the cascaded double-Rician backscatter fading channel for varying Rician K-factor: (a) OOK and (b) BPSK. Shaded areas denote the 90% CI of the LMMSE propagated from the closed-form posterior covariance.

converts (69) into a single finite-interval integral of the SNR moment-generating function (MGF),

$$\bar{P}_b^{\mathrm{OOK}} = \frac{1}{\pi}\int_0^{\pi/2} \mathcal{M}_\gamma\left(\frac{1}{4\sin^2\theta}\right) d\theta \,, \tag{70}$$

which provides an exact single-integral expression under the double-Rician model adopted in (1). For coherent BPSK with perfect CSI, the conditional error probability is $P_e^{\mathrm{BPSK}}(\gamma) = Q\left(\sqrt{2\gamma}\right)$. The same Craig-form reduction yields

$$\bar{P}_b^{\mathrm{BPSK}} = \frac{1}{\pi}\int_0^{\pi/2} \mathcal{M}_\gamma\left(\frac{1}{\sin^2\theta}\right) d\theta \,. \tag{71}$$

Expressions (70) and (71) are exact for arbitrary Rician K-factors K_{tt}, K_{tr}, depending on the channel only through the MGF $\mathcal{M}_\gamma(s)$, which is expressed as a tractable single integral in Appendix A. The derivations of (70) and (71) are provided in Appendices A and B, respectively.

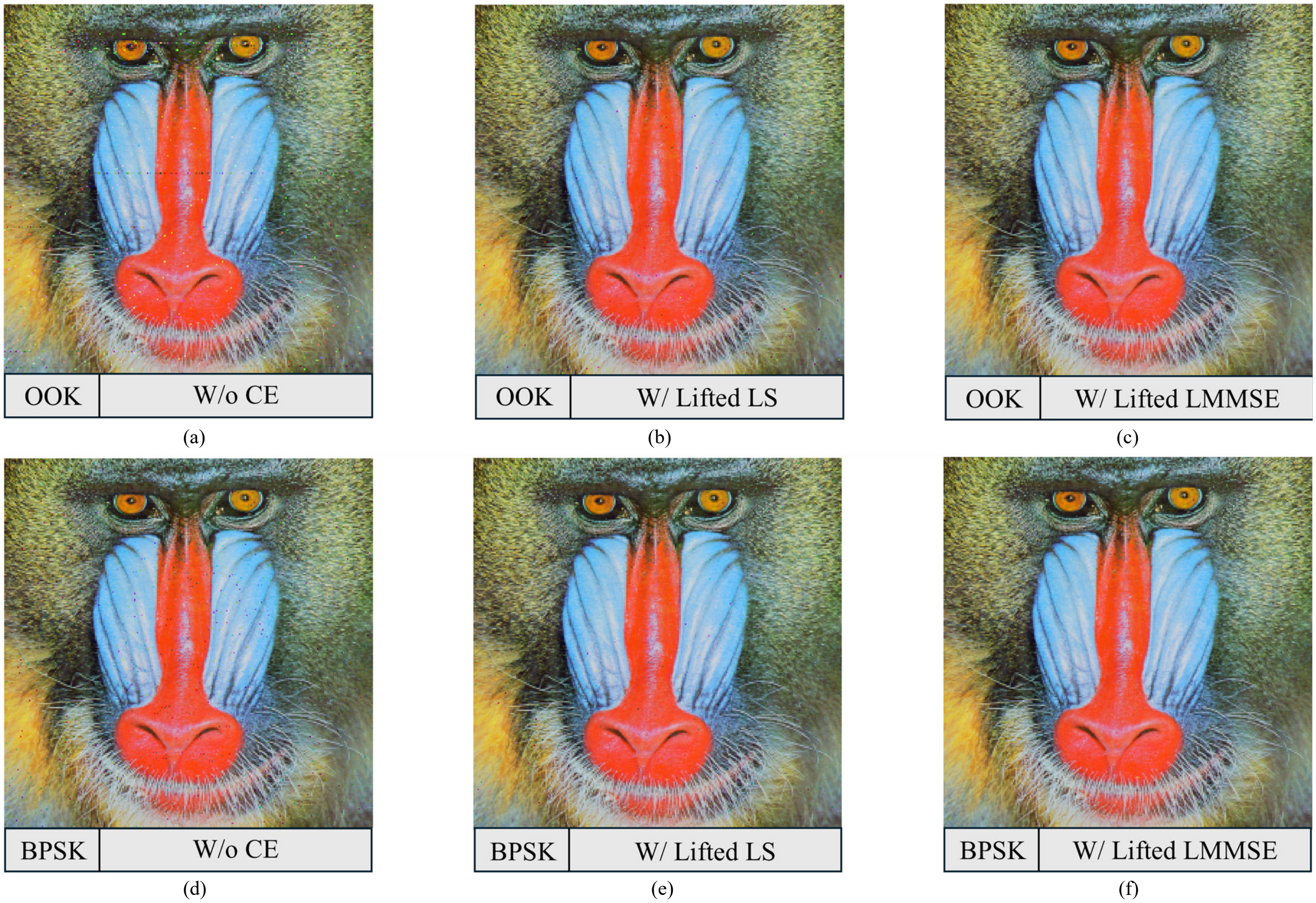


Fig. 36. OTA image reconstruction at a fixed operating point, $\bar{\gamma}_{\mathrm{eff}} = 15$ dB, showing the quantitative impact of CE on end-to-end reliability: (a) OOK without CE at BER $= 9.3 \times 10^{-4}$, (b) OOK with the lifted LS at BER $= 5.1 \times 10^{-5}$, (c) OOK with the lifted LMMSE at BER $< 10^{-9}$, (d) BPSK without CE at BER $= 9.1 \times 10^{-5}$, (e) BPSK with the lifted LS at BER $= 1.4 \times 10^{-6}$, and (f) BPSK with the lifted LMMSE at BER $< 10^{-9}$.

TABLE VII
SYSTEM-LEVEL PERFORMANCE COMPARISON OF BACKSCATTER SYSTEMS

Ref.	**Config.**	**Modulation Scheme**	**Energy Efficiency**	**Data Rate**	**Range**	**Channel Estimation**	**End-to-End Multimedia OTA**
[48]	Monostatic (MIMO)	BPSK	~10² pJ/bit	N. A.	60 cm	LS	✗
[49]	Monostatic	ASK	100 nJ/bit	67 kbps	5 m	✗	Grayscale image
[50]	Bistatic	OOK, FSK	20 µJ/bit	1 kbps	130 m	✗	✗
[51]	Ambient	4-PSK	N. A.	20 kbps	0.76 m	✗ (Var. Est.)	✗
[52]	Bistatic	DBPSK	1.8 nJ/bit	1 Mbps	15 m	✗	✗
This work	**Monostatic**	**OOK, BPSK**	**20 pJ/bit (SW), 22.4 nJ/bit (SW+MCU)**	**500 kbps**	**1 m**	**Lifted LS, LMMSE**	**Full-color image**

Monte Carlo simulation with 10^6 realizations confirms that both modulations exhibit BER well above the AWGN benchmark due to the multiplicative channel, with the curves shifting rightward relative to the theoretical limits. The Rician K-factor, denoting the specular-to-scattered power ratio, is swept over $\{0, 7, 10, 14\}$ dB, where $K = 0$ recovers Rayleigh fading; as K increases, the BER improves through reinforcement of the specular component, yet the gap to AWGN persists because both links must be simultaneously strong. ML detection with perfect CSI is plotted for each K as a reference baseline.

To quantify the benefit of CE, the effective SNR during the ID phase is obtained by substituting the estimator-specific error variance σ_e^2 directly into (7). Using the LS variance $\sigma_{e,\mathrm{LS}}^2$ from (32) and substituting into (7) gives

$$\bar{\gamma}_{\mathrm{eff}}^{\mathrm{LS}} = \frac{\bar{\gamma}_{\mathrm{eff}}}{1 + \bar{\gamma}_{\mathrm{eff}} \sigma_w^2 s_2 / (\sigma_h^2 \Delta)}, \tag{72}$$

and using the LMMSE posterior variance $\sigma_{e,\mathrm{LM}}^2$ from (40) gives

$$\bar{\gamma}_{\mathrm{eff}}^{\mathrm{LM}} = \frac{\bar{\gamma}_{\mathrm{eff}}}{1 + \bar{\gamma}_{\mathrm{eff}} b_{11} / (\sigma_h^2 \Delta_{\mathrm{B}})}. \tag{73}$$

Fig. 35 decomposes the resulting BER gains into three distinct contributions. First, introducing CE elevates the effective SNR enough to shift the BER curve by approximately 4–5 dB for both OOK in Fig. 35(a) and BPSK in Fig. 35(b). Second, the SRRs suppress Tx–Rx self-interference at the antenna interface, reducing the leakage-induced colored noise residual and lowering the effective noise floor, yielding an additional 1–2 dB improvement. Third, replacing the LS with the LMMSE exploits the statistical prior to further reduce the estimation error, contributing approximately 2 dB. The gains are more pronounced in BPSK because the Q-function argument $\sqrt{2\bar{\gamma}_{\mathrm{eff}}^{\mathrm{CE}}}$ makes coherent detection more sensitive to estimation quality.

The per-frame reliability beyond the mean BER is further assessed by propagating the closed-form posterior covariance in (39) through the ZF post-equalization SNR, yielding 90% CIs on both LMMSE curves as shown by the shaded bands in Figs. 35(a) and (b). The CI remains tightly centered at low $\bar{\gamma}_{\mathrm{eff}}$, where thermal noise dominates the total error budget and the estimation uncertainty contributes only a minor relative perturbation. At high $\bar{\gamma}_{\mathrm{eff}}$, the CI widens in the log-BER domain despite the posterior variance decreasing monotonically; this reflects the exponential sensitivity of the Q-function argument, whereby a sub-dB variation in $\bar{\gamma}_{\mathrm{eff}}^{\mathrm{CE}}$ maps to a large multiplicative shift in BER. The widening is therefore intrinsic to the detection nonlinearity rather than to estimation degradation.

The visual reconstructions in Fig. 36(a)–(f) provide a fixed-SNR quantitative comparison of the BER improvement enabled by the proposed CE at $\bar{\gamma}_{\mathrm{eff}} = 15$ dB. Relative to the no-CE reference, the lifted LS estimator yields approximately 18-fold and 65-fold BER reductions for OOK and BPSK, respectively, while the lifted LMMSE estimator drives the BER below 10^{-9} for both modulations, corresponding to improvements exceeding five and four orders of magnitude. The progression from distorted recovery in Fig. 36(a),(d) to visually clean reconstruction in Fig. 36(c),(f) is fully consistent with these BER reductions, confirming that the proposed phase-aware CE delivers the end-to-end reliability gains predicted by the BER analysis in Fig. 35.

Table VII provides a system-level comparison of reported backscatter communication systems. The MIMO extension of the monostatic backscatter architecture in [48] demonstrates BPSK with pilot-based LS estimation at close range. The ASK-based system in [49] achieves 67 kbps with energy consumption of 100 nJ/bit over a 5 m range, while the bistatic system in [50], employing OOK and frequency-shift keying (FSK), supports 1 kbps over 130 m but requires 20 μJ/bit, making it unsuitable for energy-constrained IoT nodes. Ambient backscatter approaches demonstrate varied trade-offs: the 4-PSK system in [51] achieves only 20 kbps over 0.76 m without pilot-based estimation, while the differential BPSK (DBPSK) design in [52] supports a higher throughput of 1 Mbps over 15 m but at a much higher energy cost of 1.8 nJ/bit. In contrast, the proposed platform demonstrates a balanced and scalable design, achieving 500 kbps at 1 m with a tag-side energy efficiency of 20 pJ/bit at the RF switch. While the inclusion of the general-purpose MCU brings the total energy consumption to 22.4 nJ/bit, the intrinsic backscatter modulation power remains at the μW level. These results establish the practicality of the proposed platform as one of the first multimedia-capable backscatter IoT system that jointly achieves high throughput, ultra-low energy consumption, and estimation robustness.

The reported 22.4 nJ/bit tag-side efficiency was dominated by the unoptimized general-purpose MCU, which accounts for approximately 99.9% of the total tag power, while the RF switch consumes only 10 μW. Therefore, this value reflects controller overhead rather than the intrinsic backscatter modulation cost, represented by the 20 pJ/bit switch-level efficiency. The overhead can be reduced either by replacing the general-purpose MCU with a duty-cycled commercial ultra-low-power MCU, yielding tag-side efficiency on the order of a few hundred pJ/bit [53], or by integrating the load-modulation switch and control logic into a custom backscatter application-specific integrated circuit (ASIC), approaching the intrinsic switch-level bound [54]. Because the proposed estimation and resource-allocation framework is implemented at the reader-side baseband, these tag-side substitutions require no modification to the signal-processing pipeline.

Although the prototype was validated at 1 m, longer-range behavior can be interpreted using the monostatic product-channel model. The received backscatter power follows the monostatic two-way free-space scaling d^{-4}, so increasing d reduces the operating SNR and shifts the BER curves in Fig. 35 to the right. The pilot-allocation rule in (12)–(15) increases the training length to maintain reliability until the frame-budget constraints become active, while the receiver chain remains unchanged. Beyond the feasible SNR region, range is limited by the link budget rather than by the proposed signal processing [55], so extension requires complementary improvements in antenna gain, Tx–Rx isolation, tag reflection efficiency, or transmit power within regulatory limits.

VII. Conclusion and Future Directions

This work demonstrates that reliable multimedia data communication is feasible in ultra-low-power backscatter settings, as required by modern IoT applications. A cost-effective and energy-efficient monostatic system is presented, integrating a custom tag, an SDR-based monostatic reader, a

2×1 planar Yagi–Uda array, and a resonator network that enhances isolation without degrading radiation performance. The signal chain is rigorously derived and fully realized in practice: pre-processing at the reader stabilizes the carrier and suppresses deterministic offsets, while post-processing at the host applies the proposed lifted LS/LMMSE and ZF algorithms prior to ID and IR. The proposed estimators mitigate residual phase-slope across each frame under the double-Rician fading structure, thereby enabling robust OOK and BPSK operation across the targeted operating SNR range. End-to-end experiments confirmed the alignment among the theoretical model, hardware design, and software pipeline for consistent multimedia recovery.

A promising future direction is to enhance the lifted estimators using lightweight deep learning. This refinement would further suppress residual deterministic phase errors under large phase variations, without sacrificing the interpretability or low overhead of the proposed framework. At the system level, the same architecture also points to several practical extensions. Energy harvesting at the tag could eliminate batteries and support sustained multimedia links under practical power budgets. Higher-order modulation can be enabled by replacing the SPDT switch with a calibrated multi-state load network supporting constellations such as quadrature phase-shift keying (QPSK) or 16-QAM. The core lifted CE framework remains applicable when the pilot symbols and reflection states are known, while the main practical constraints become stricter antenna isolation, load-state calibration, and EVM requirements.

Multi-antenna readers and tags could further enhance performance through spatial diversity and interference rejection, achieving MIMO-level reliability and throughput while keeping the reader compact. Furthermore, the per-tag estimation and resource allocation framework extends to multi-tag scenarios through pilot-domain separation (Section IV-B). Specifically, time-division pilot access allows each tag's CE to be addressed using the single-tag lifted model without modifying the underlying estimator structure. Alternatively, frequency-division separation using per-tag subcarrier offsets provides a viable option when programmable switching rates are supported.

Practical deployment challenges, including near-far power imbalance, successive interference cancellation, and medium access control design, remain important extensions of this work. Multi-tag operation further requires frame-level synchronization with guard-time control to prevent pilot-slot overlap, as well as per-tag isolation calibration to suppress leakage from nonideal matched-load termination of inactive tags. Addressing these issues would extend the proposed framework to broader IoT deployments while preserving the cost-effective and energy-efficient design philosophy of the present system.

APPENDIX

A. Derivation of Equation (70)

Let $\gamma = (E_s/N_0)\left|h_{eq}\right|^2 = \bar{\gamma}_{\text{eff}}\,|h_{tt}|^2|h_{tr}|^2$ denote the instantaneous SNR, where the normalization $\mathbb{E}[|h_i|^2] = 1$ is adopted so that $\bar{\gamma}_{\text{eff}} = \mathcal{E}_s/N_0$ per (4). For OOK detection with perfect CSI, the conditional BER is

$$P_e^{\text{OOK}}(\gamma) = Q\left(\sqrt{\frac{\gamma}{2}}\right), \tag{74}$$

and the average BER follows from (5) as

$$\bar{P}_b^{\text{OOK}} = \int_0^\infty Q\left(\sqrt{\frac{\gamma}{2}}\right) f_\gamma(\gamma)d\gamma\,. \tag{75}$$

Direct substitution of the double-Rician envelope density (1) into (75) involves the infinite double series in (1) together with an integral of $Q(\cdot)$ against the modified Bessel functions, which does not reduce to a standard tabulated form. The MGF approach circumvents this difficulty by decoupling the Q-function from the envelope density. Craig's alternative representation of the Gaussian Q-function,

$$Q(x) = \frac{1}{\pi}\int_0^{\pi/2} \exp\left(-\frac{x^2}{2\sin^2\theta}\right)d\theta\,, \qquad x \geq 0, \tag{76}$$

upon substitution into (75) and exchange of integration order, reduces the average BER to a single finite-interval integral of the SNR MGF $\mathcal{M}_\gamma(s) \triangleq \mathbb{E}[e^{-s\gamma}]$:

$$\bar{P}_b^{\text{OOK}} = \frac{1}{\pi}\int_0^{\pi/2} \mathcal{M}_\gamma\left(\frac{1}{4\sin^2\theta}\right)d\theta\,. \tag{77}$$

The MGF admits a tractable form by exploiting the independence of the two hops. Let $u \triangleq |h_{tt}|^2$. Under the normalization $\mathbb{E}[|h_{tr}|^2] = 1$, the MGF of the second hop power follows a non-central chi-squared distribution with Rician K-factor K_{tr} [56] and is given by,

$$\mathbb{E}[e^{-s|h_{tr}|^2}] = \frac{K_{tr}+1}{K_{tr}+1+s}\exp\left(-\frac{K_{tr}s}{K_{tr}+1+s}\right). \tag{78}$$

Averaging (78) over the first-hop envelope-squared density

$$\begin{aligned} f_u(u) &= (K_{tt}+1)\exp(-K_{tt}-(K_{tt}+1)u) \\ &\quad \times I_0\left(2\sqrt{K_{tt}(K_{tt}+1)u}\right), \qquad u \geq 0, \end{aligned} \tag{79}$$

where $I_0(\cdot)$ denotes the modified Bessel function of the first kind, yields the double-Rician SNR MGF

$$\begin{aligned} \mathcal{M}_\gamma(s) = \int_0^\infty & \frac{K_{tr}+1}{K_{tr}+1+s\bar{\gamma}_{\text{eff}}u} \\ & \times \exp\left(-\frac{K_{tr}s\bar{\gamma}_{\text{eff}}u}{K_{tr}+1+s\bar{\gamma}_{\text{eff}}u}\right) f_u(u)du. \end{aligned} \tag{80}$$

Equations (77) and (80) together constitute the exact average BER of OOK under double-Rician fading. The outer integral in (77) is over the finite interval $[0, \pi/2]$; the inner integral in (80) has effective support on $[0, \mathcal{O}(1/(K_{tt}+1))]$ due to the exponential decay of (79). Both are directly amenable to standard numerical quadrature.

B. Derivation of Equation (71)

For coherent BPSK detection with perfect CSI, the conditional BER is

$$P_e^{\mathrm{BPSK}}(\gamma) = Q\left(\sqrt{2\gamma}\right), \quad (81)$$

and the average BER over the γ is

$$\bar{P}_b^{\mathrm{BPSK}} = \int_0^{\infty} Q\left(\sqrt{2\gamma}\right) f_\gamma(\gamma) d\gamma. \quad (82)$$

Applying the same Craig-form reduction used in Appendix A gives,

$$\bar{P}_b^{\mathrm{BPSK}} = \frac{1}{\pi}\int_0^{\pi/2} \mathcal{M}_\gamma\left(\frac{1}{\sin^2\theta}\right) d\theta\,, \quad (83)$$

where $\mathcal{M}_\gamma(\cdot)$ is the double-Rician SNR MGF specified in (80). Equations (80) and (83) jointly define the exact average BER of BPSK under double-Rician fading. Together with the OOK counterpart in Appendix A, they provide the analytical benchmarks against which the Monte Carlo simulations in Fig. 35 are compared.

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